

This Could Be Anywhere

Winning Geoguessr using Convolutional Neural Networks

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GitHub: <https://github.com/890tsugua/AppliedML2026>

Data (Google Drive): <https://drive.google.com/drive/folders/1SV1wbInxP3pNCU2XCWl14cqe2YSgLwjH?usp=sharing>

UNIVERSITY OF COPENHAGEN



Where did Troels go on vacation?



The dataset

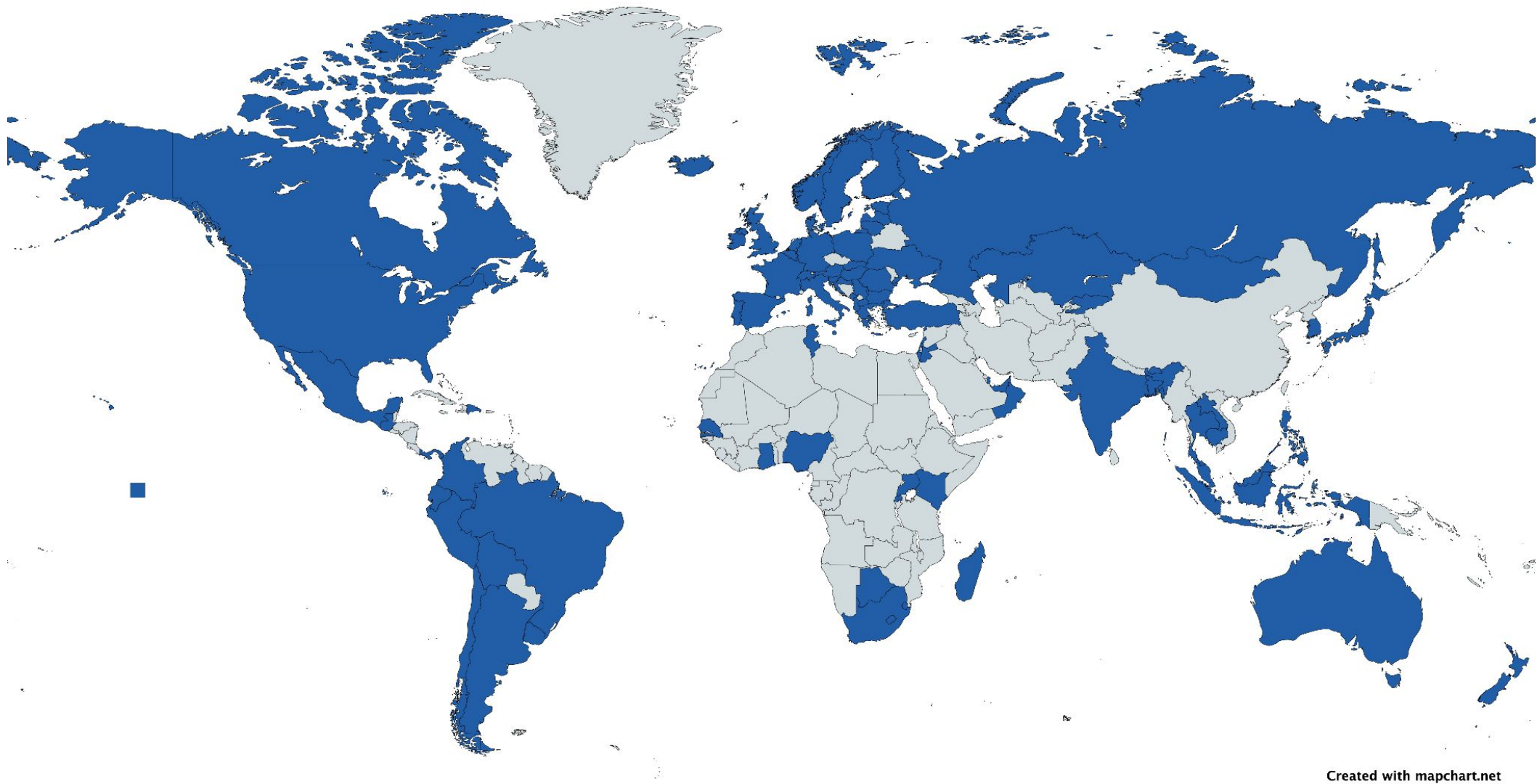
- Images from around the world
- Scraped from Geoguessr*
- ~ 300 images per country
- 1600 x 900 image size
- ~ 60 GB
- Two folders: test and train (80/20)



*Credit Pepijn van Wijk (deboradum on GitHub):

<https://huggingface.co/datasets/deboradum/GeoGuessr-countries>

Countries in the dataset



Example images



Colombia

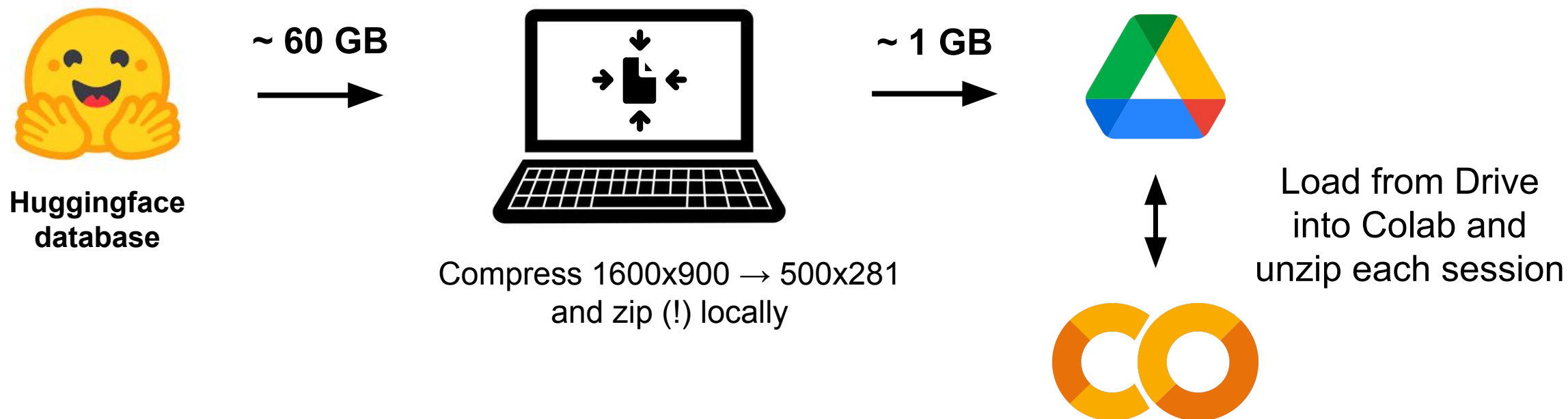


Denmark



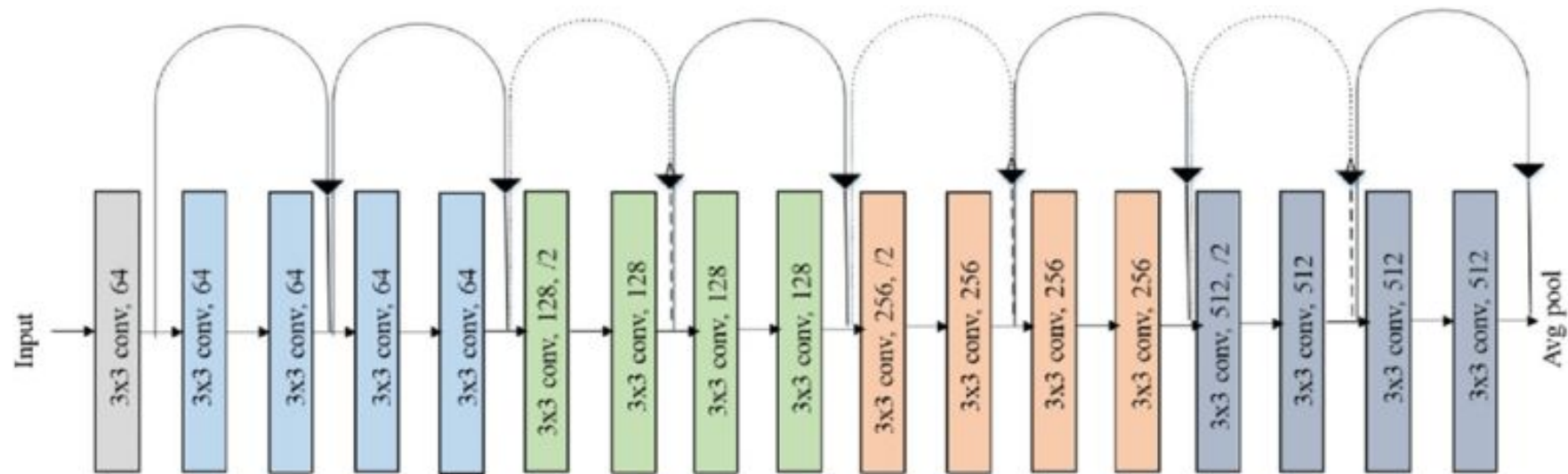
South Korea

Preparing the dataset

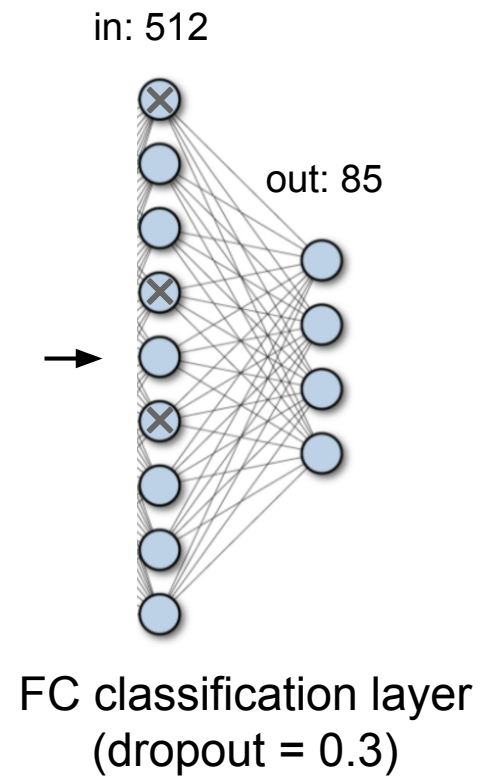


- We used the same Google account across PCs
- Gave us a space to share files, results and access to good GPU
- Would mostly edit in VScode and push to Github, then clone repo into Colab.

Pretrained ResNets



Resnet18
(4 trainable layers)



Loss function: **Cross-entropy loss**

Data augmentation

- Artificially create new data samples for training
- Involves random cropping, rotation, color jitter
- Random data augmentation → better performance.
- Pre-computed augmentation → severe overtraining



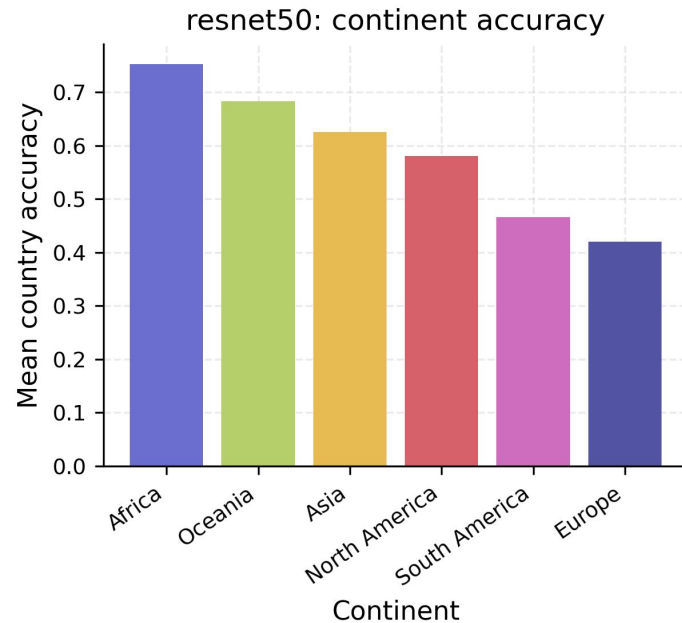
Somewhere in Albania...



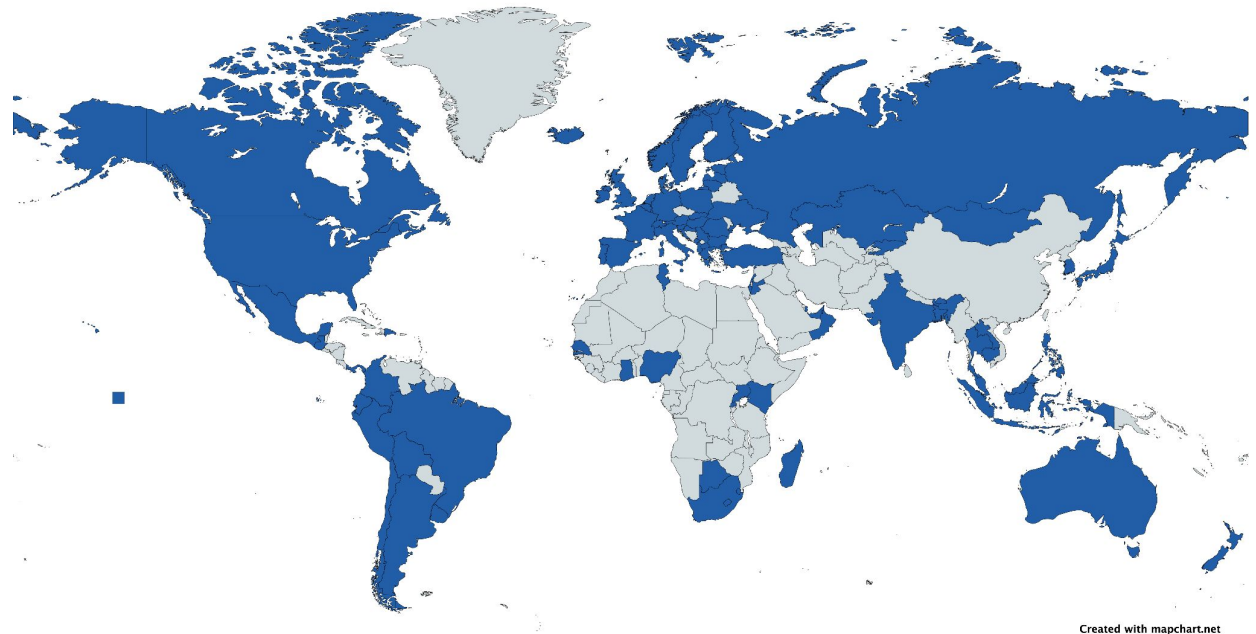
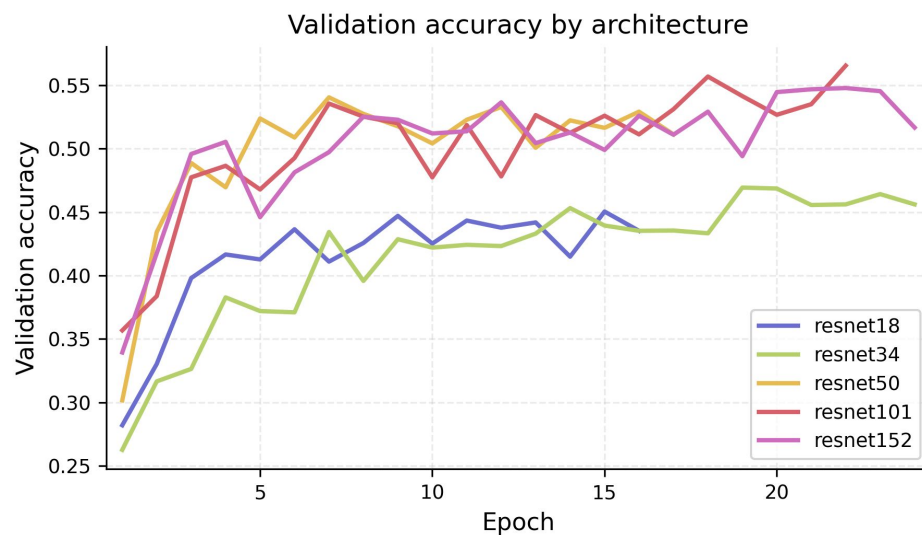
GPUMaxxing

- Colab gives you 100 Compute Units. Use wisely!
 - Do *not* choose a more powerful GPU than necessary. Consider how heavy your task is and check with `nvidia-smi`.
 - Usage starts already from connection, *not* when you start running code.
- CPU bottleneck: Image preprocessing / augmentation can be expensive.
 - Think about your transforms
- Dataloader settings:
 - `num_workers = 8 (!)`
 - `prefetch_factor = 2` or `4`

Initial results with dropout and simple augmentation



- Europe hard. Unique countries easy.
- Dataset imbalance and image artefacts



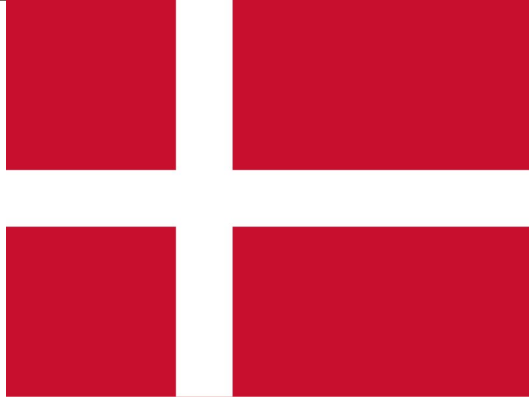
Why cluster countries?

Motivation



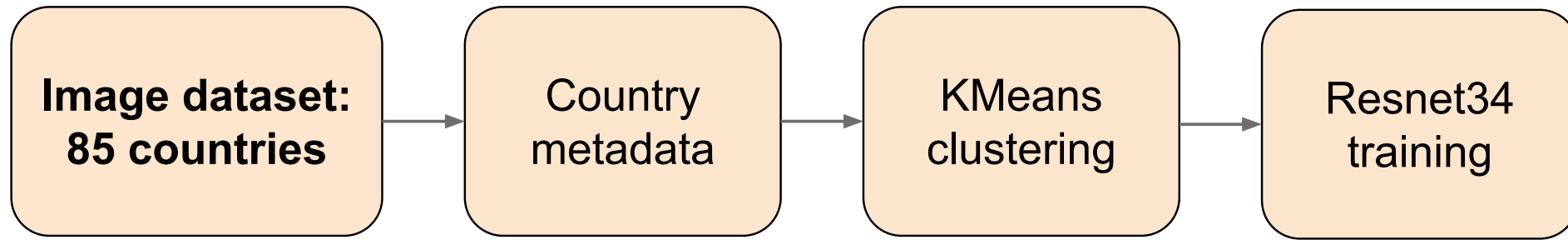
Why cluster countries?

Motivation



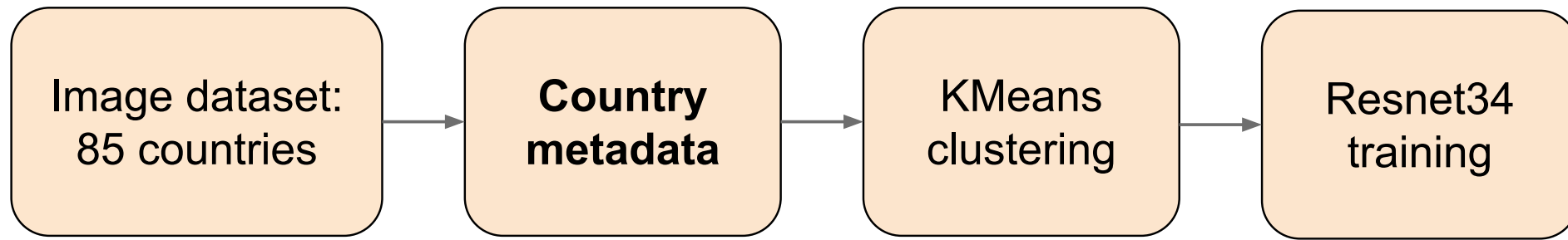
Cluster Based Country Image Classification

Setup



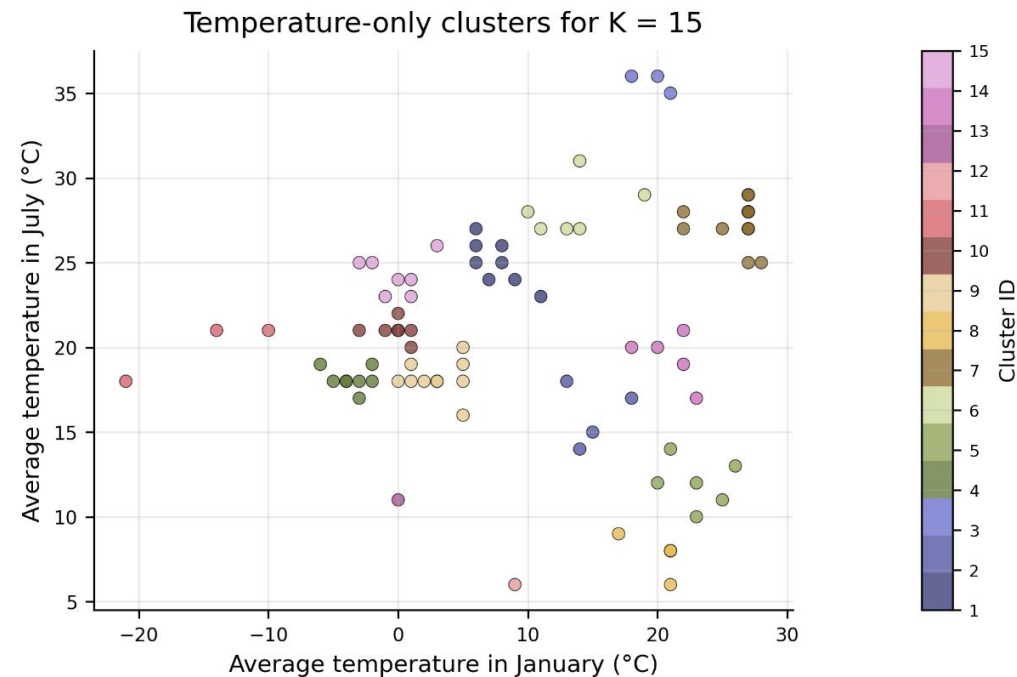
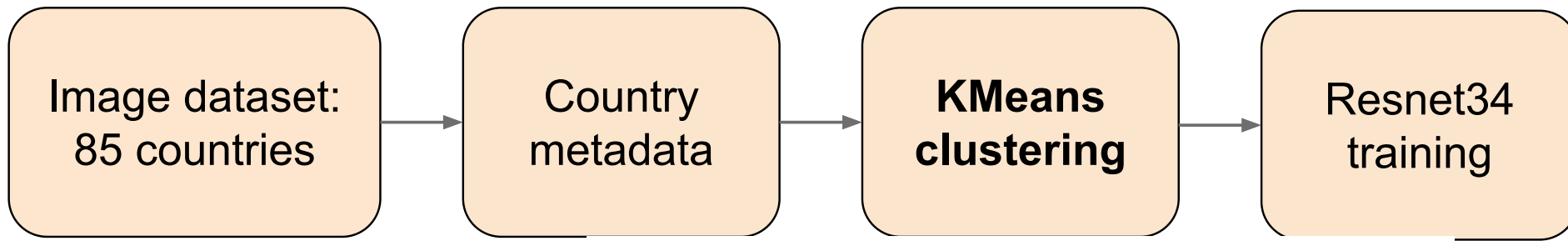
Cluster Based Country Image Classification

Setup



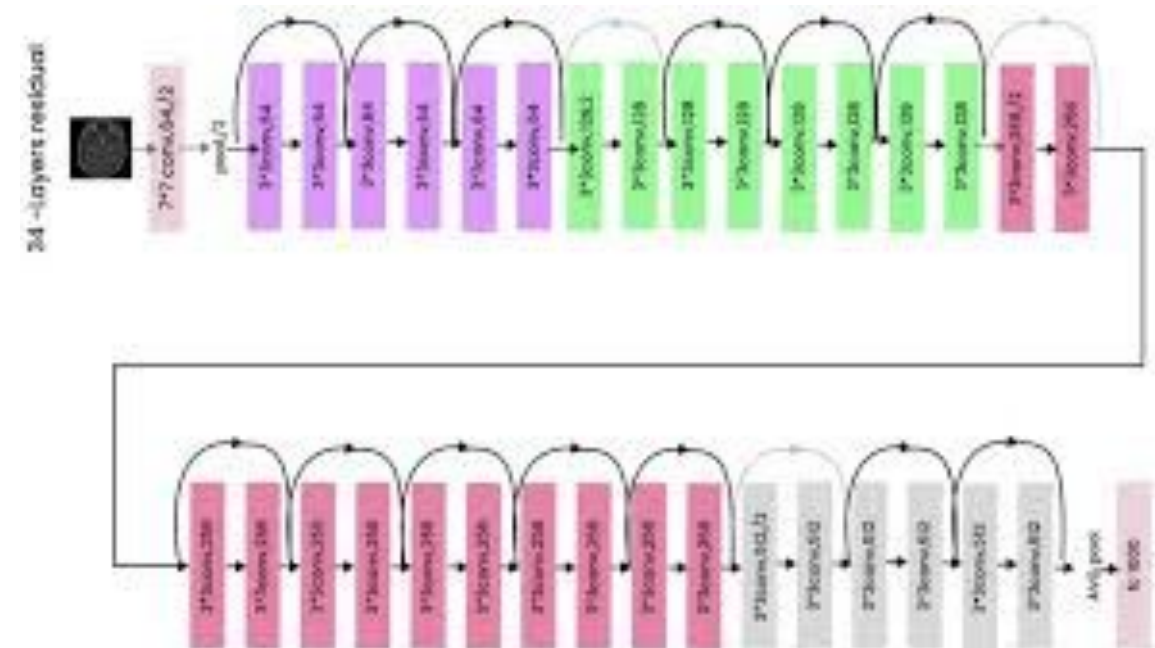
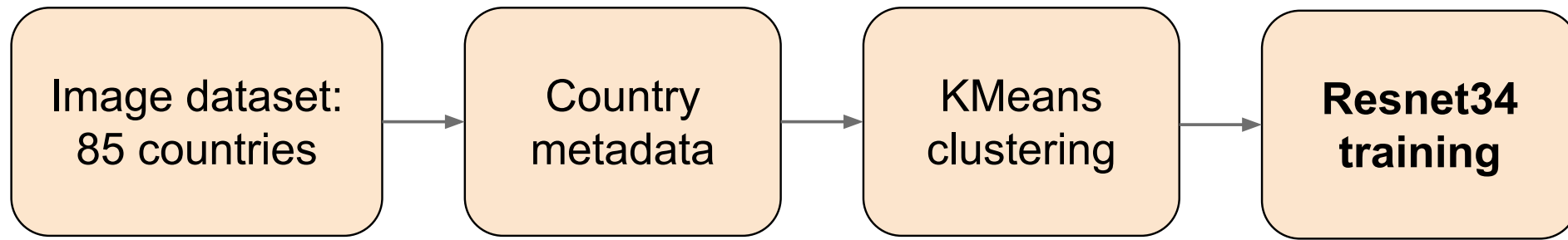
Cluster Based Country Image Classification

Setup



Cluster Based Country Image Classification

Setup



Cluster Based Country Image Classification

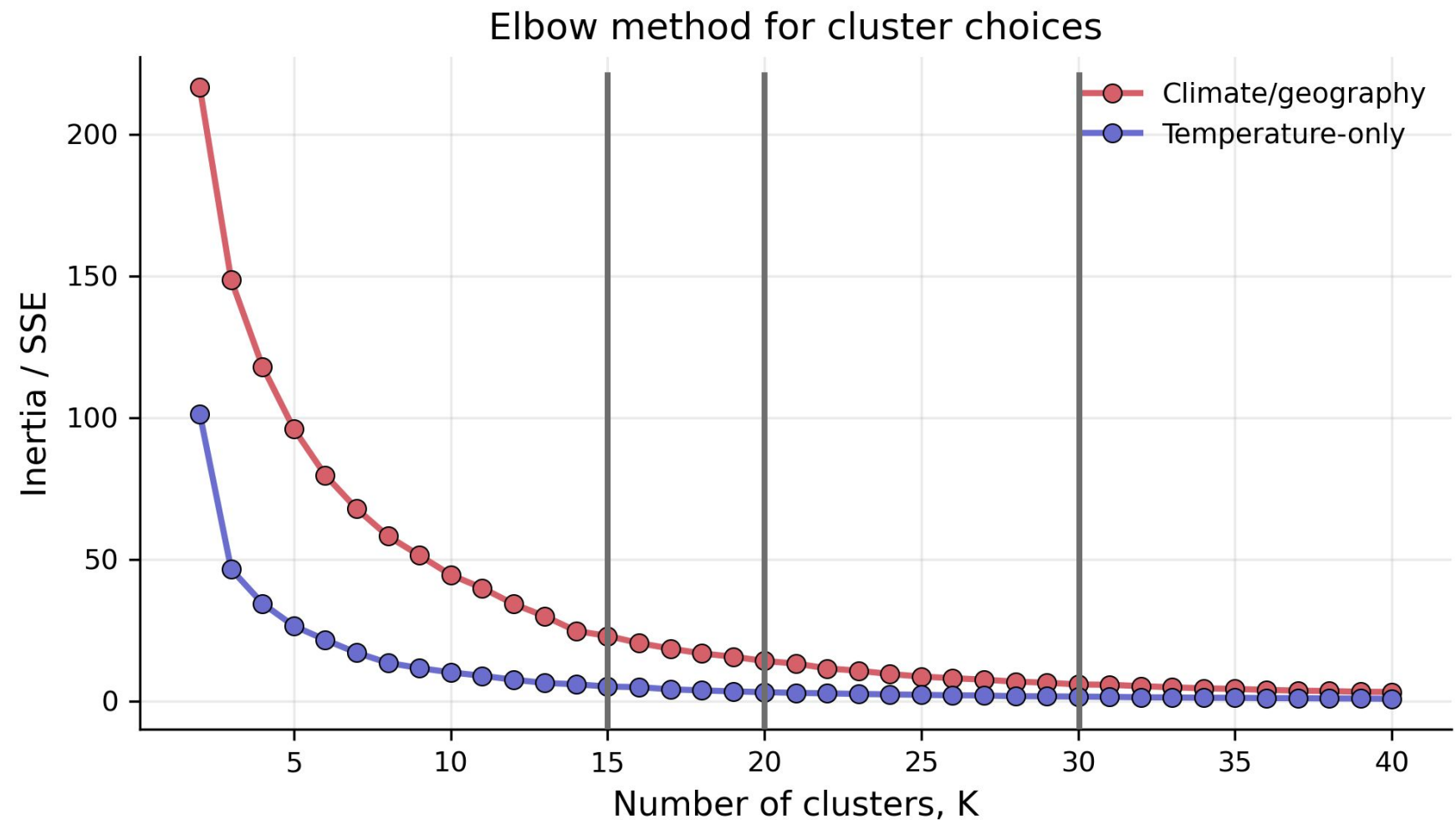
Two sets of features used for clustering

- **Temperature-only based clustering:**
 - Jan, July temperature (2 features)
- **Climate/Geography based clustering:**
 - Jan, July temp. + latitude and longitude (4 features)



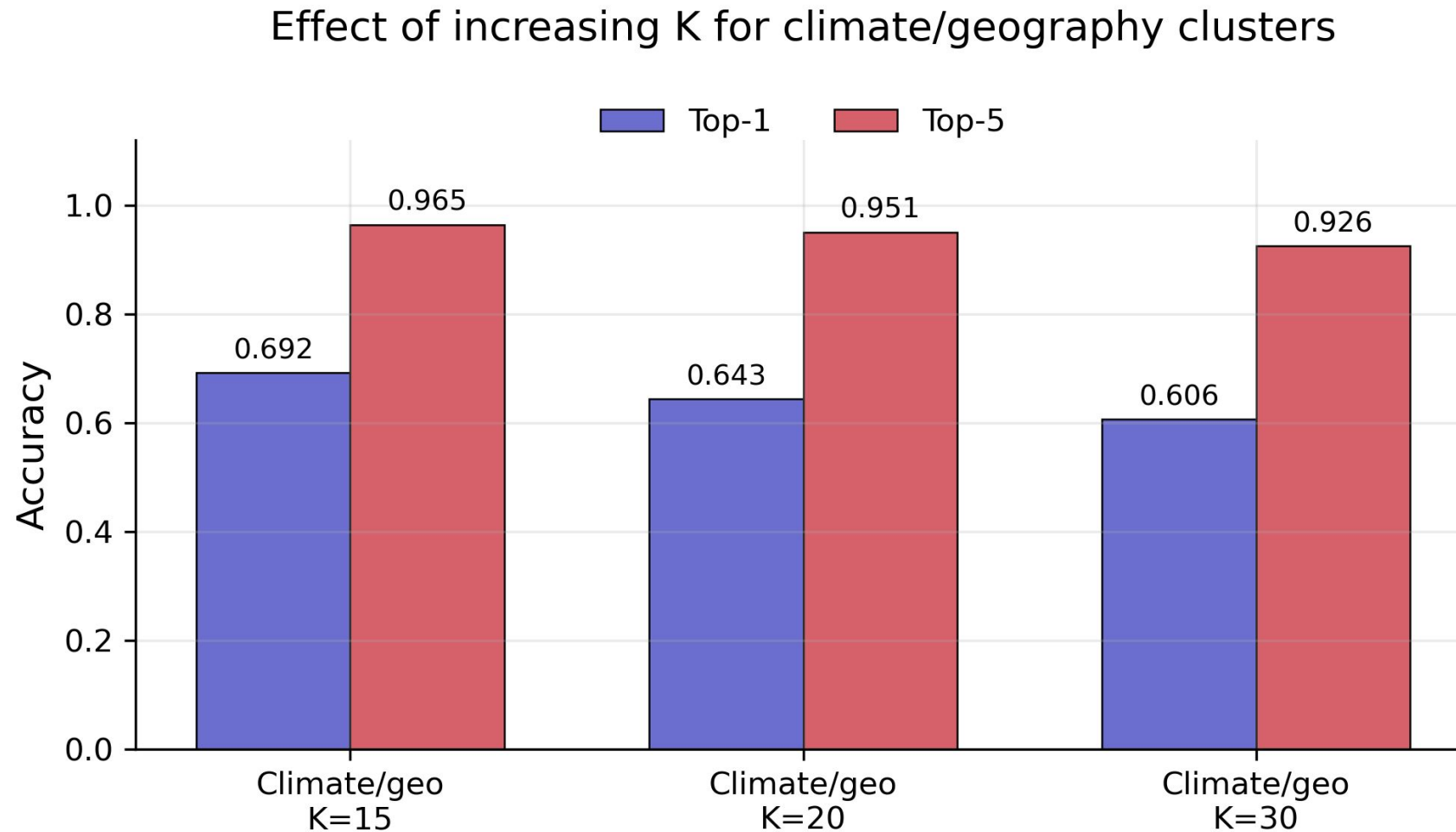
Cluster Based Country Image Classification

Choosing the number of clusters



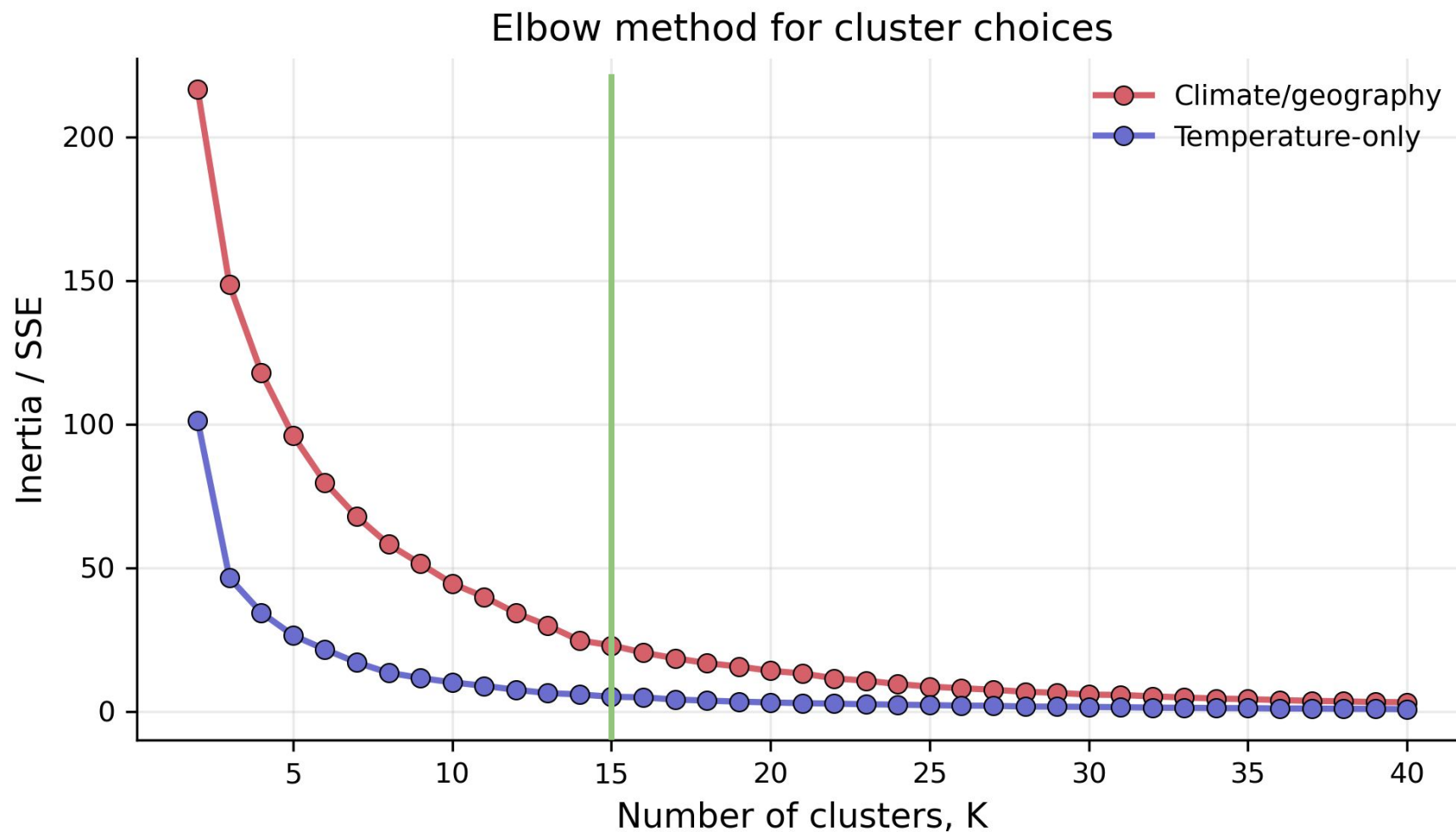
Predicting clusters from images using ResNet-34

Accuracy of prediction decreases with number of clusters

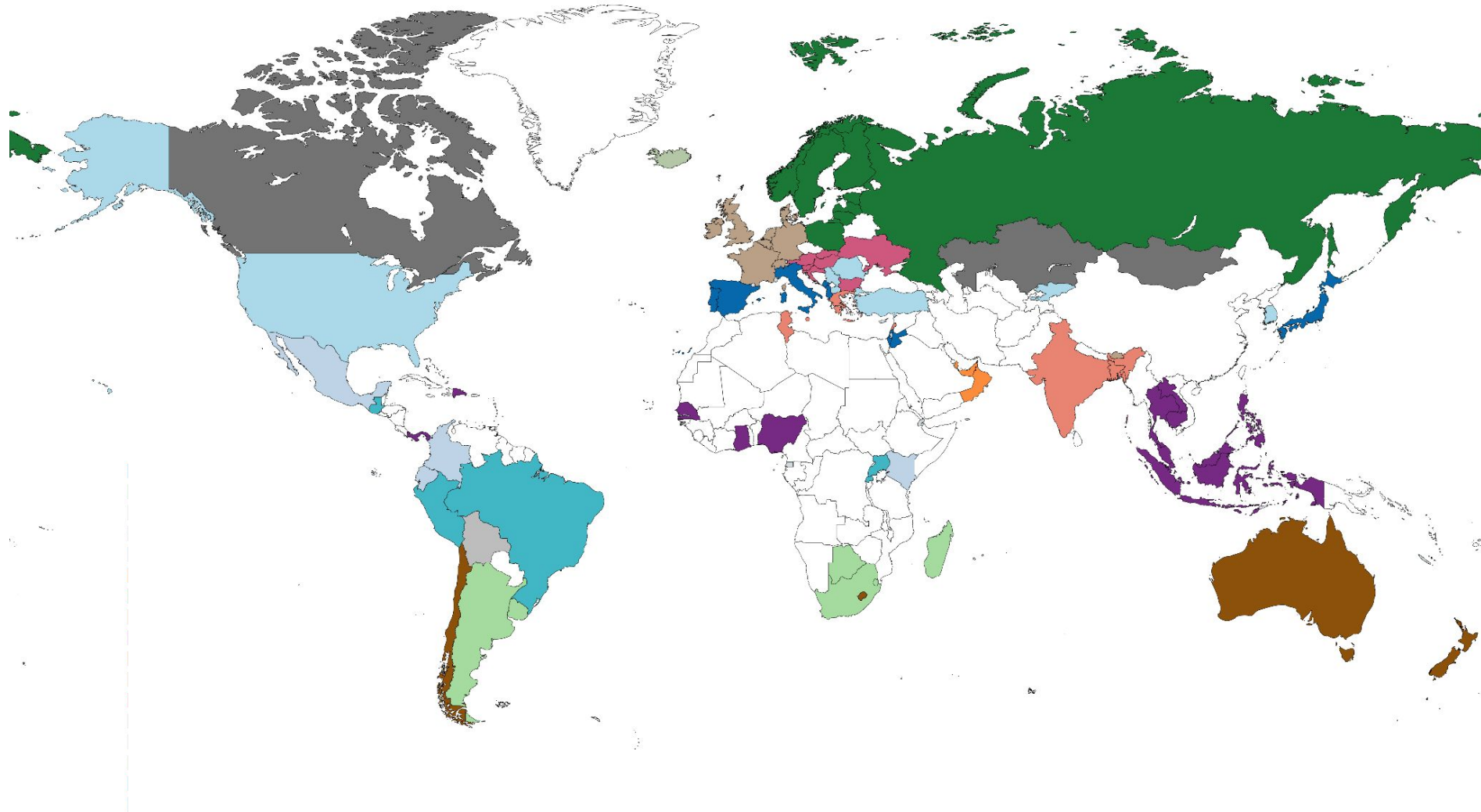


Cluster Based Country Image Classification

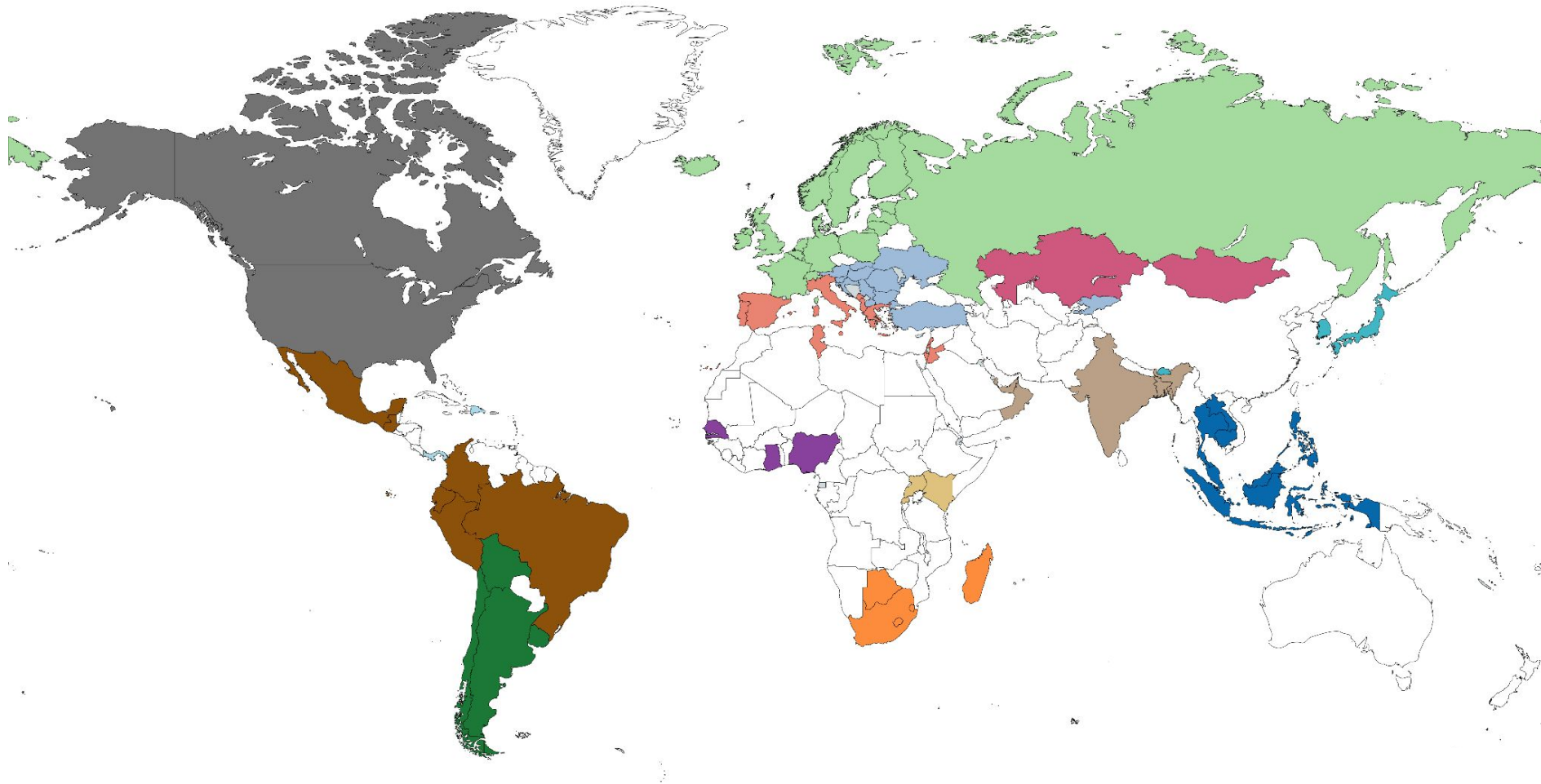
We will continue with 15 clusters



Temperature-only: Clusters are spread geographically

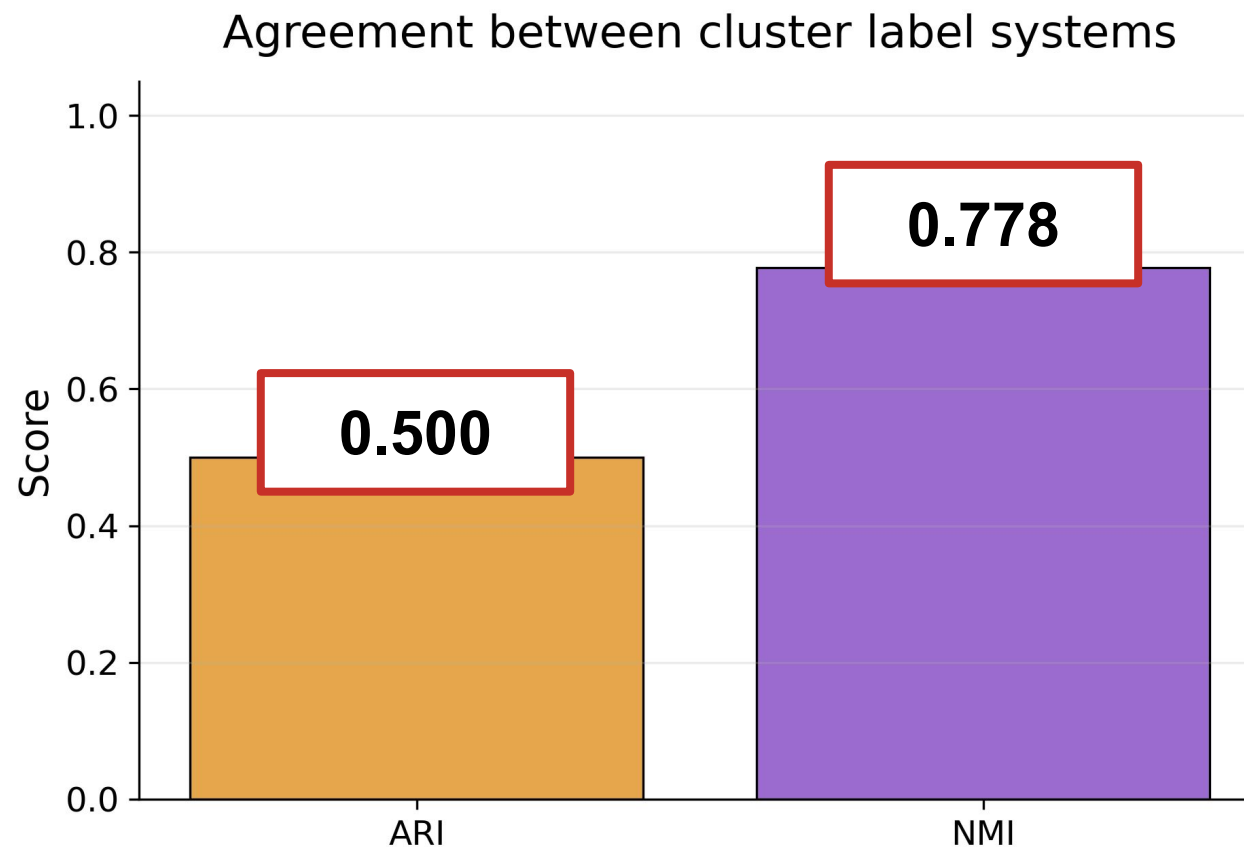


Climate/Geography: Clusters are grouped geographically



Comparing clustering for temperature-only vs. climate/geography

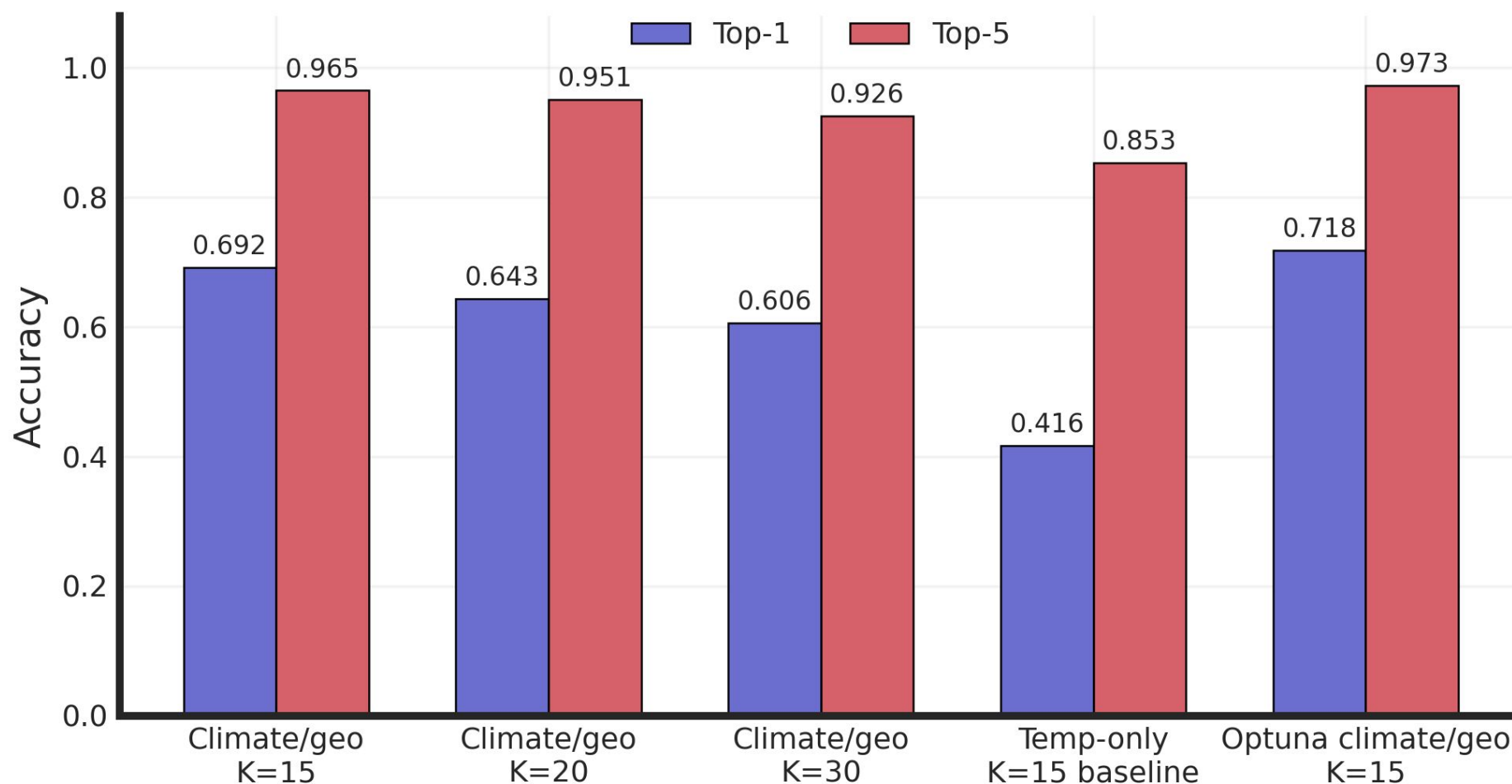
Adjusted Rand Index (ARI) and Normalized Mutual information (NMI)



Predicting clusters from images using ResNet-34

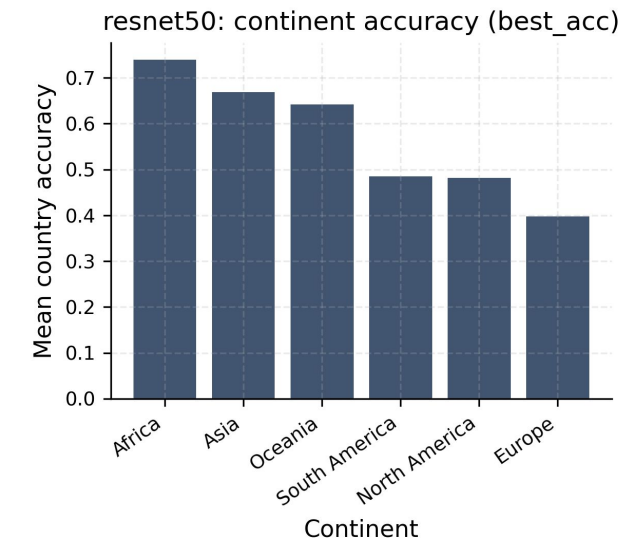
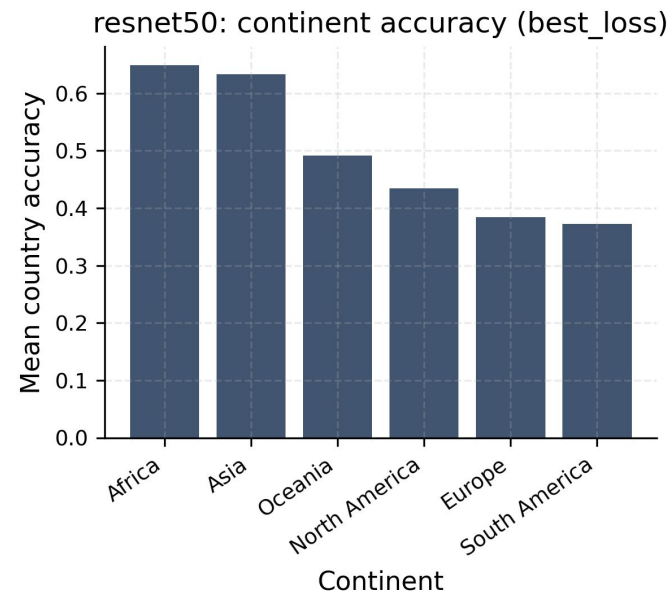
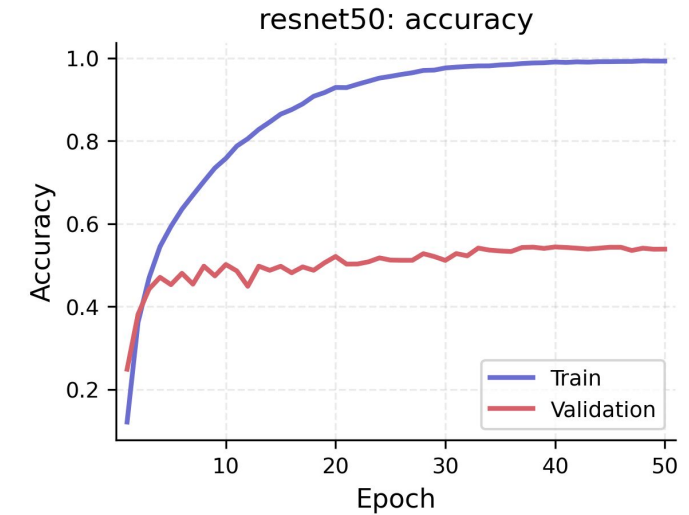
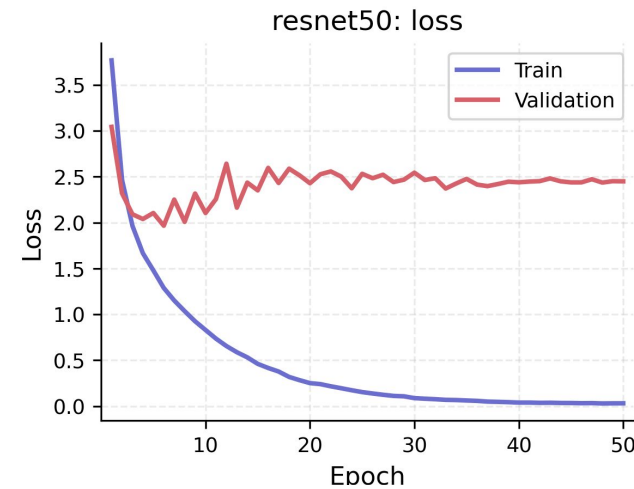
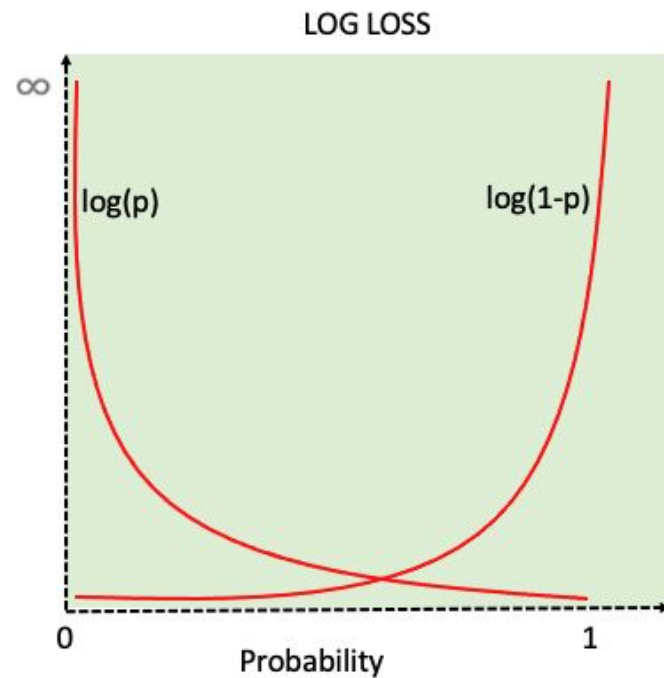
Prediction works best on climate/geography based clusters. Optuna helps.

Comparison of cluster label systems



Added weights to cross entropy loss

$$L = - \sum_{k=1}^K y_k \log(p_k)$$

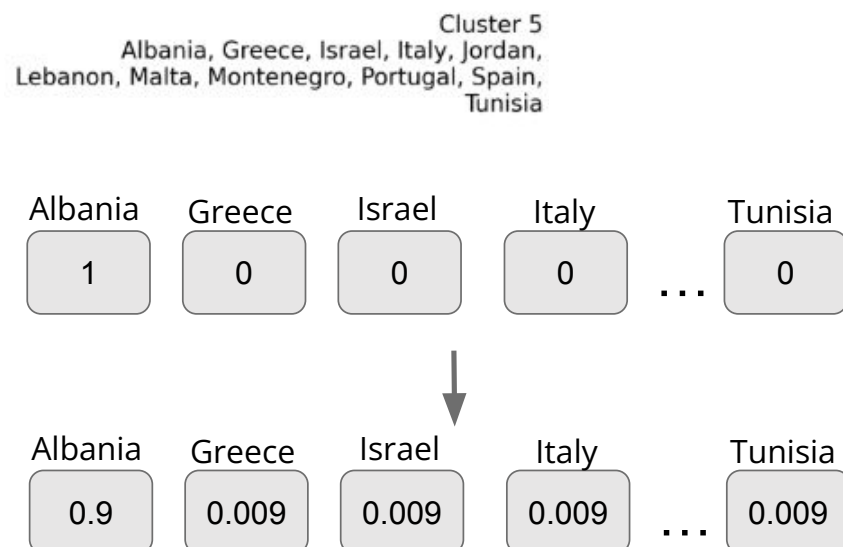


Label Smoothing

$$L = - \sum_{k=1}^K y_k \log(p_k)$$

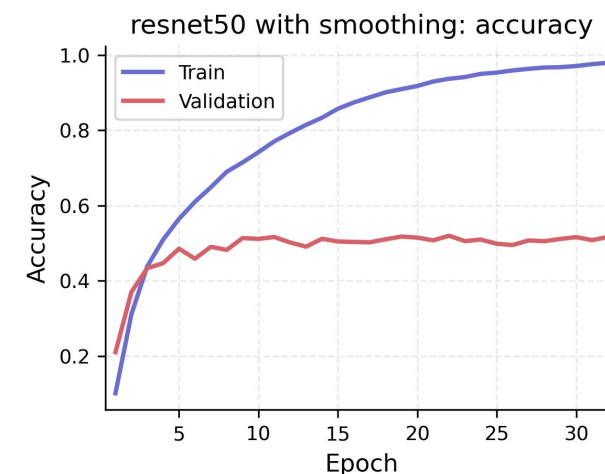
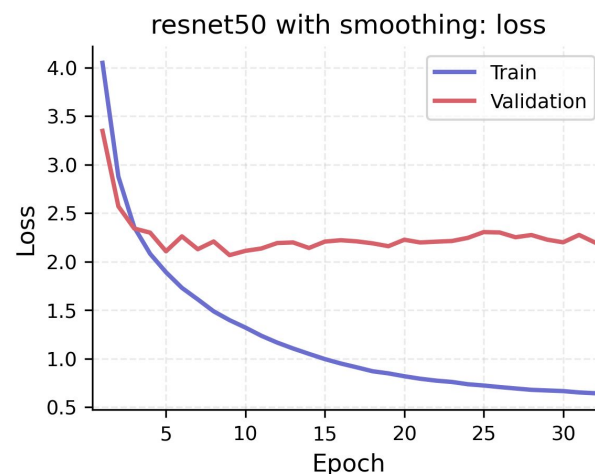
The k=15 clusters was used
With prob 0.9

Example (Albania in Cluster 5)

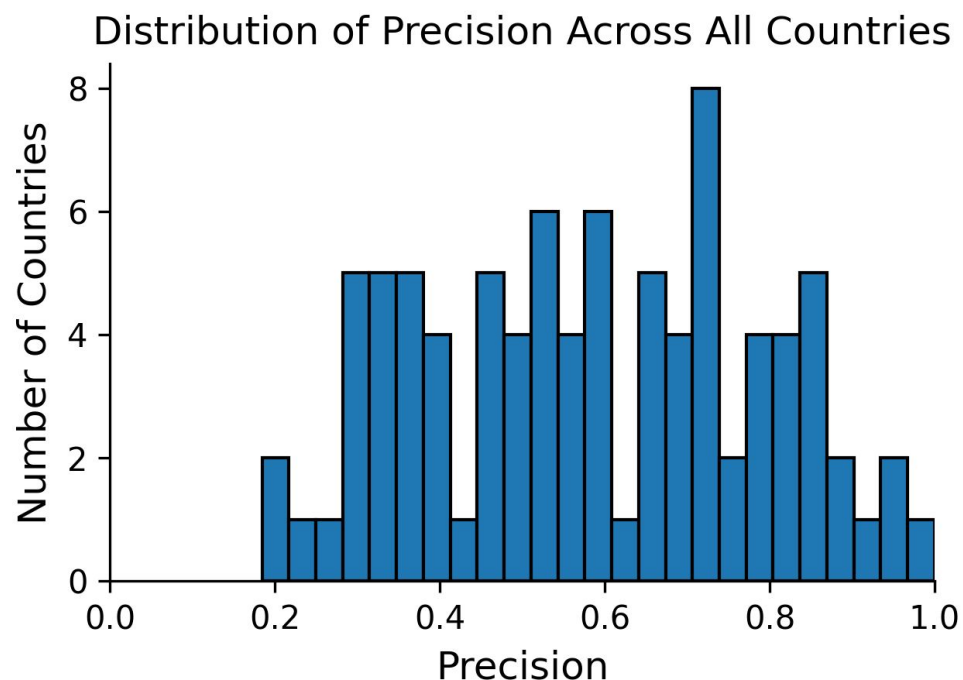


The clusters were probably too large, so the smoothing probability was distributed across too many countries.

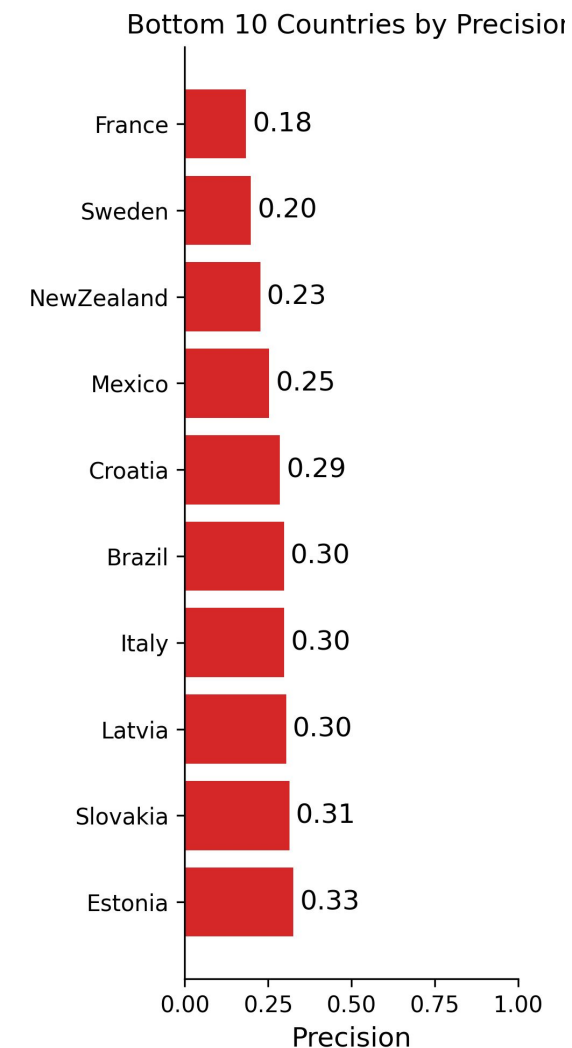
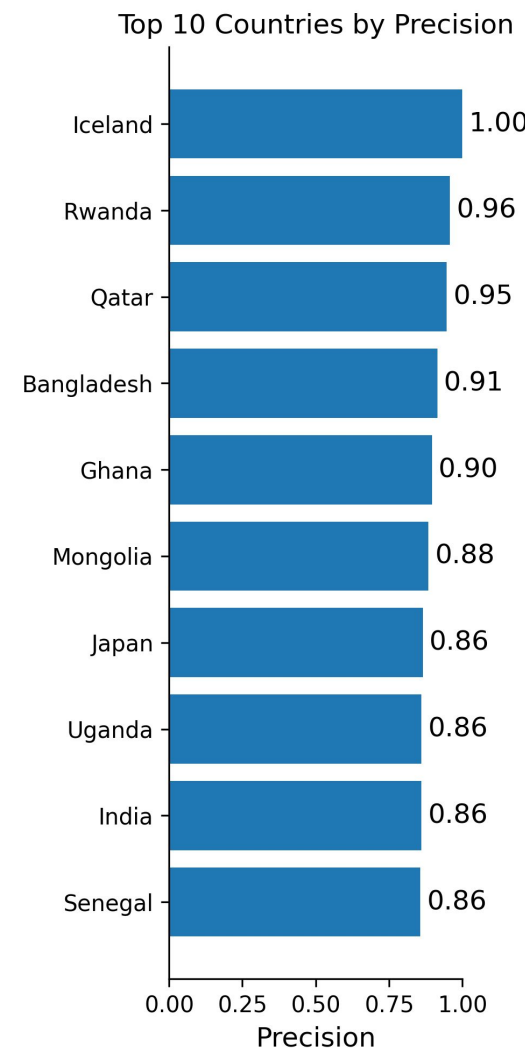
As a result, each extra country received very little probability, making the smoothed labels almost identical to normal one-hot labels.



Accuracy of prediction depends on the country



- Europe hard. Unique countries easy.
- Dataset imbalance and image artefacts



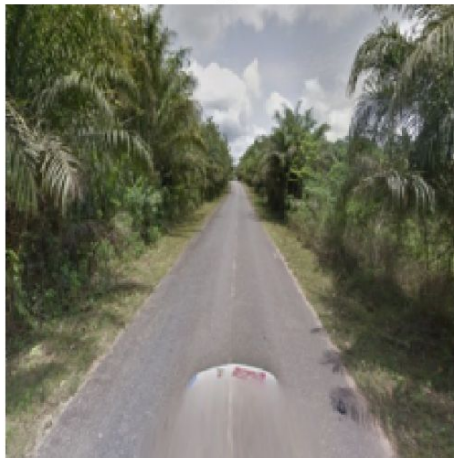
Dataset pollution: Car artefacts

Example images from Ghana

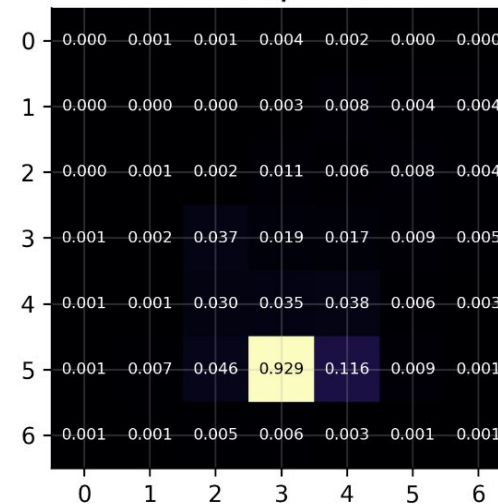


Network has learned the car hoods of specific countries...

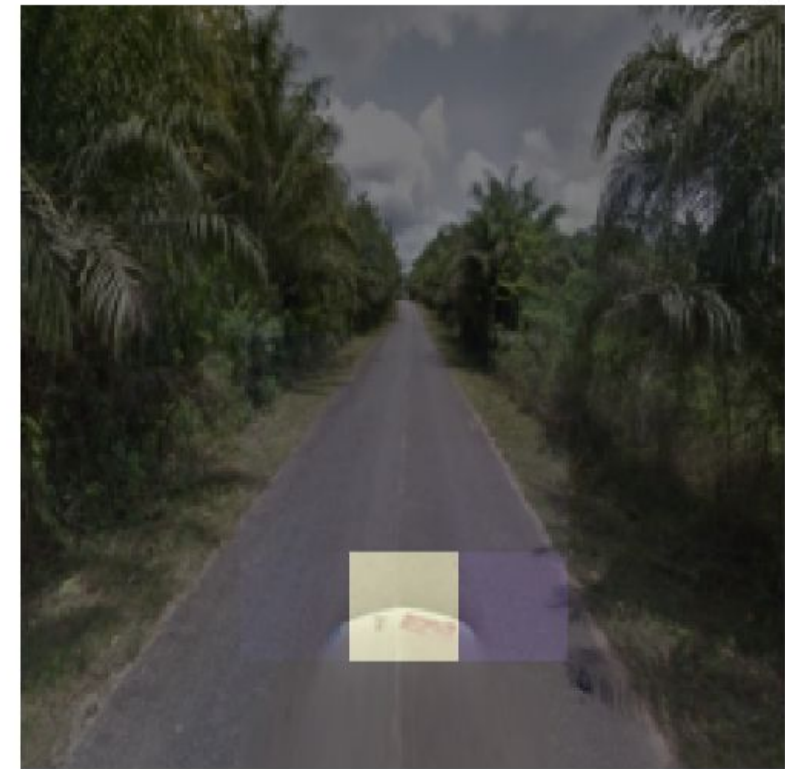
Original



Patch importance



Patch importance overlay



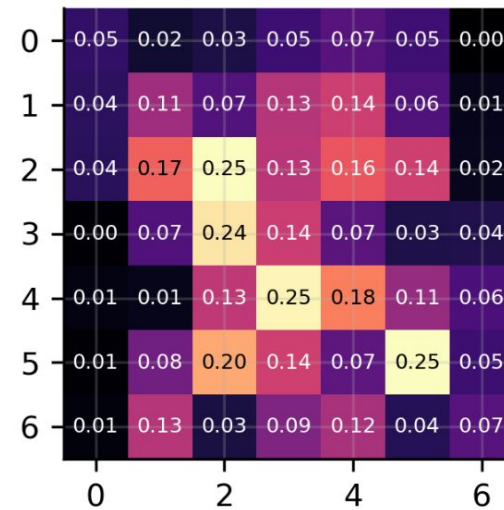
... but not always!

Correctly predicted Norway's cluster without car artefact.

Original



Patch importance

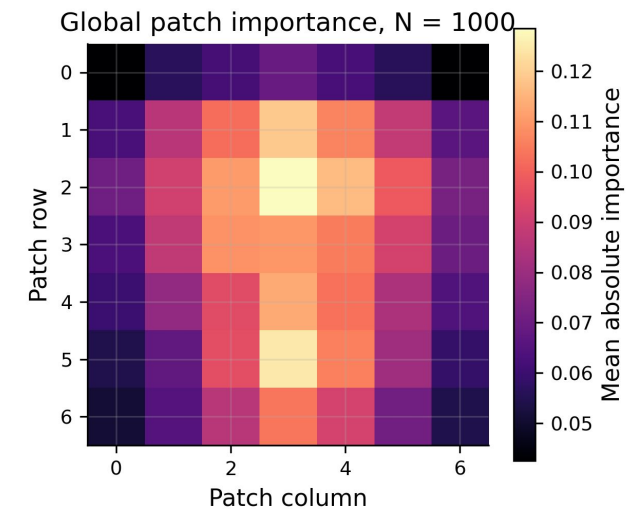
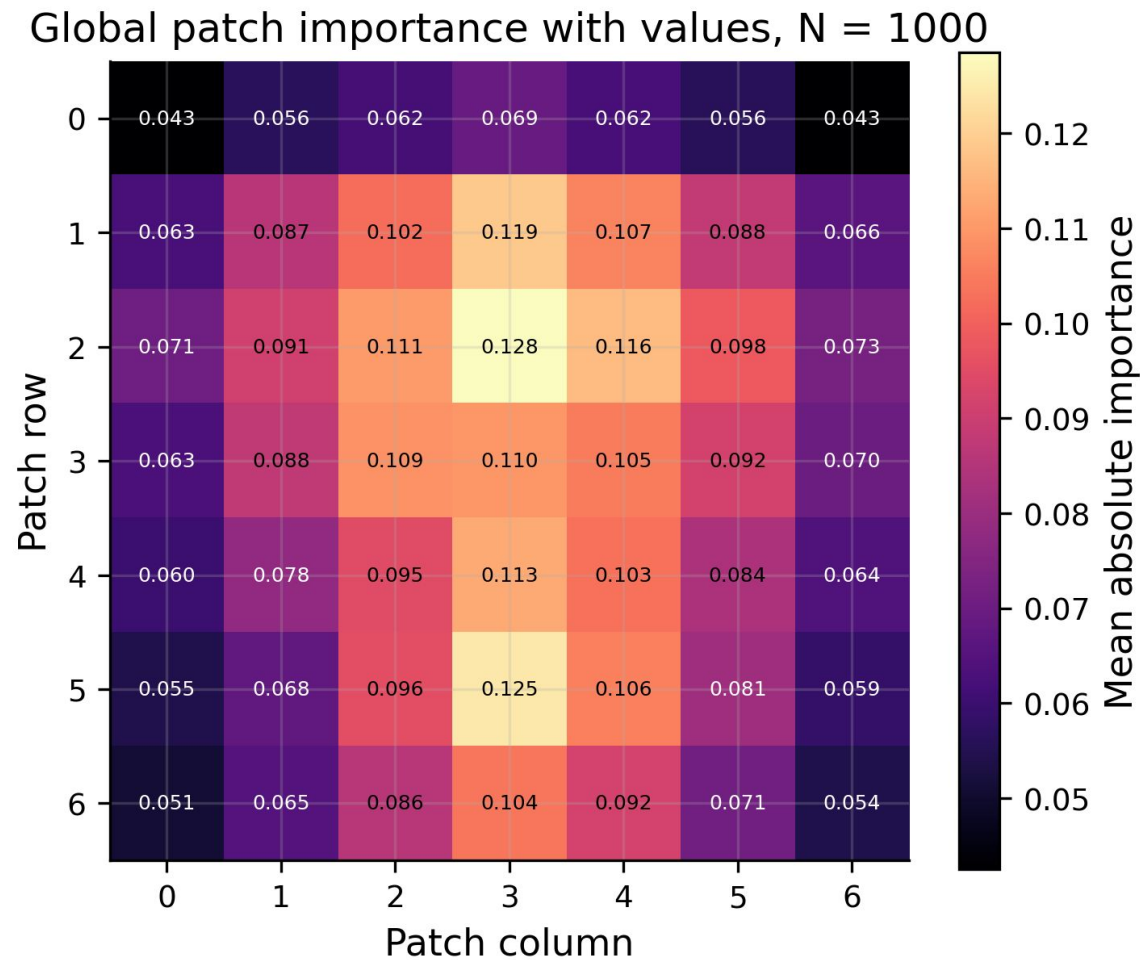


Overlay + patch IDs



Prediction relies not only on car artefacts

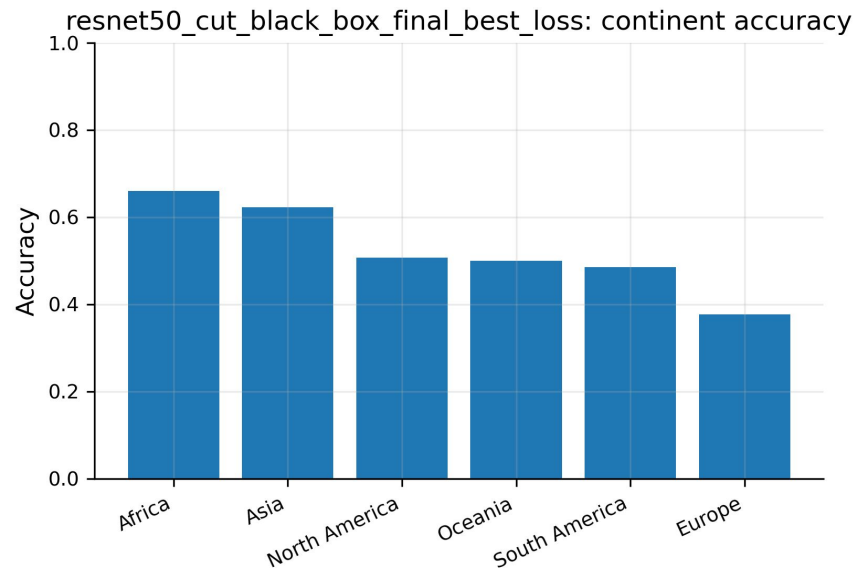
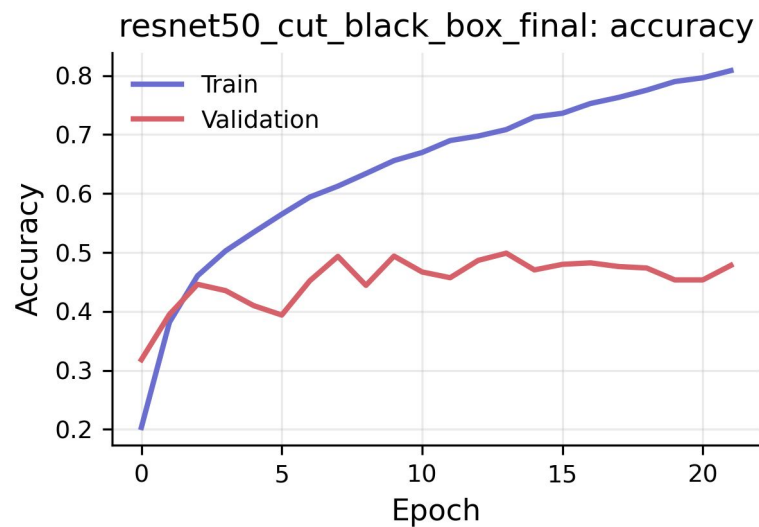
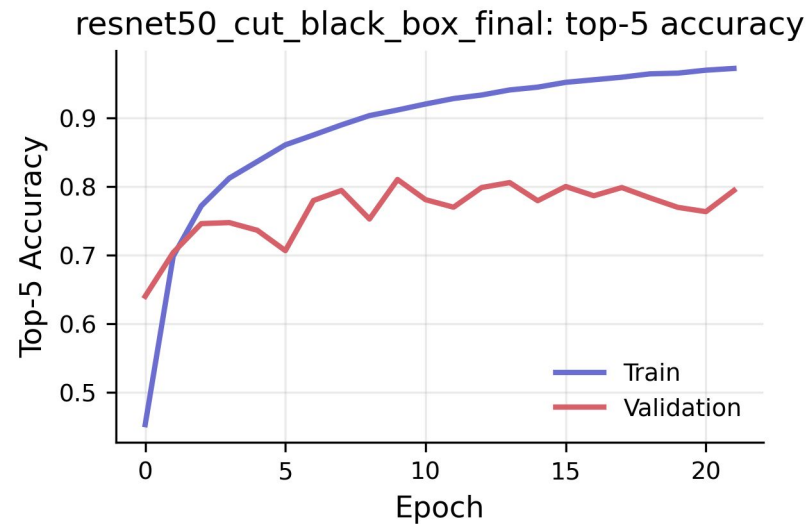
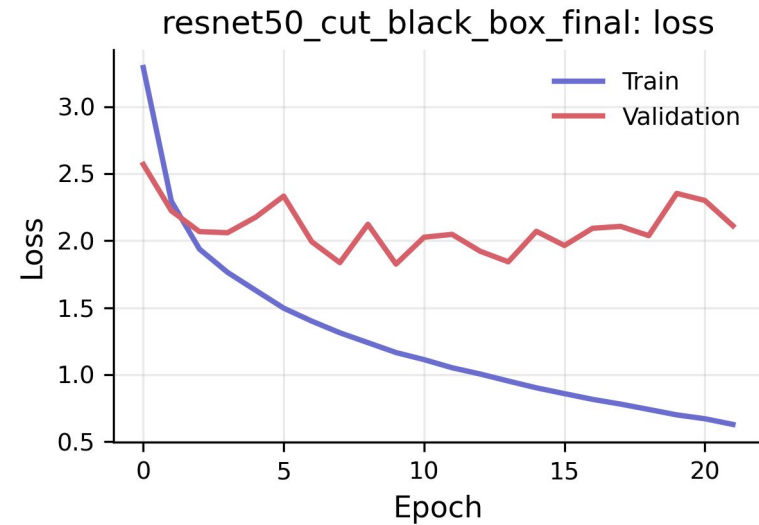
... but definitely relies heavily on car artefacts



Cutting out the cars: Inserting a black box



Last experiment with black box



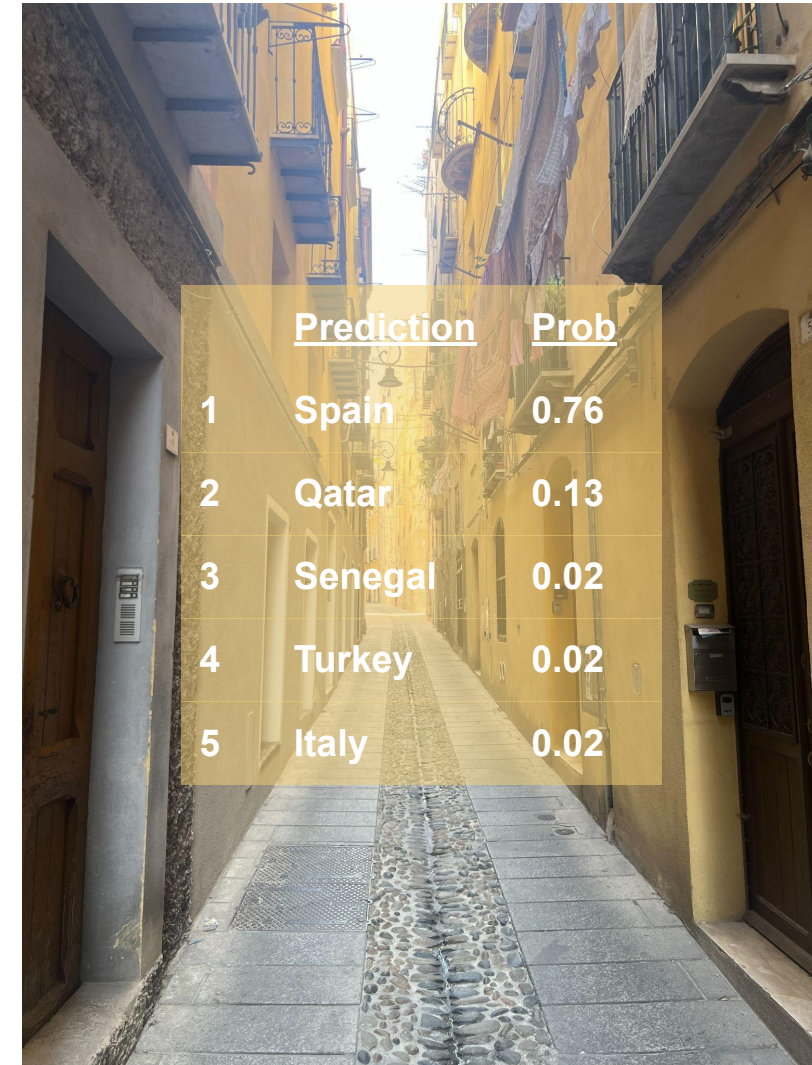
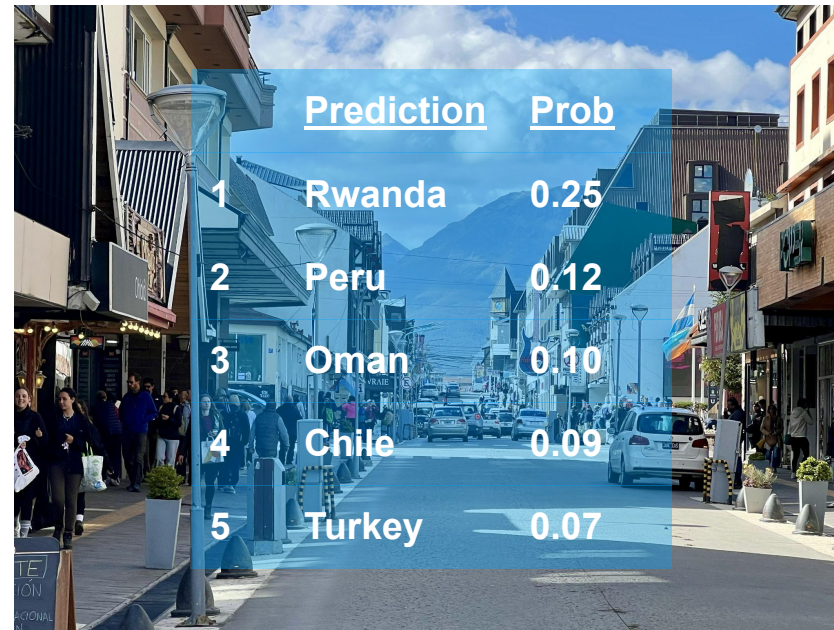
ResNet50 Test Accuracy

Top 1	Top 5
0.502	0.813

Where did Troels go on vacation?



Where did Troels go on vacation?

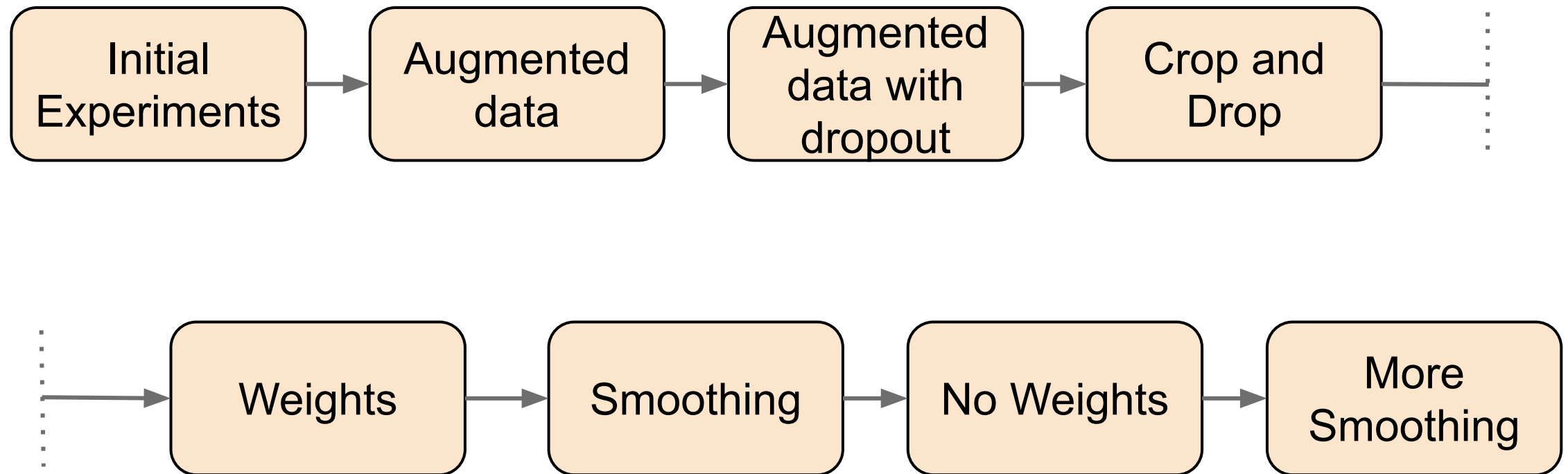


Future Directions

- Bigger data set (5 million images) with coordinates and more (<https://osv5m.github.io/>)
- *Really* deep and long training to learn details of road signs etc.

APPENDIX

Model Development Timeline



1. Augmentation and initial optimization

Optuna on ResNet34 with 4 trainable layers
20 trials with 5 epoch

Results:

Best loss value: 3.703

Best parameters:

lr: 0.0019

weight_decay: 9.26e-06

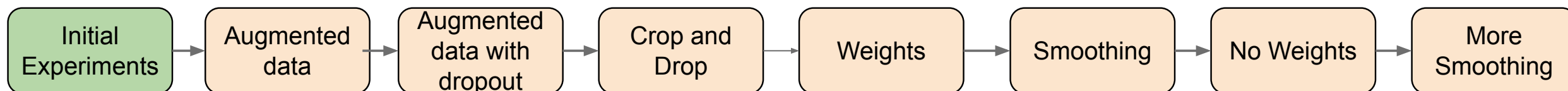
Same HP was used to train a ResNet50 model

ResNet50 Test Accuracy

Top 1	Top 5
0.487	0.779

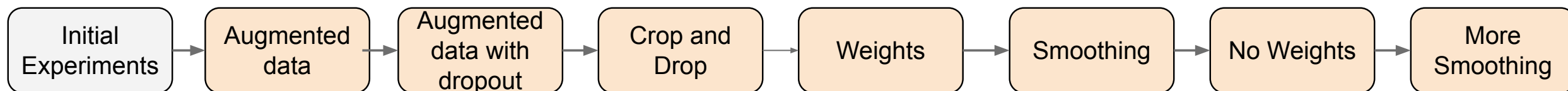
Test Accuracy

After training the best model is used to predict countries on the test data. The accuracy on this is the test accuracy



CPU Bottleneck

- Data augmentation caused a CPU bottleneck for smaller resnets and few trainable layers.
- Crop + Resize was the most expensive operation, due to interpolation.
- So we just cropped a 224x224 random region from our compressed 500x281 images.
- JPG compression helped reduce size of dataset substantially.
- JPG → tensor conversion was just done on-the-fly during training, since it was relatively cheap.
- Used torch.profiler, nvidia-smi and timing the dataloaders to investigate CPU bottleneck.



Augmentation data

We tried to avoid the CPU completely by augmenting data once and for all, so it could be loaded straight onto RAM. We knew it could have an effect on performance. Same HP were used.

The conclusion was clear: High degree of overfitting, data augmentation is key for performance!

It makes sense: Our dataset is not huge, only ~300 datapoints per class.

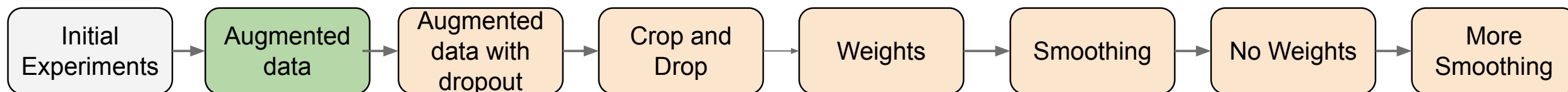
We introduced dropout to reduce overfitting and decided to move forward with on-the-fly augmentation: Crop, color jitter and rotation.

ResNet50 Test Accuracy

Top 1	Top 5
0.464	0.744

Top 5 Accuracy

The model gives a probability to every country. If the true country is in the top 5 highest probabilities it is in the top 5 accuracy.

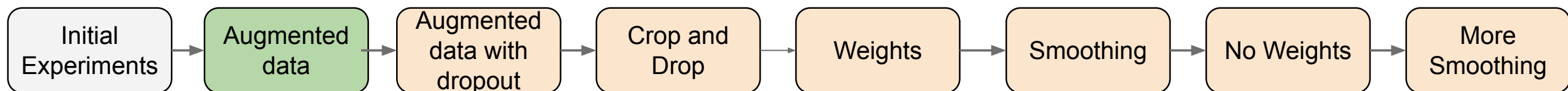
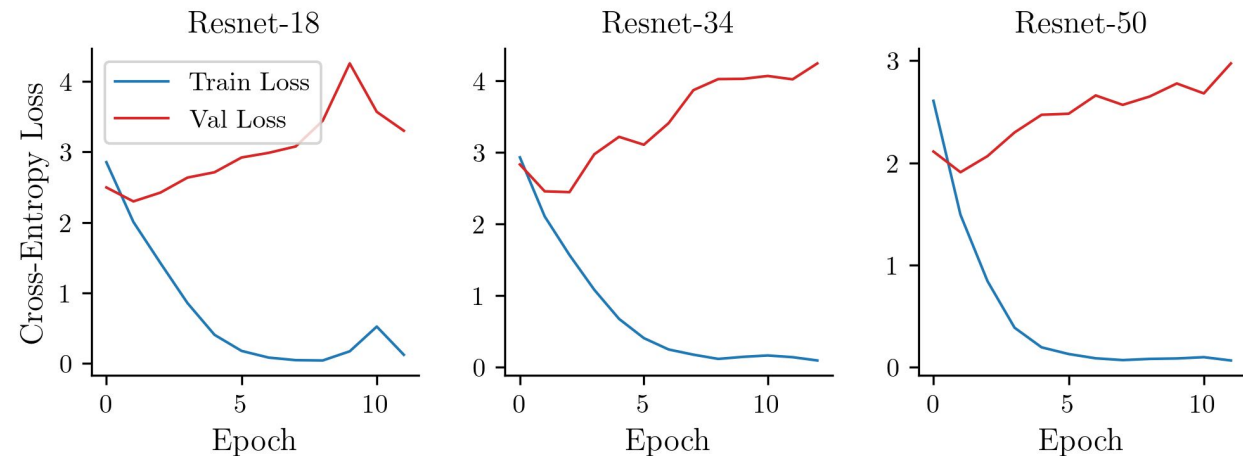


Random data augmentation is key for performance

- Each image were cut into three regions.
- 75,000 images for instant loading, instead of on-the-fly augmentation.
- Pre-computed augmentation → severe overtraining



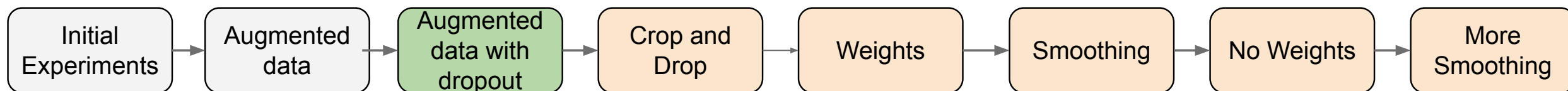
Model training using one-time data augmentation



Augmentation data + Dropout

- Tried to fix the over training with drop out
- Used dropout 0.3
- The accuracy did not really improve
 - + Overfitting was still an issue

ResNet50 Test Accuracy	
Top 1	Top 5
0.473	0.757



Augmentation on the go + Dropout

Did not work as hoped
Went back to model where we cropped
along the way and now also included
dropout

Same HP parameters was used

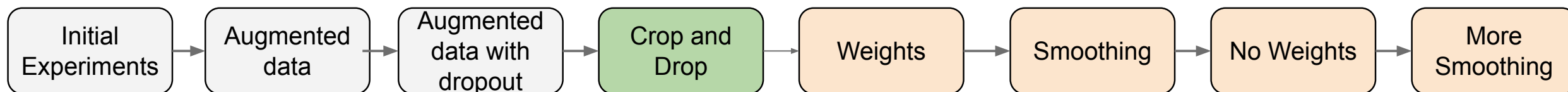
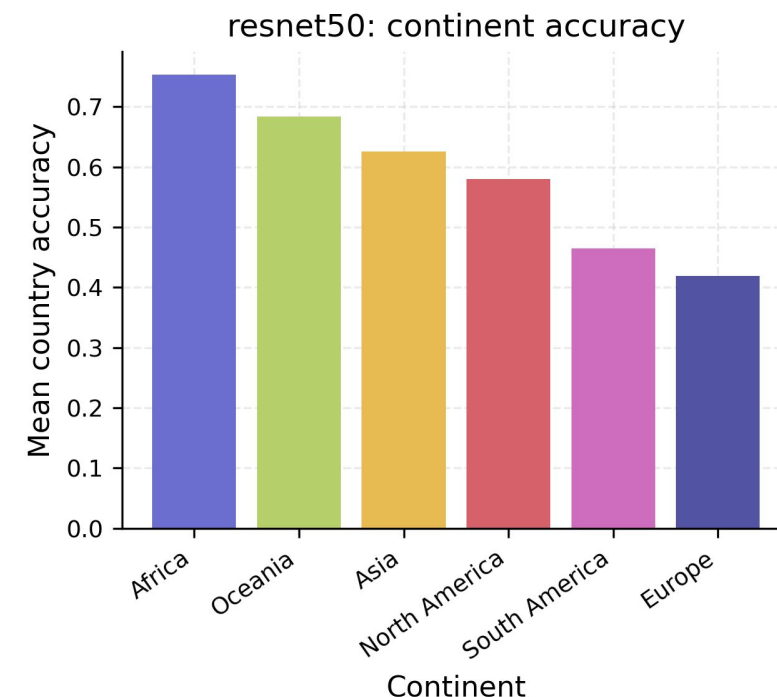
Conclusion:

Dropout is good

European countries is hard to predict

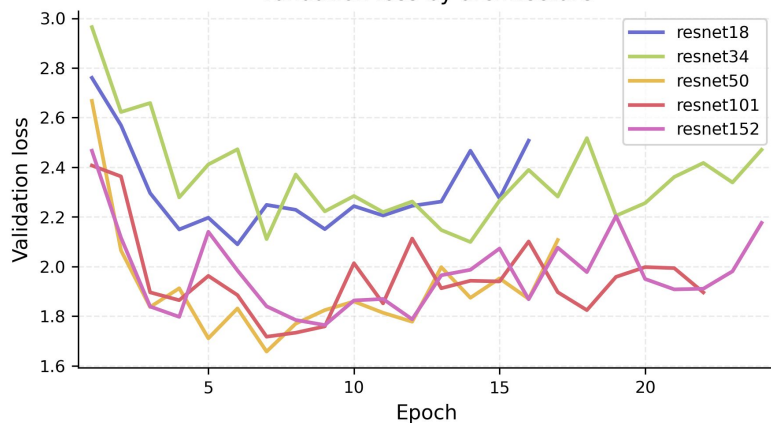
ResNet50 Test Accuracy

Top 1	Top 5
0.541	0.838

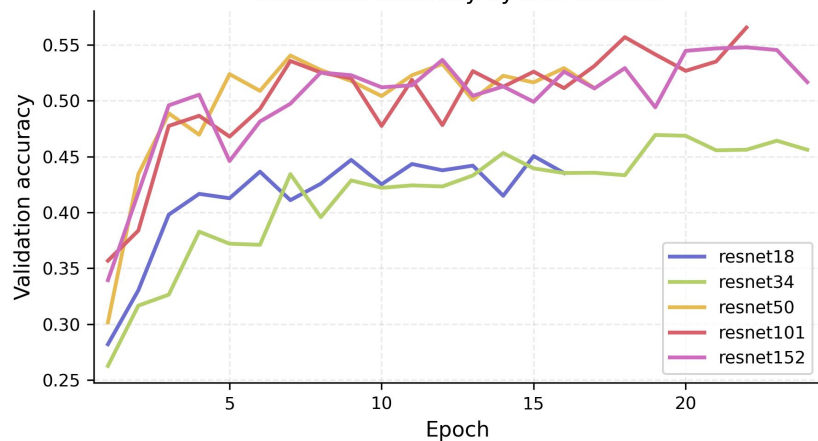


Augmentation on the go + Dropout - Training plots

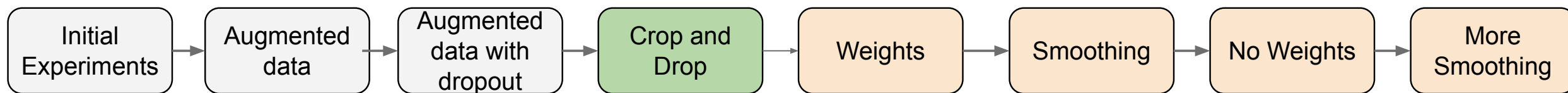
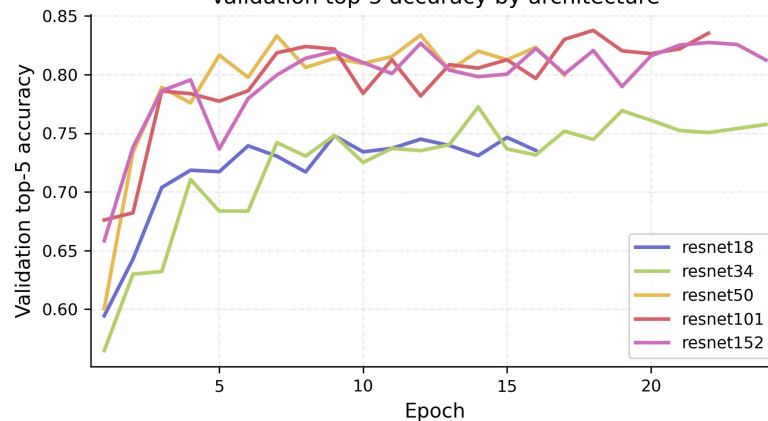
Validation loss by architecture



Validation accuracy by architecture



Validation top-5 accuracy by architecture



Adding Class Weights

Some info here

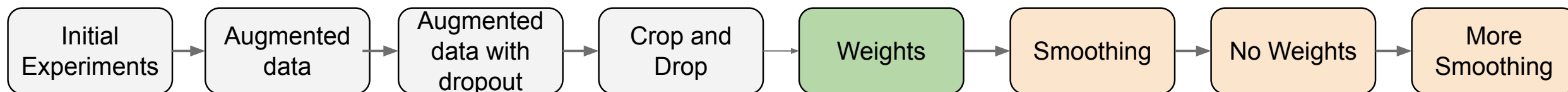
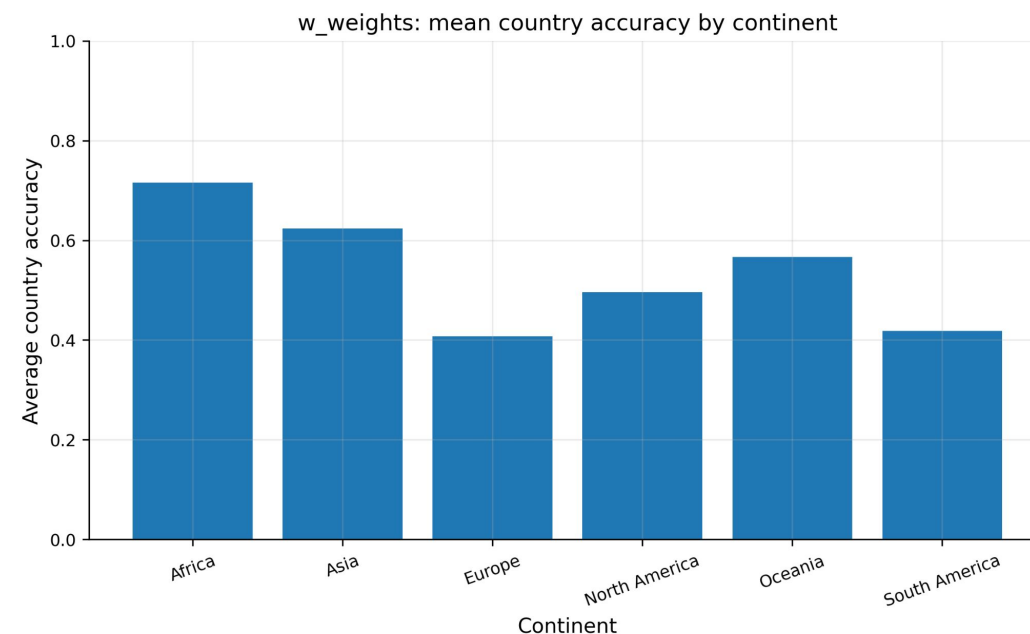
ResNet50 Test Accuracy

Top 1	Top 5
0.518	0.817

Results:

Accuracy got a bit worse

Avg accuracy for Europe did not improve



Adding Class Weights - Optimization

Grid search:

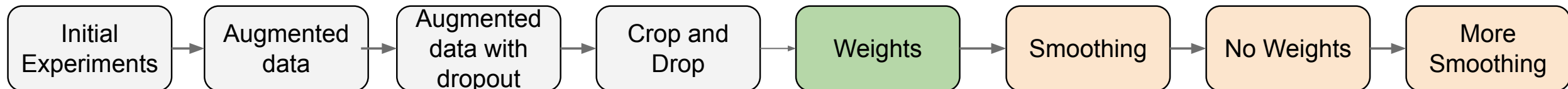
ResNet50 Test Accuracy	
Top 1	Top 5
0.542	0.838

model_name	batch_size	learning_rate	weight_decay	trainable_layers	color_jitter	rotation	horizontal_flip	best_val_loss	best_val_acc	best_epoch_loss	test_top1	test_top5
resnet50	256	0.001	0.0001	1	TRUE	TRUE	TRUE	1.802716344	0.5024925224	5	0.5178044438	0.82038164

And then Optuna

run_name	test_top1	test_top5	checkpoint	report	confusion_matrix	predictions	learning_rate	weight_decay	trainable_layers
trial_9_resnet50_bs2	0.5418060422	0.8382844925	/content/AppliedML2	/content/drive/MyDrive	/content/drive/MyDrive	/content/drive/MyDrive	0.0008701655484	1.00E-05	1

Seemed like it was getting better



With Class Weights - Optimization Loop

Motivation:

To evaluate which ResNet is best

First tried with ResNet50's HP but the results seemed biased

model	test_top1	test_top5
resnet18	0.4467833936	0.7631320357
resnet34	0.4646862149	0.7654927969
resnet50	0.5107220411	0.8170371652
resnet101	0.5075742602	0.8101514578
resnet152	0.5221325755	0.8152665496

ResNet50 Test Accuracy

Top 1	Top 5
0.518	0.817

Execution was looping over ResNet(34, 50, 101, 152)

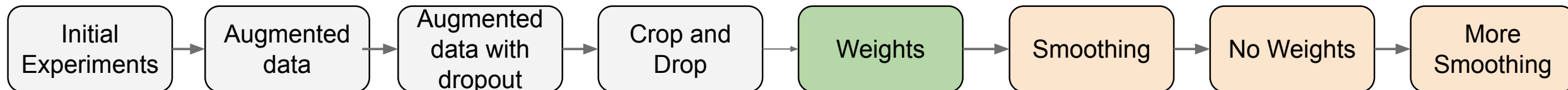
Tested multiple ResNet models with weighted loss and Optuna-tuned:

- learning rate: $1e-4$ – $2e-3$
- weight decay: $1e-6$ – $1e-4$
- trainable layers: 0, 1, 2

Then retrained each model with the best settings and evaluated both:

- best validation loss checkpoint
- best validation accuracy checkpoint

Saved test accuracy, top-5 accuracy, confusion matrices, predictions, and continent accuracy for both a best loss model and a best accuracy model.

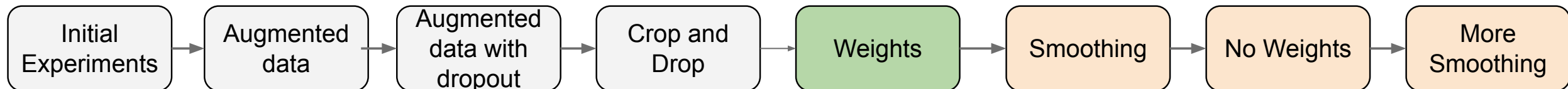
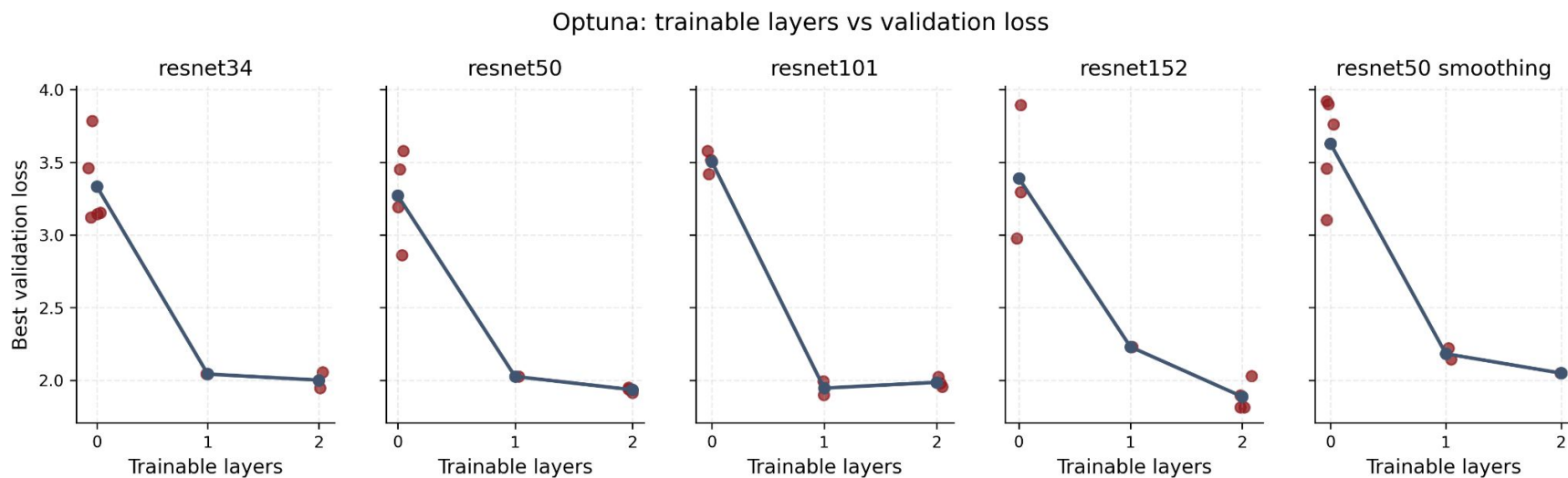


With Class Weights - Optuna Results

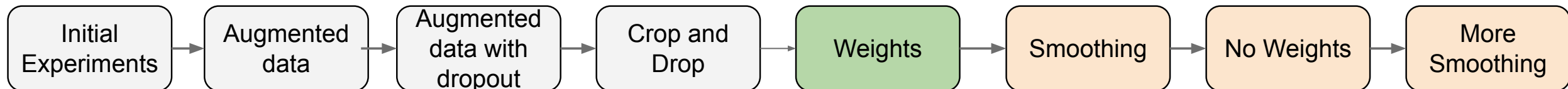
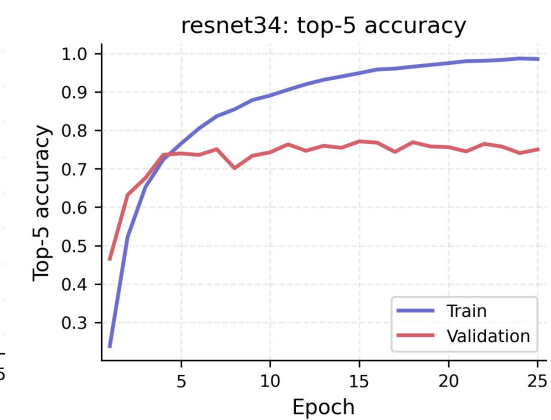
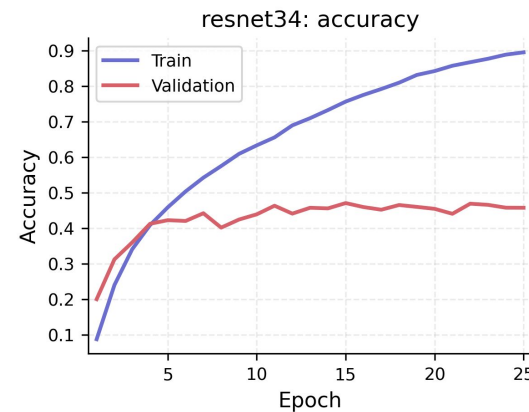
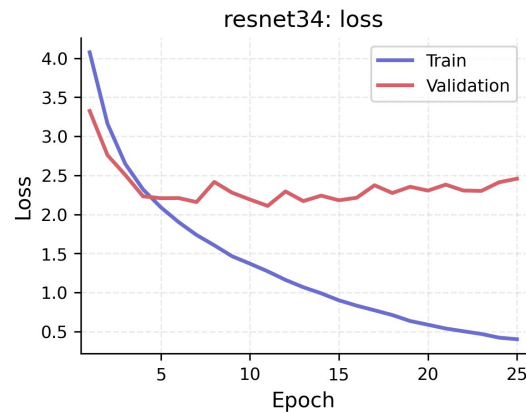
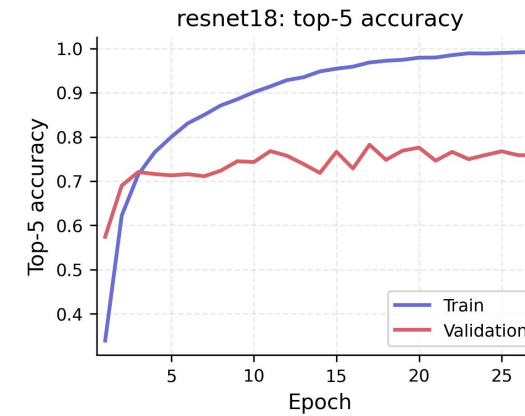
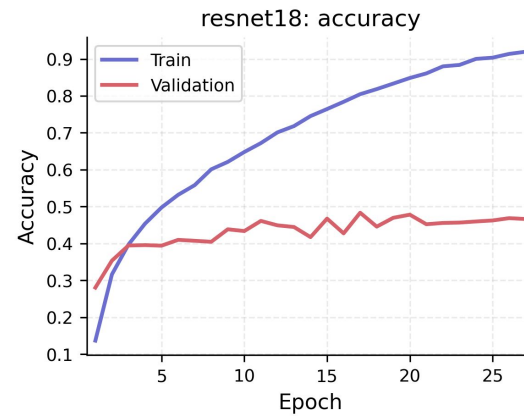
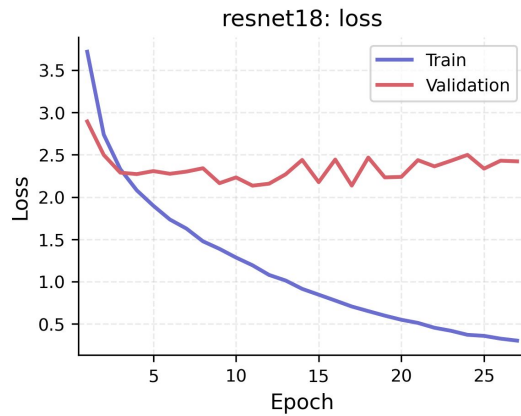
model	best_val_loss	learning_rate	weight_decay	trainable_layers
resnet101	1.8983	7.55e-04	1e-06	1
resnet152	1.8118	9.98e-04	1e-05	2
resnet34	1.9447	1.23e-04	5e-05	2
resnet50	1.9145	3.95e-04	1e-06	2
resnet50 smoothing	2.0483	2.32e-04	1e-06	2

The previous optuna suggested that one layer was best. Which was why only [0, 1, 2] was tested.

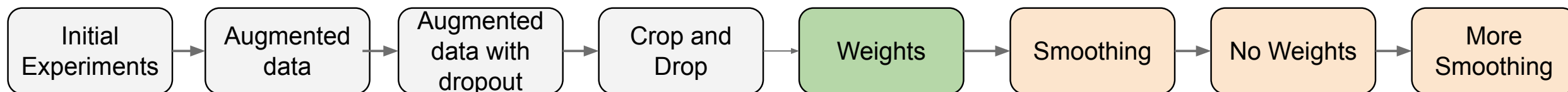
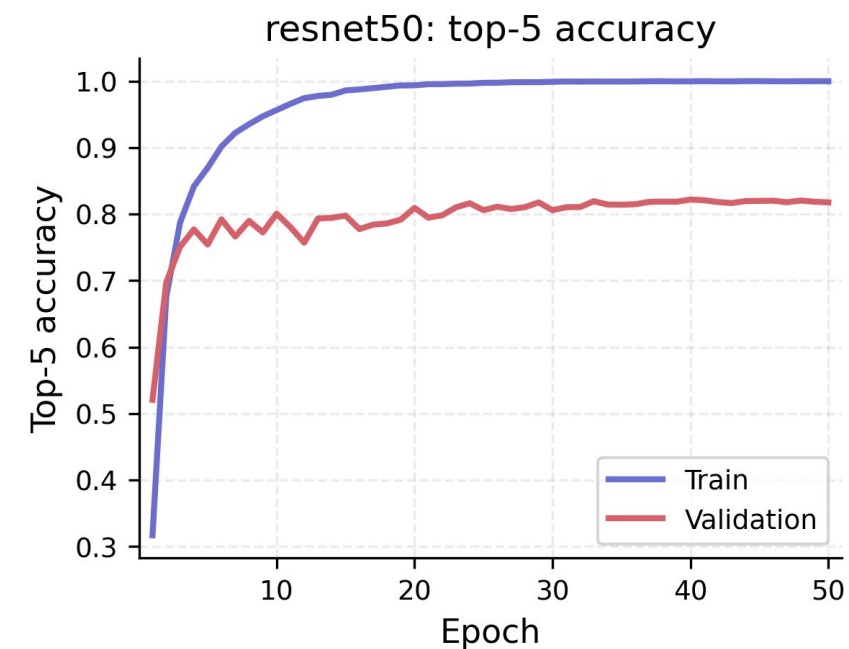
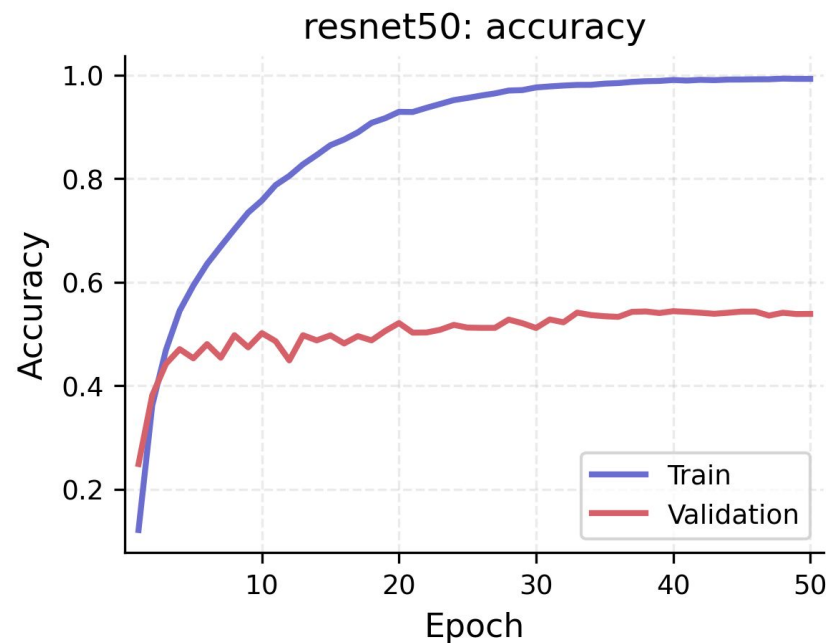
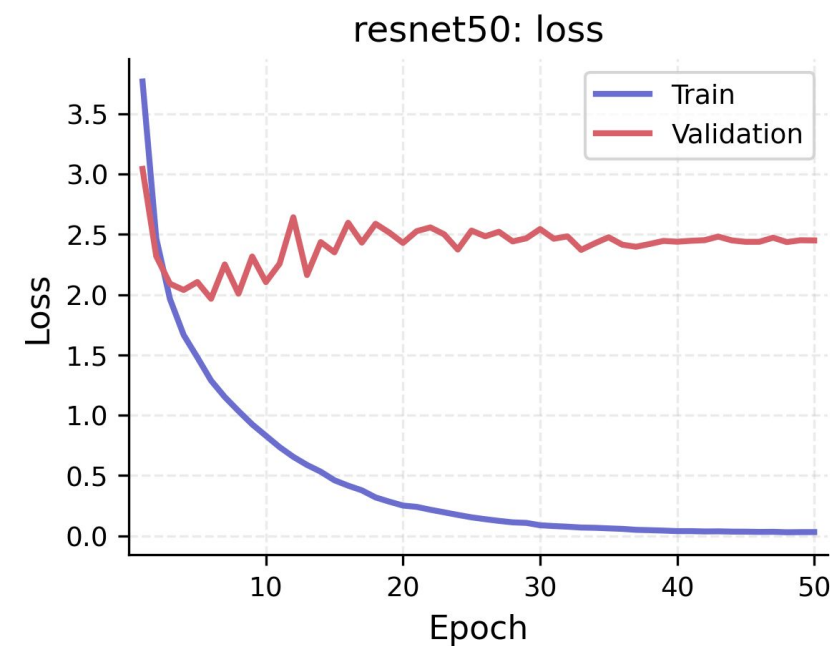
In hindsight, trainable layer setting 3 should likely also have been tested.



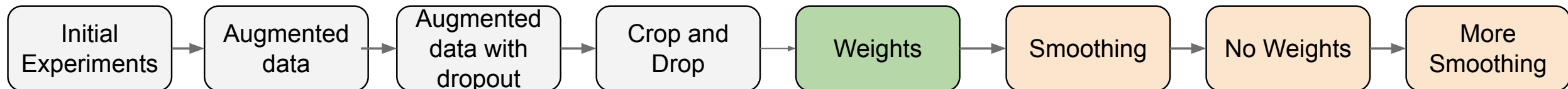
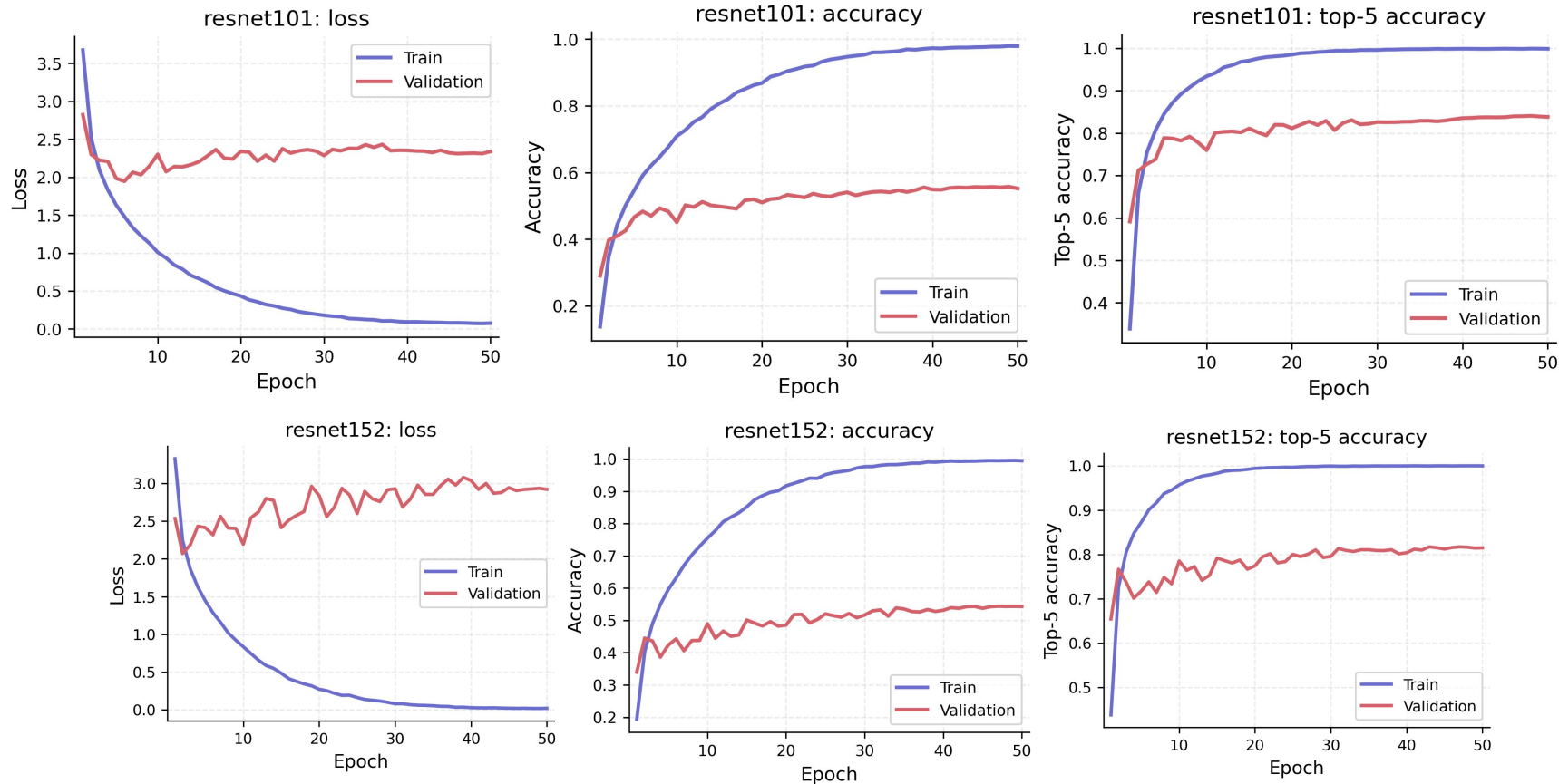
With Class Weights - ResNet(18 + 34)



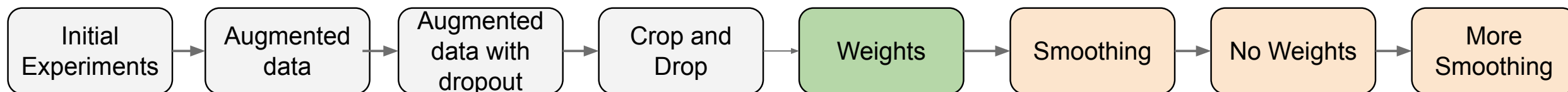
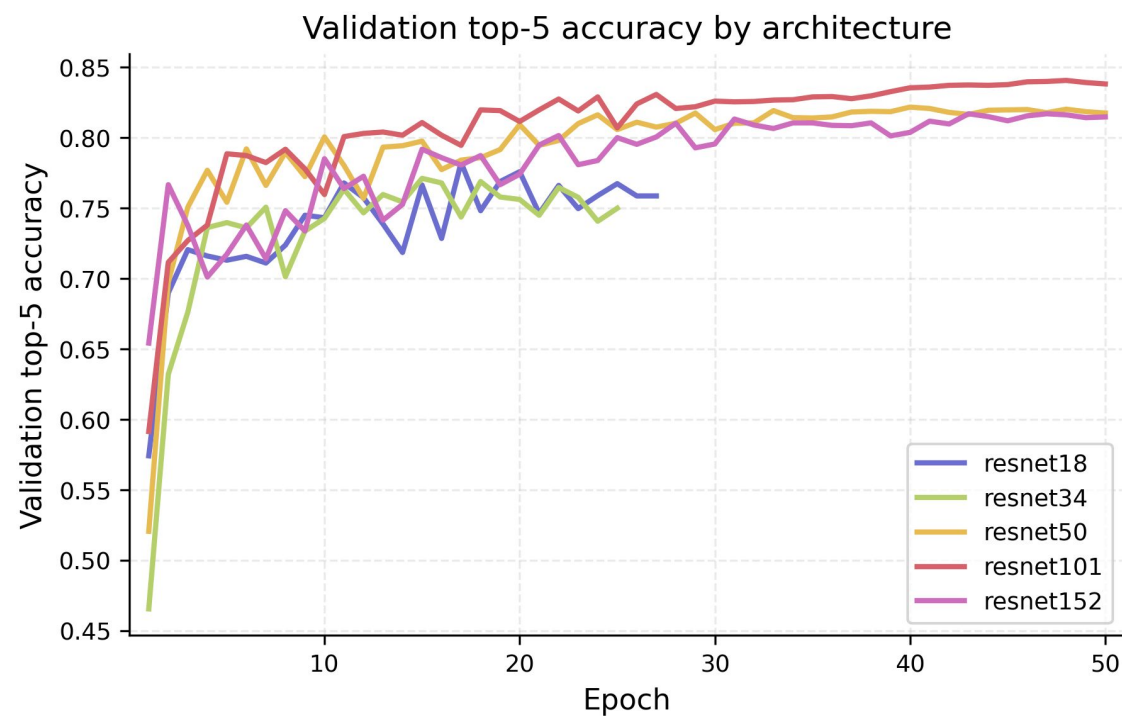
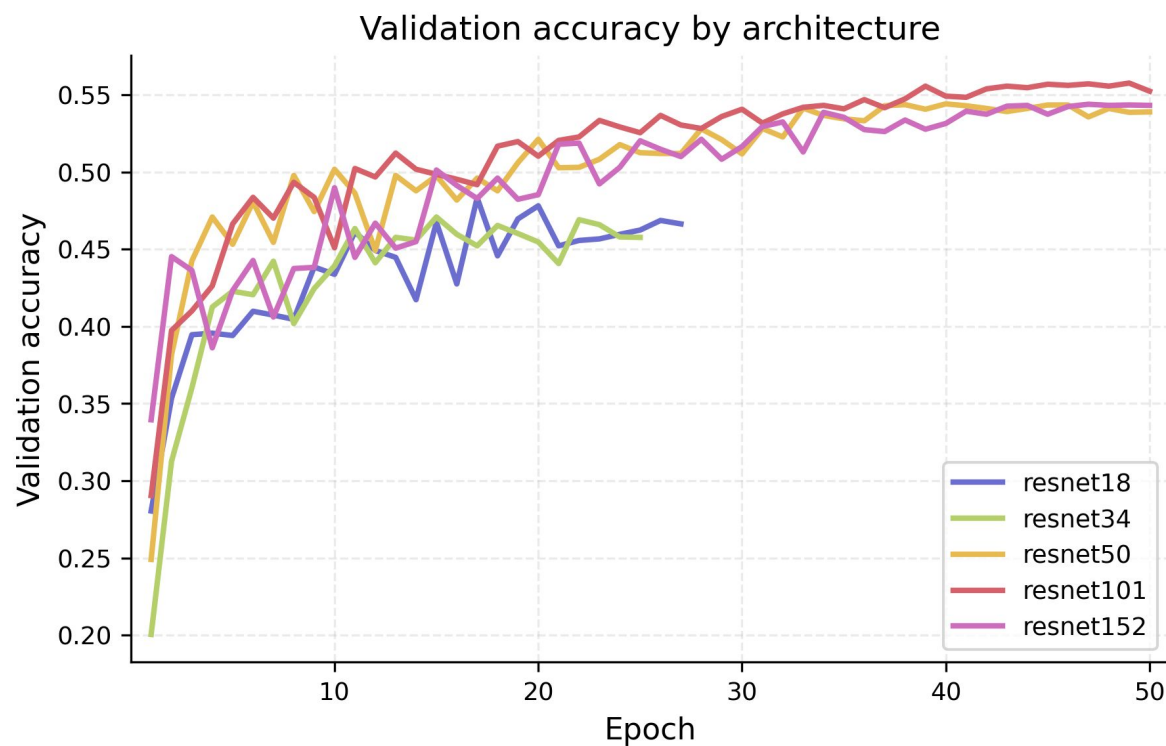
With Class Weights - Resnet 50



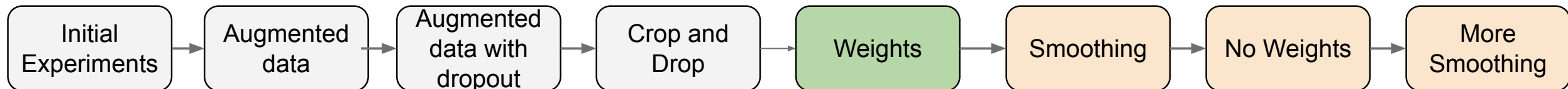
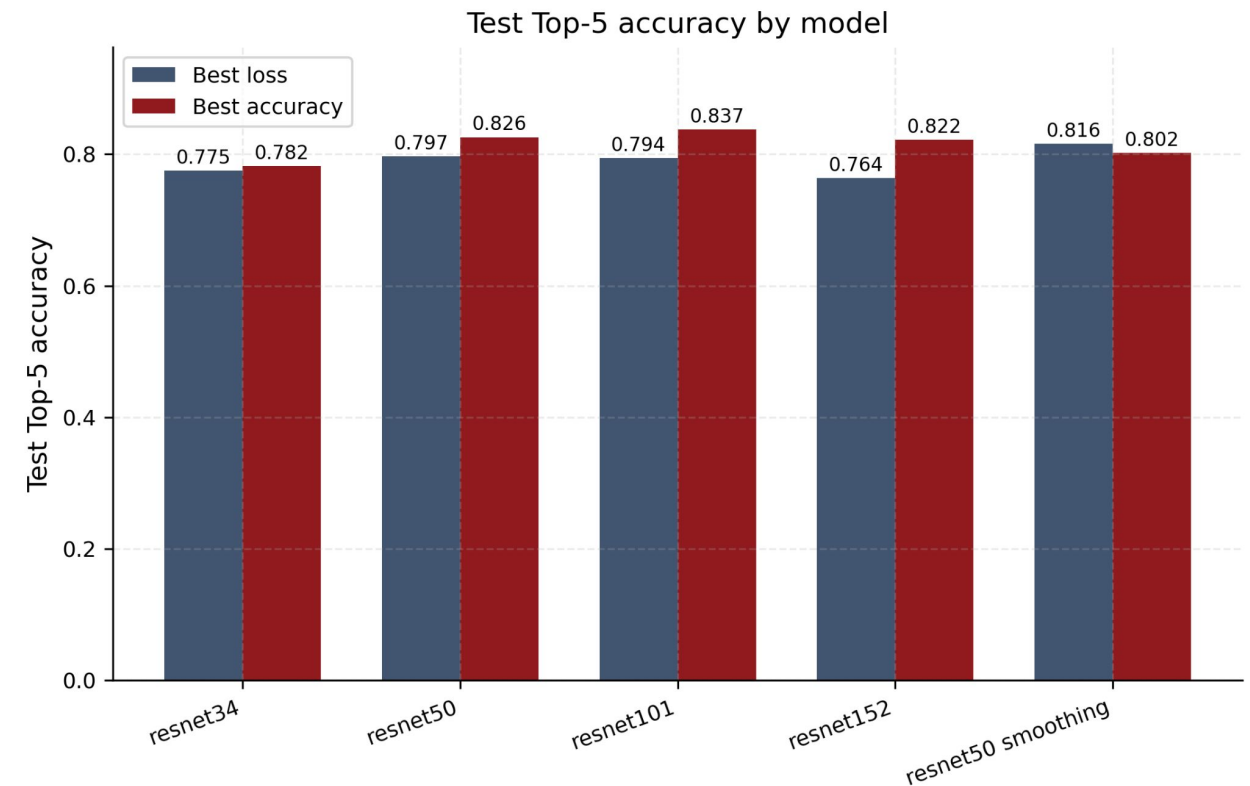
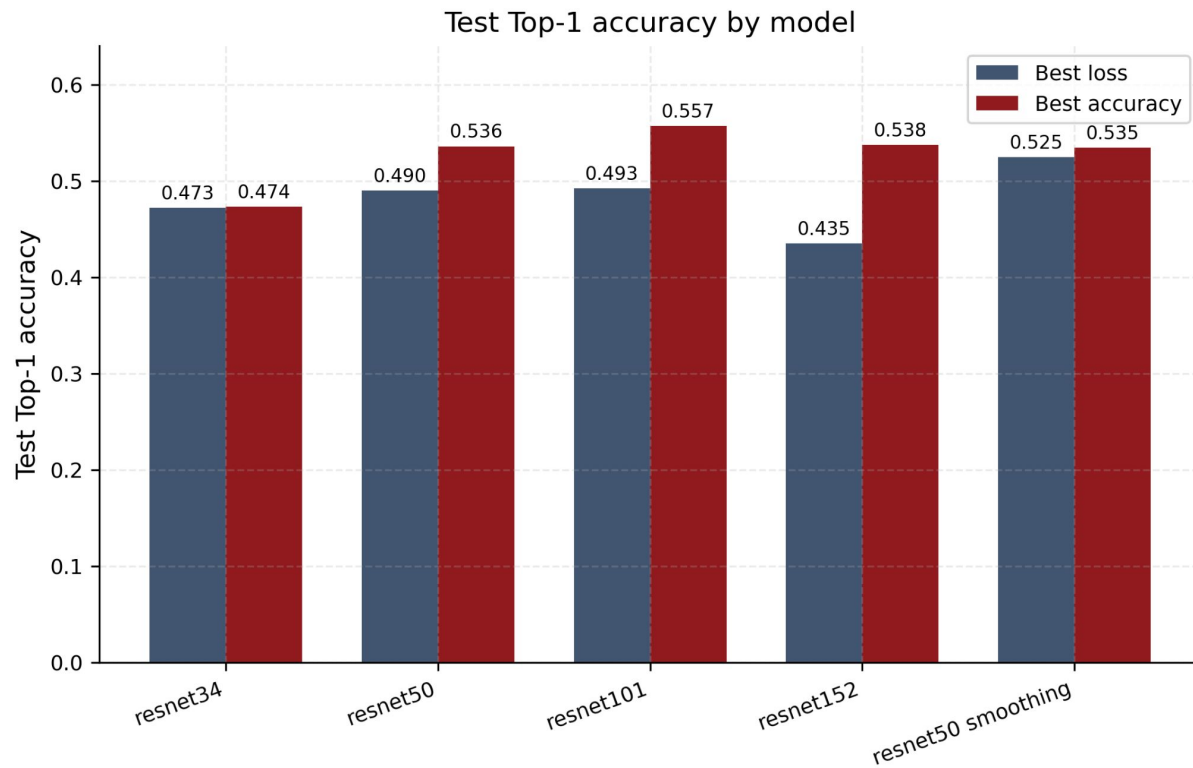
With Class Weights - ResNet(101+152)



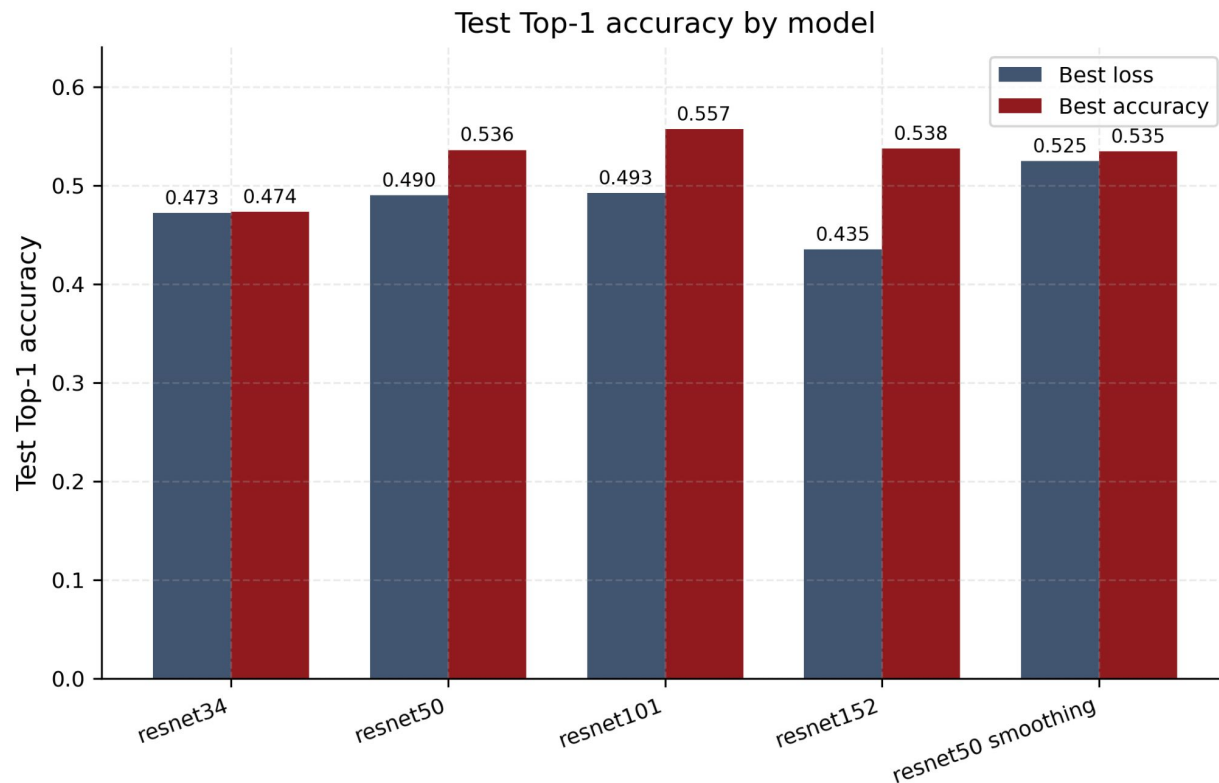
With Class Weights - Accuracy for all ResNet



With Class Weights - Test Accuracy for all ResNet



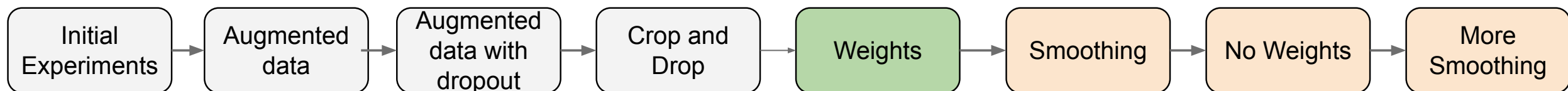
With Class Weights - Conclusion



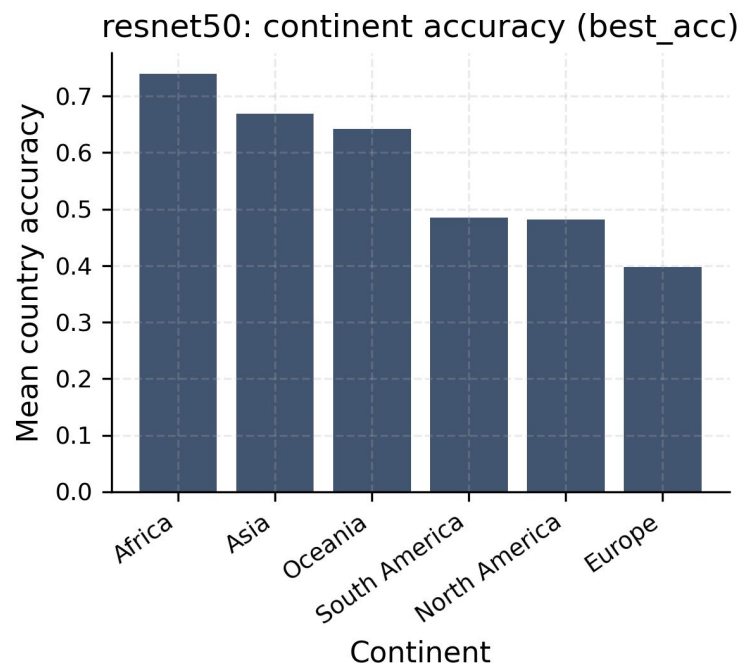
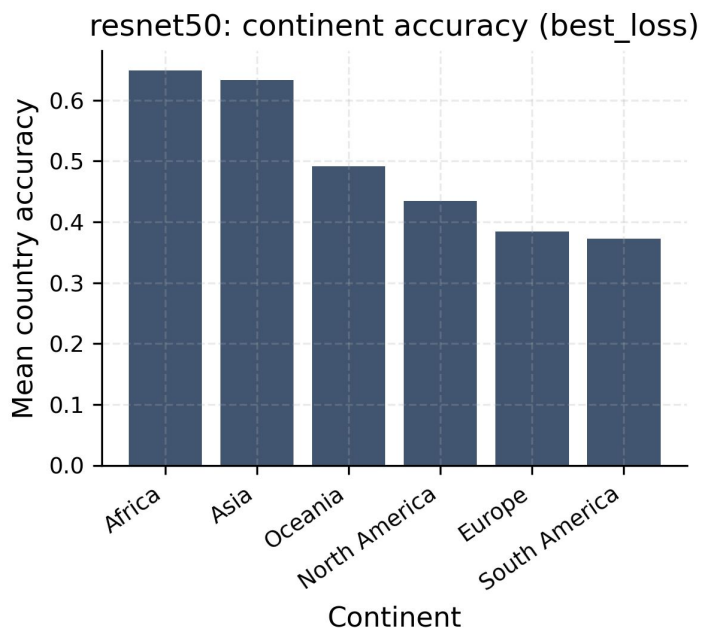
From these results it is clear that when training the the accuracy keeps increasing while loss quickly finds minima.

The Crop and Drop did not experience this. But we did not do HP tuning on this, which maybe could explain the differences.

However it could also be because Europe is hard to learn which penatalizes the loss a lot.



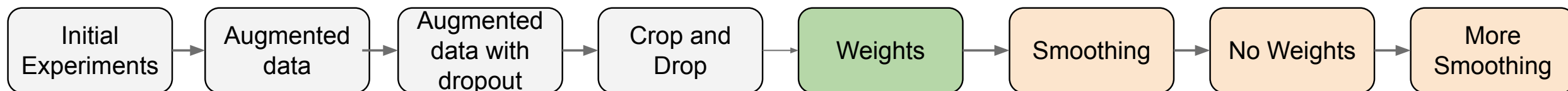
With Class Weights - Conclusion



Here we see that using the best validation loss checkpoint results in relatively low average accuracy across all continents compared to Crop and Drop.

Using the best validation accuracy checkpoint instead gives much better test accuracy for all continents except Europe.

This suggests that the weighted loss may no longer be a reliable metric for selecting the best model.

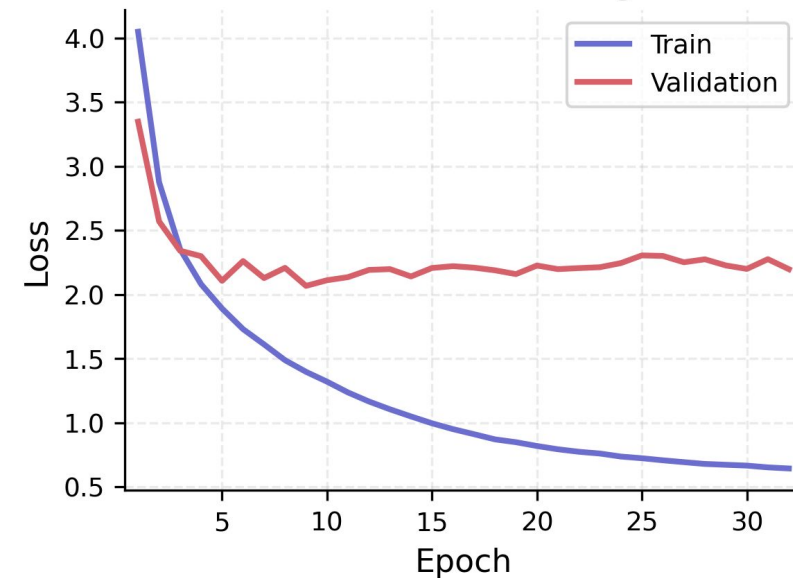


Smoothing with ResNet50

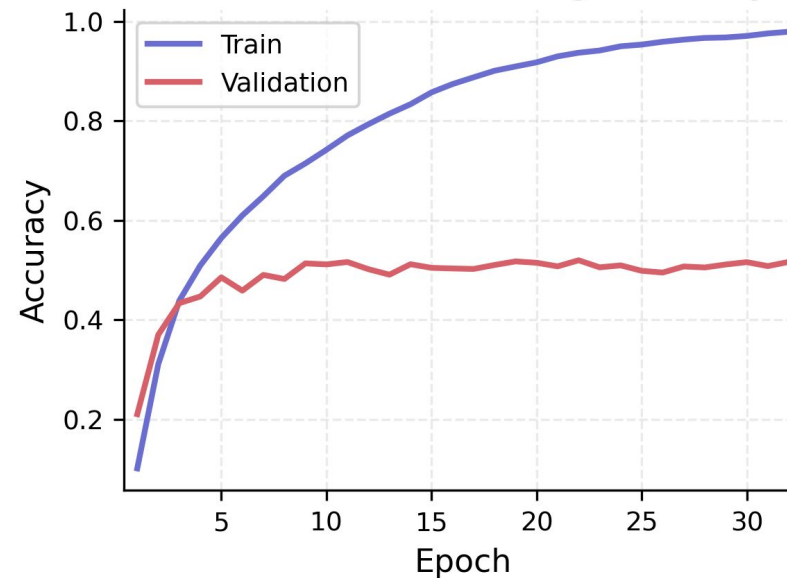
Added smoothing by using cluster
($k=15$) as explained in slide X.
Optuna was run like with the weights

ResNet50 Test Accuracy	
Top 1	Top 5
x	x

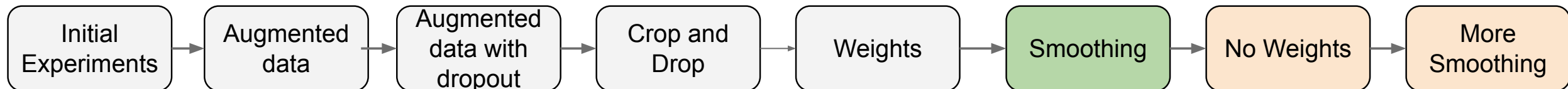
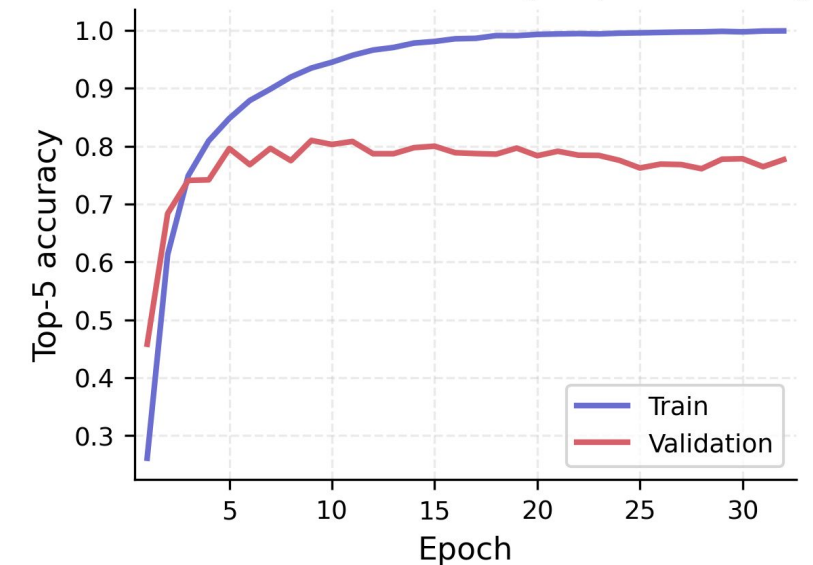
resnet50 with smoothing: loss



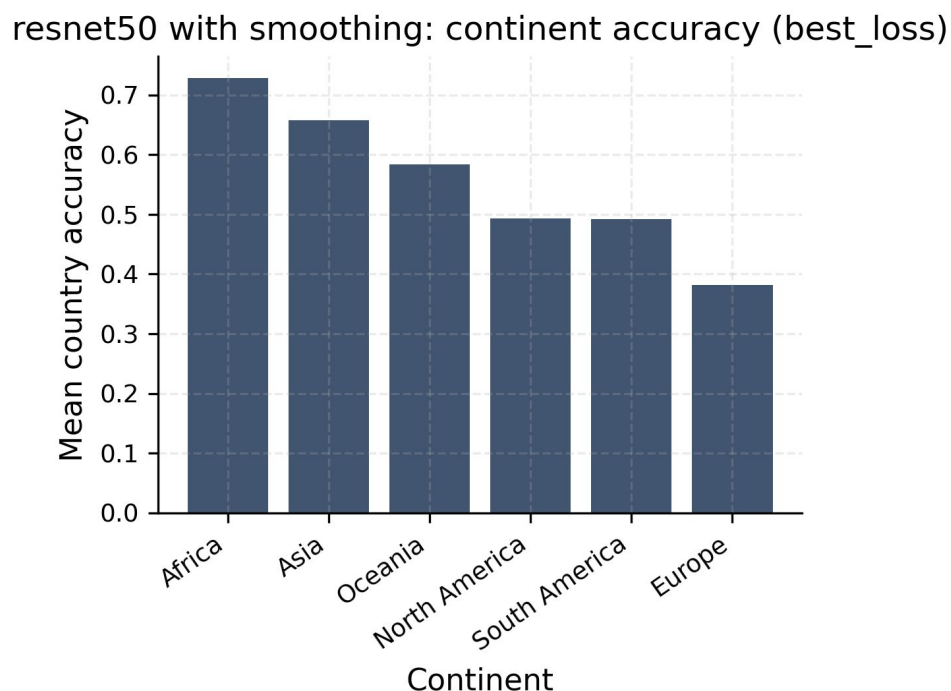
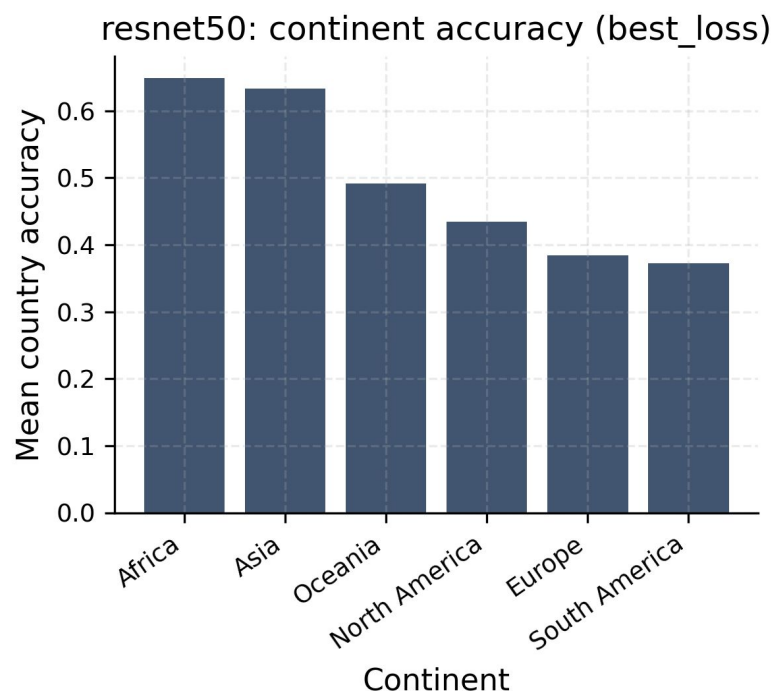
resnet50 with smoothing: accuracy



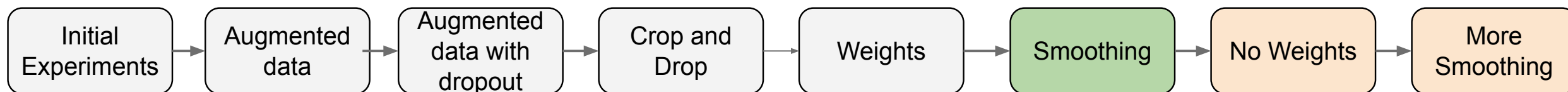
resnet50 with smoothing: top-5 accuracy



Smoothing with ResNet50 - weights vs smoothing

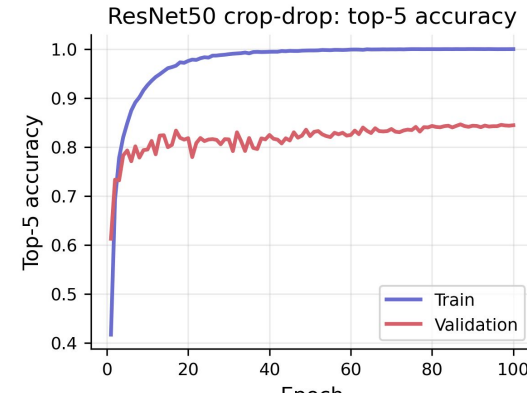
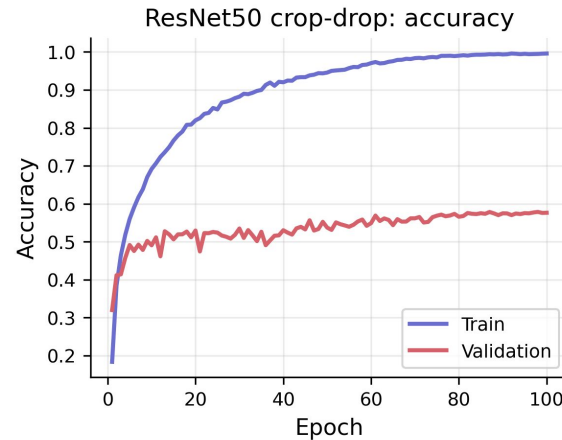
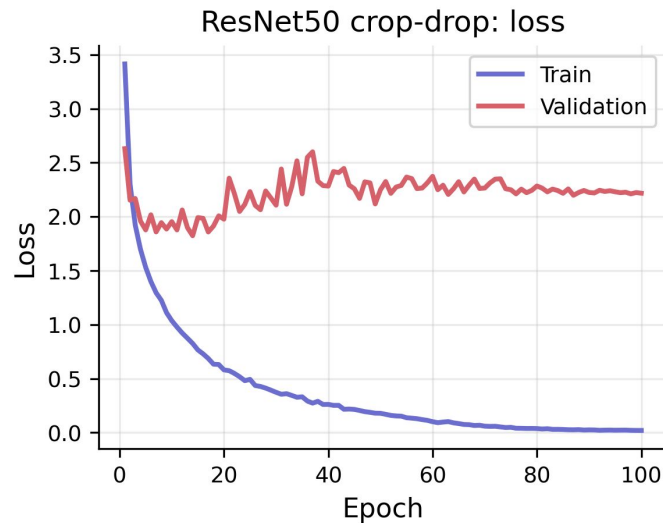


Smoothing has best training around 10 epoch both if we look at loss and accuracy which makes the two models almost identical when evaluating compared to the added weights models



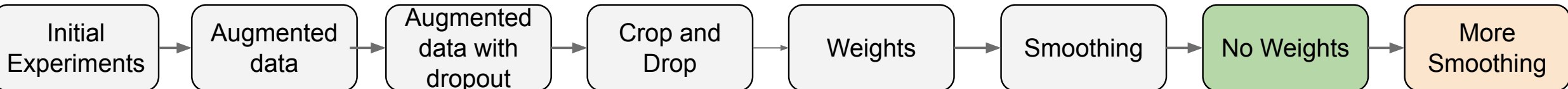
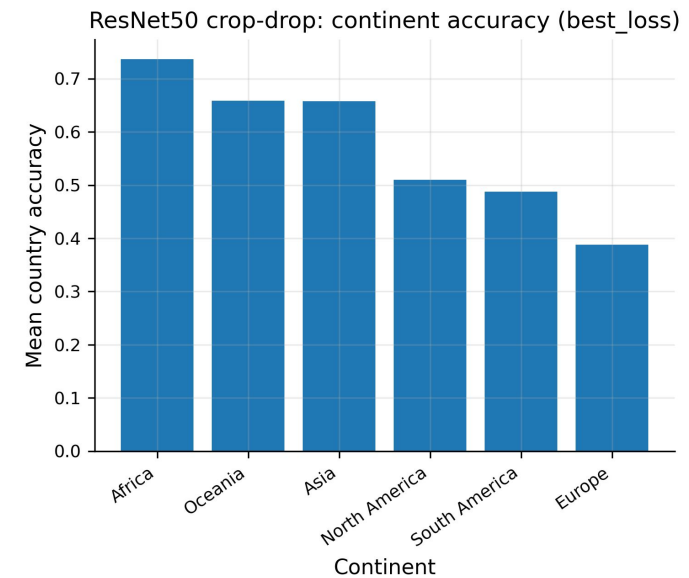
Removing Weights - Initial

Used HP's from old grid search



ResNet50 Test Accuracy

Top 1	Top 5
0.518	0.817



Removing Weights - HP tuning

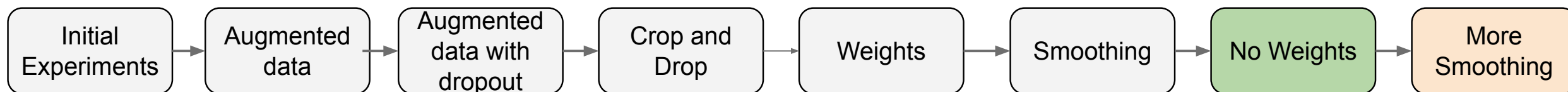
HP tuning with optuna for
ResNet50 and ResNet 101
8 trials of 15 epoch
Training with 50 epoch and
early stopping 8

Note:
Maybe not big
enough HP search

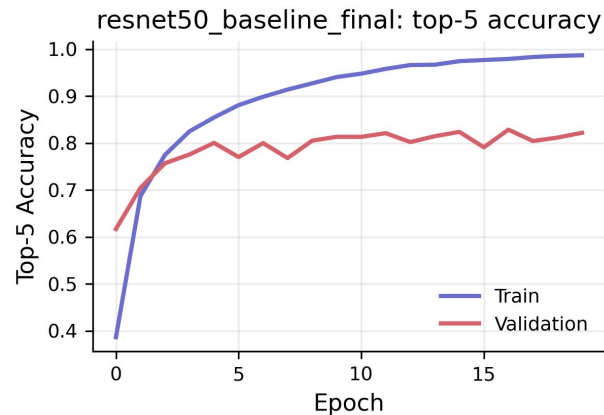
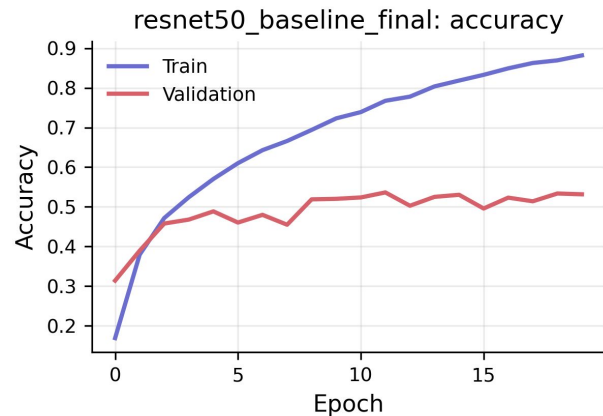
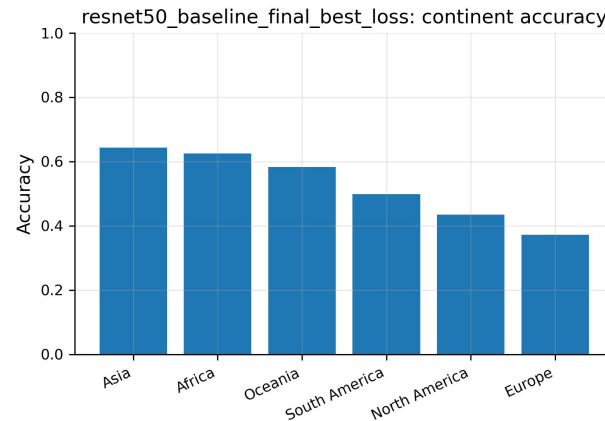
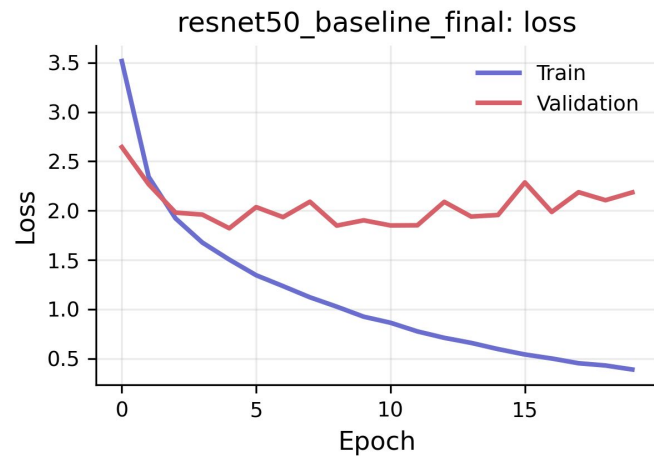
ResNet50 Test Accuracy

Top 1	Top 5
0.499	0.804

experiment	model	checkpoint_type	test_top1	test_top5	learning_rate	weight_decay	trainable_layers
baseline	resnet50	best_loss	0.4989179671	0.8046429157	0.00084449129418	5.00E-06	1
baseline	resnet50	best_acc	0.5453472137	0.8235294223	0.00084449129418	5.00E-06	1
baseline	resnet101	best_loss	0.5374778509	0.8310053349	0.0004069368009	5.00E-05	3
baseline	resnet101	best_acc	0.5508558154	0.8347432613	0.0004069368009	5.00E-05	3

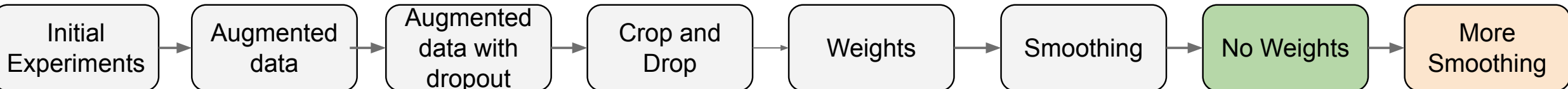


Removing Weights - HP tuning Results ResNet50



ResNet50 Test Accuracy

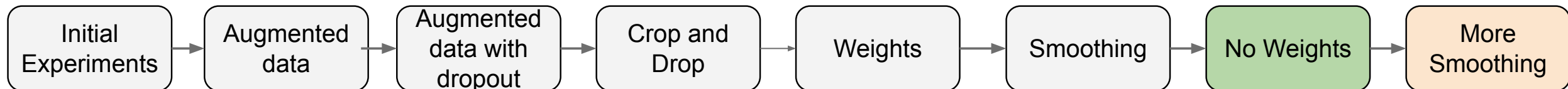
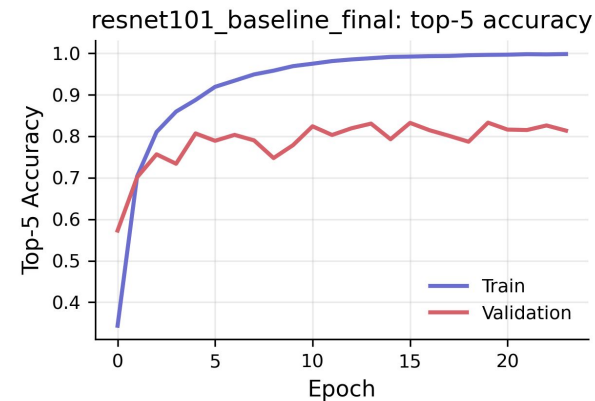
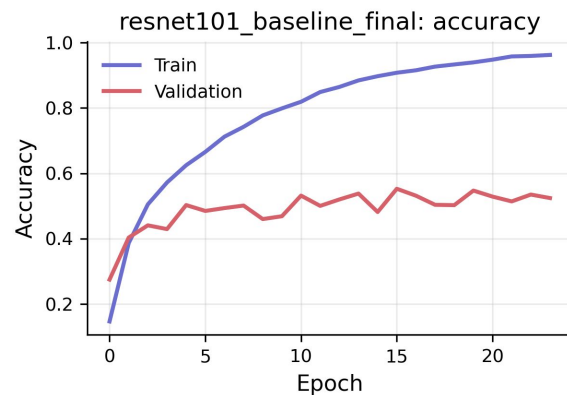
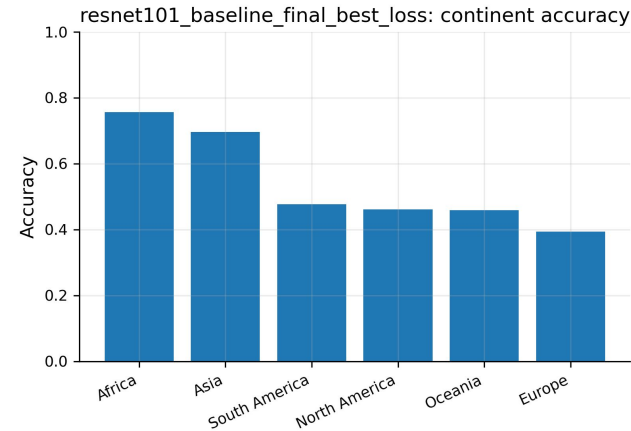
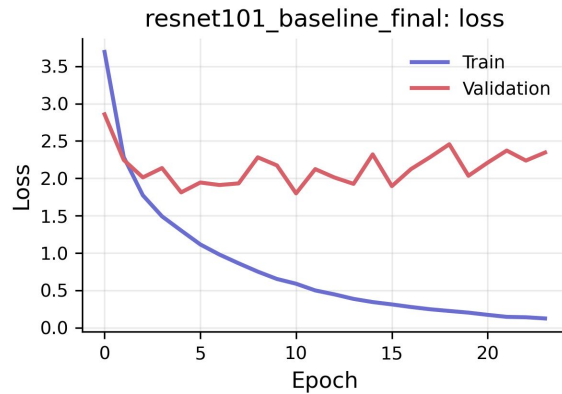
Top 1	Top 5
0.499	0.804



Removing Weights - HP tuning Results ResNet101

ResNet101 Test Accuracy

Top 1	Top 5
0.537	0.831

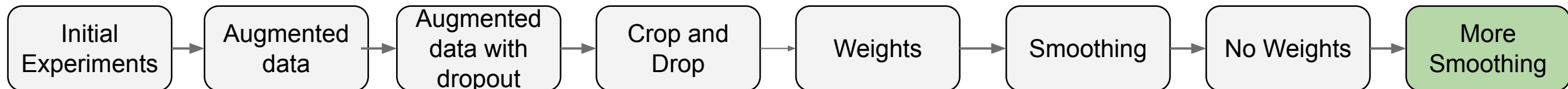


Trying Smoothing again

Different prob and Clusters

Did not improve

run_name	model	cluster_k	correct_prob	checkpoint_type	test_top1	test_top5	learning_rate	weight_decay	batch_size	trainable_layers	num_epochs	patience	rank
resnet50_K30_p09	resnet50	30	0.9	best_acc	0.5284280777	0.7704111934	0.0002316833773	1.00E-06	256	2	100	15	
resnet50_K30_p08	resnet50	30	0.8	best_acc	0.5225260854	0.7645091414	0.0002316833773	1.00E-06	256	2	100	15	
resnet50_K15_p08	resnet50	15	0.8	best_acc	0.5195750594	0.7570332289	0.0002316833773	1.00E-06	256	2	100	15	
resnet50_K30_p08	resnet50	30	0.8	best_loss	0.5085579157	0.7889041901	0.0002316833773	1.00E-06	256	2	100	15	
resnet50_K15_p08	resnet50	15	0.8	best_loss	0.4855400324	0.7851662636	0.0002316833773	1.00E-06	256	2	100	15	
resnet50_K30_p09	resnet50	30	0.9	best_loss	0.4855400324	0.7898878455	0.0002316833773	1.00E-06	256	2	100	15	



What have we learned from these “experiments”

Model/training:

Dropout seems to be good

- (could maybe be optimized also)
- Optimizing HP is important for models before comparing them
- The experiments also suggest that hyperparameter optimization is important for finding models where both validation loss and validation accuracy perform well.

Workflow:

- Important to save all tries
(we needed to go back and luckily we saved the results)
- We could probably have saved some expensive compute units by having a better plan and thinking and testing what could go wrong earlier
- Could also have saved units on not optimizing all models in a loop, but worked with ResNet50 until we were confident about what happened and then maybe do a loop

Why cluster countries?

Country level classification is difficult because many countries can look visually similar.

Instead of predicting 85 individual countries, we grouped countries into broader labels based on metadata,

Here:

- Temperature only
 - Average January temperature
 - Average July temperature
- Climate/geography
 - Average January temperature
 - Average July temperature
 - Absolute Latitude (Because distance from the equator is more relevant for climates structure than north/south direction)
 - Longitude

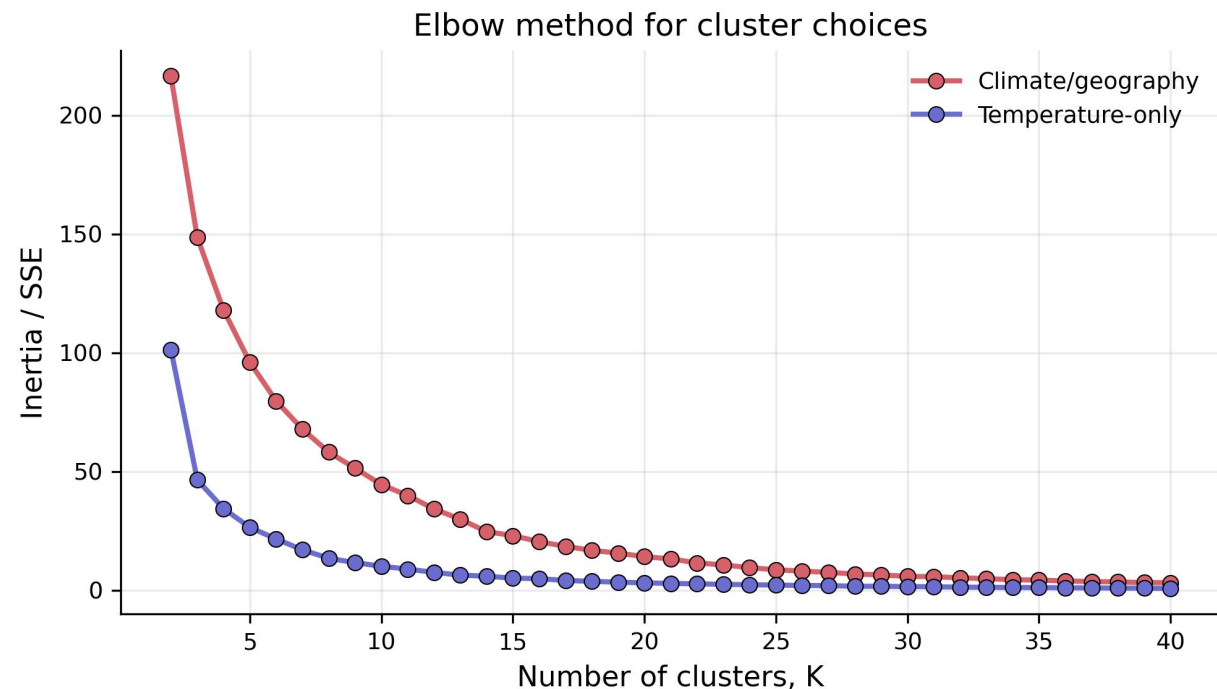
This reduces the number of classes and allows us to test whether regional climate/geography structure is easier for the CNN to learn.

Elbow method / choosing K

The elbow plot was used as a guideline for choosing the number of clusters. For temperature only clustering, the curve flattens quite early. This suggests that a smaller value of (K) could probably have been sufficient, because the data only constraints two features; Jan and July temp.

However we still used (K=15) for temperature-only clustering in the comparison, because the main climate/geography model was also tested with (K=15). This made the comparison between the two label systems more controlled.

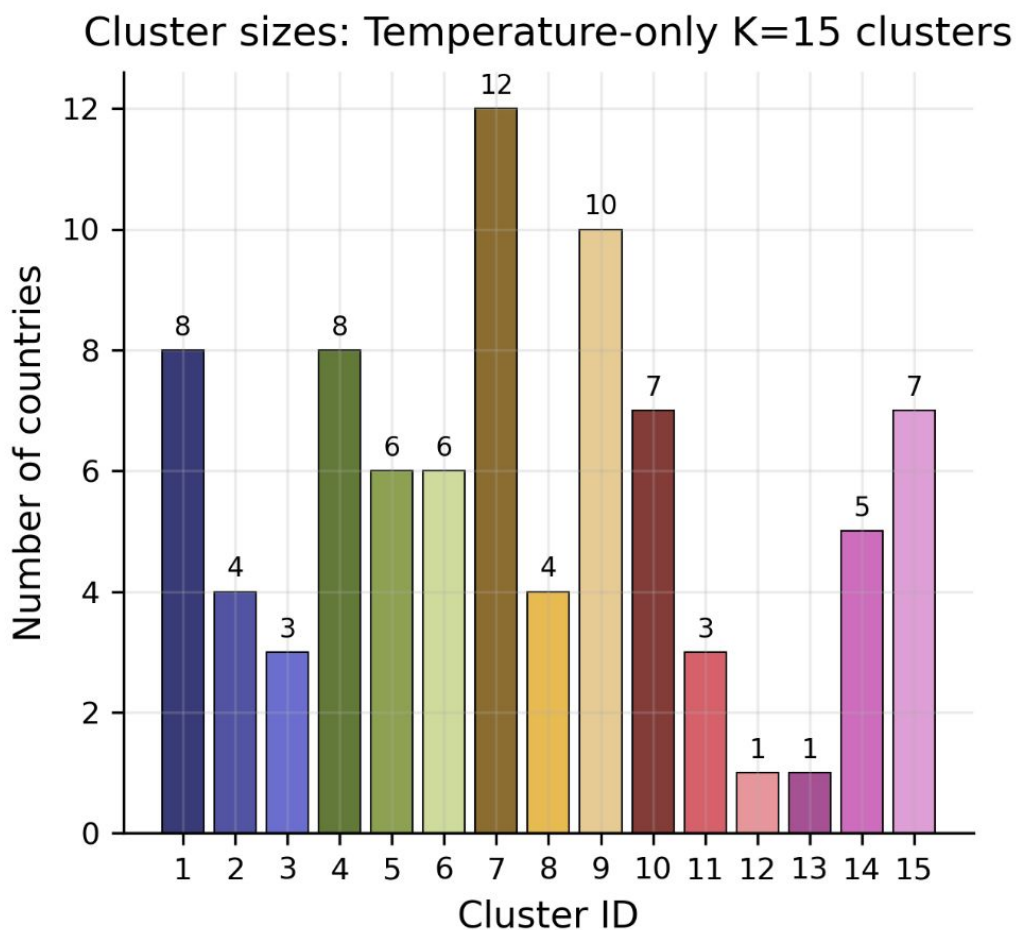
So, one limitation is that temperature-only clustering might have been better evaluated with fewer clusters. But for our main comparison, we kept (K) fixed to focus on the effect of **adding geography**



Temperature-only cluster size overview

Countries clustered using only January and July temperature. The plot shows the size of each cluster, while the overview on the right shows which countries are assigned to each group.

The cluster sizes are uneven, some temp. Patterns are shared by many countries, while others are more specific.



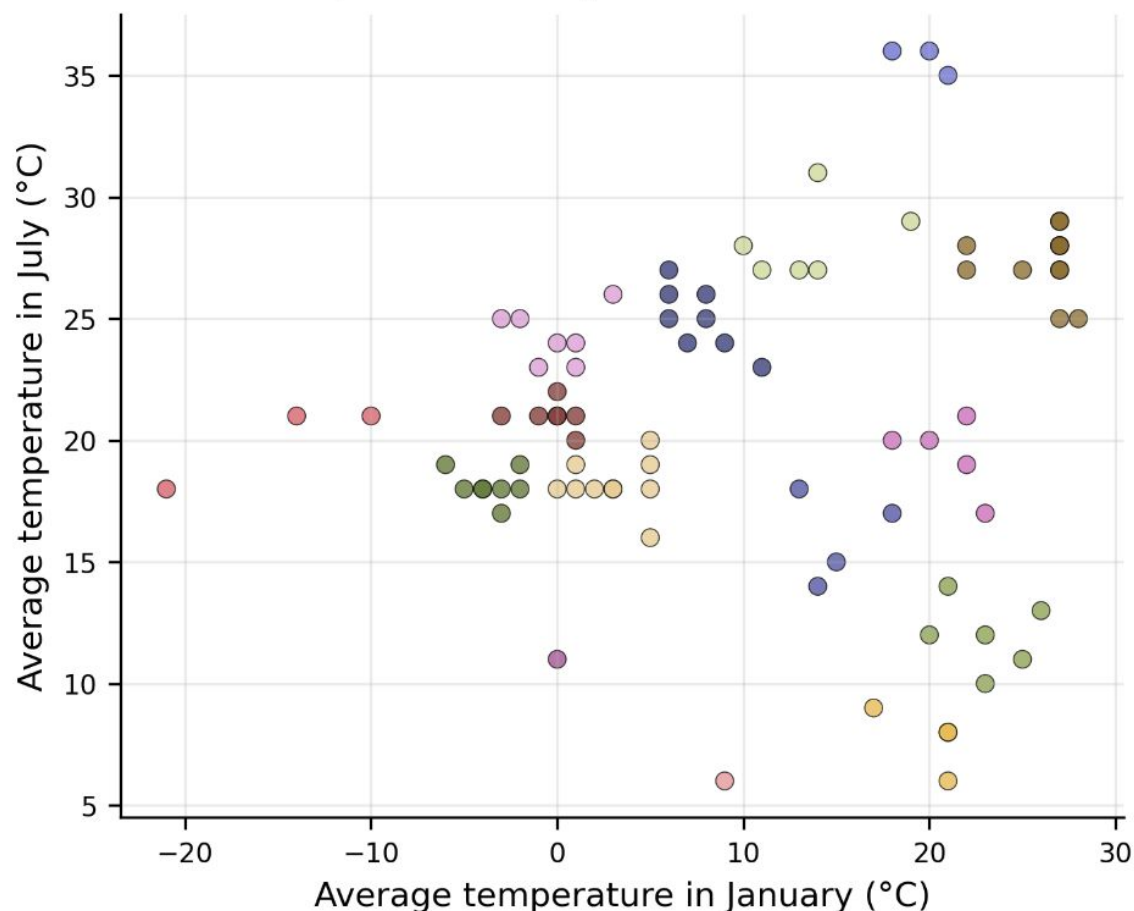
Cluster overview

- 1: AL, ES, IL, IT, JO, JP, ME PT
- 2: CO, EC, KE, MX
- 3: AE, OM, QA
- 4: EE, FI, LT, LV, NO, PL, RU SE
- 5: AR, BW, MG, SZ, UY, ZA
- 6: BD, GR, IN, LB, MT, TN
- 7: DO, GH, ID, KH, LA, MY, NG PA, PH, SG, SN, TH
- 8: AU, CL, LS, NZ
- 9: BE, BT, CH, DE, DK, FR, IE LU, NL, UK
- 10: AT, BG, HR, HU, SI, SK, UA
- 11: CA, KZ, MN
- 12: BO
- 13: IS
- 14: BR, GT, PE, RW, UG
- 15: KG, KR, MK, RO, RS, TR, US

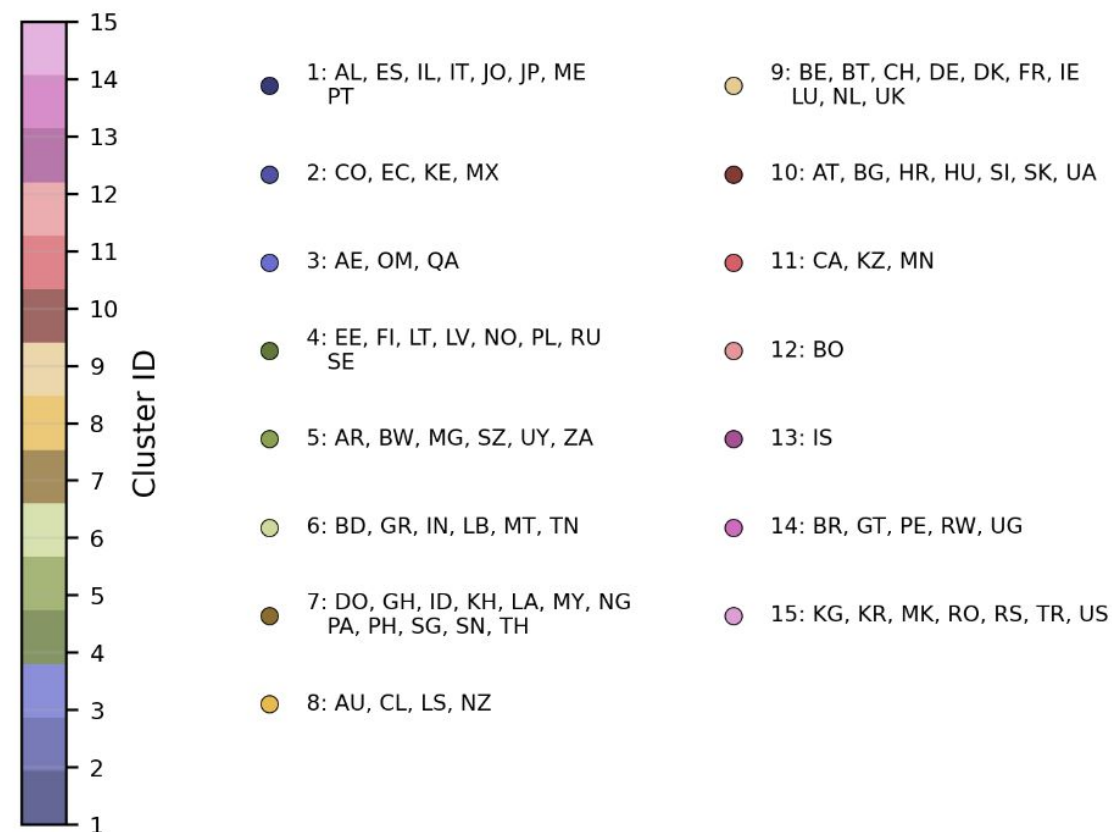
Temperature only clusters in feature space

Each point represents one country. The color indicates the assigned temperature-only cluster. Countries with similar seasonal temperature patterns are grouped together, even if they are geographically far apart

Temperature-only clusters for K = 15



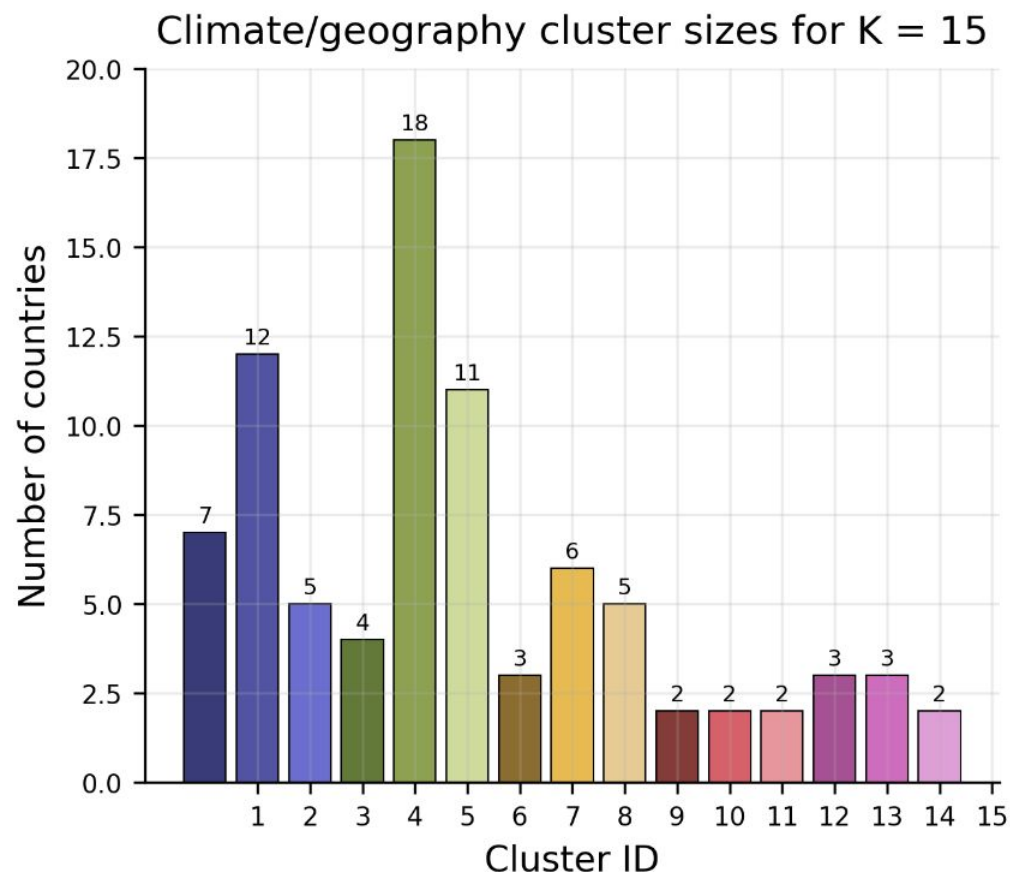
Cluster overview



Climate/geography cluster size overview

Countries clustered using latitude, longitude, Jan.- and July temp. The plot shows the size of each cluster, while the overview on the right shows which countries are assigned for each group.

Adding geographic information changes the grouping compared to temp.-only clustering. The cluster sizes are still uneven.



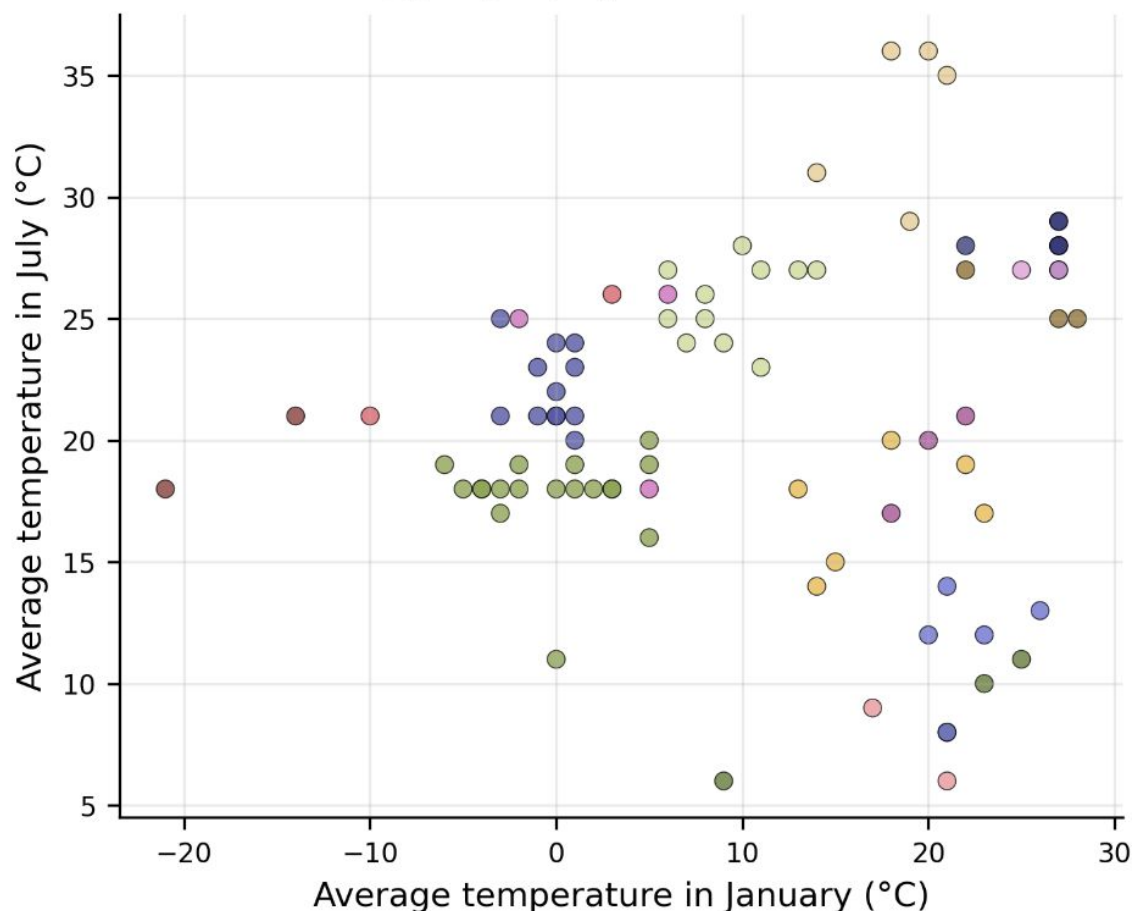
Cluster overview

- | | |
|---|-----------------------|
| 1: ID, KH, LA, MY, PH, SG, TH | 9: AE, BD, IN, OM, QA |
| 2: AT, BG, HR, HU, KG, MK, RO
RS, SI, SK, TR, UA | 10: KZ, MN |
| 3: BW, LS, MG, SZ, ZA | 11: CA, US |
| 4: AR, BO, CL, UY | 12: AU, NZ |
| 5: BE, CH, DE, DK, EE, FI, FR
IE, IS, LT, LU, LV, NL, NO
PL, RU, SE, UK | 13: KE, RW, UG |
| 6: AL, ES, GR, IL, IT, JO, LB
ME, MT, PT, TN | 14: BT, JP, KR |
| 7: GH, NG, SN | 15: DO, PA |
| 8: BR, CO, EC, GT, MX, PE | |

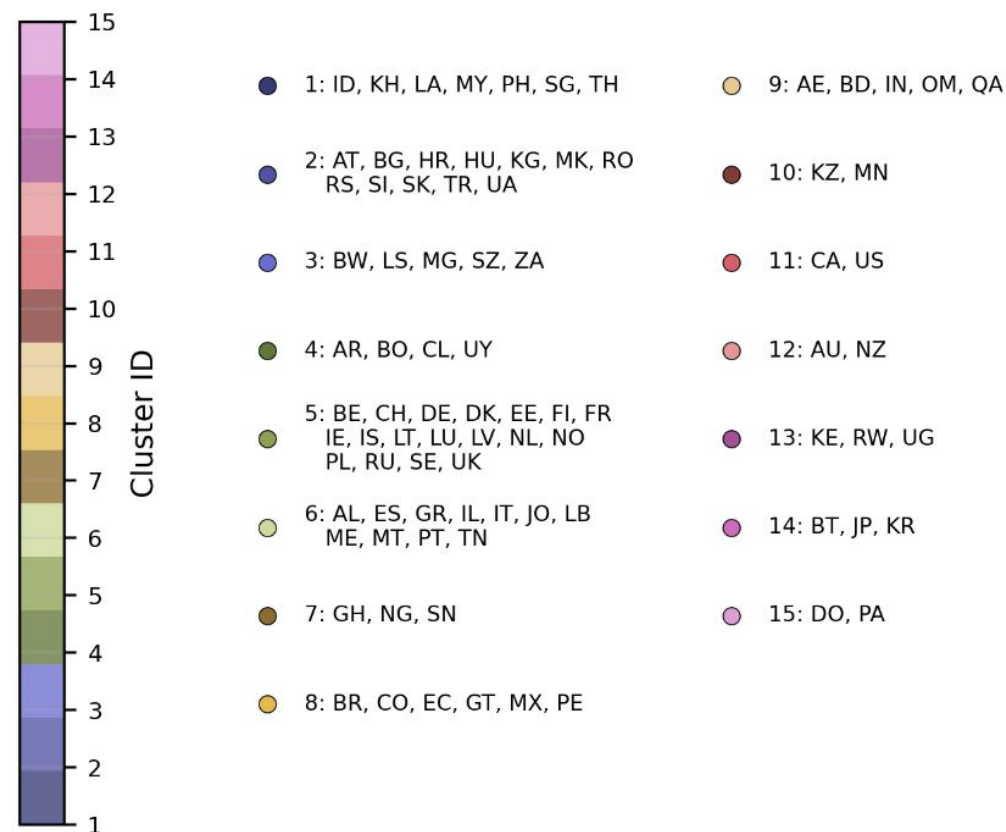
Climate/geography only clusters in feature space

Each point is still plotted by Jan and July temperature, but the clusters were created using both temp. and geography. This means countries with similar temperatures can now belong to different clusters if they differ in latitude or longitude.

Climate/geography clusters for K = 15



Cluster overview



Uneven cluster size overview

The cluster size plots show how many countries are assigned to each cluster.

Uneven cluster sizes are important because large clusters may constrain more countries and more visual variation, while small clusters may be more specific but less representative.

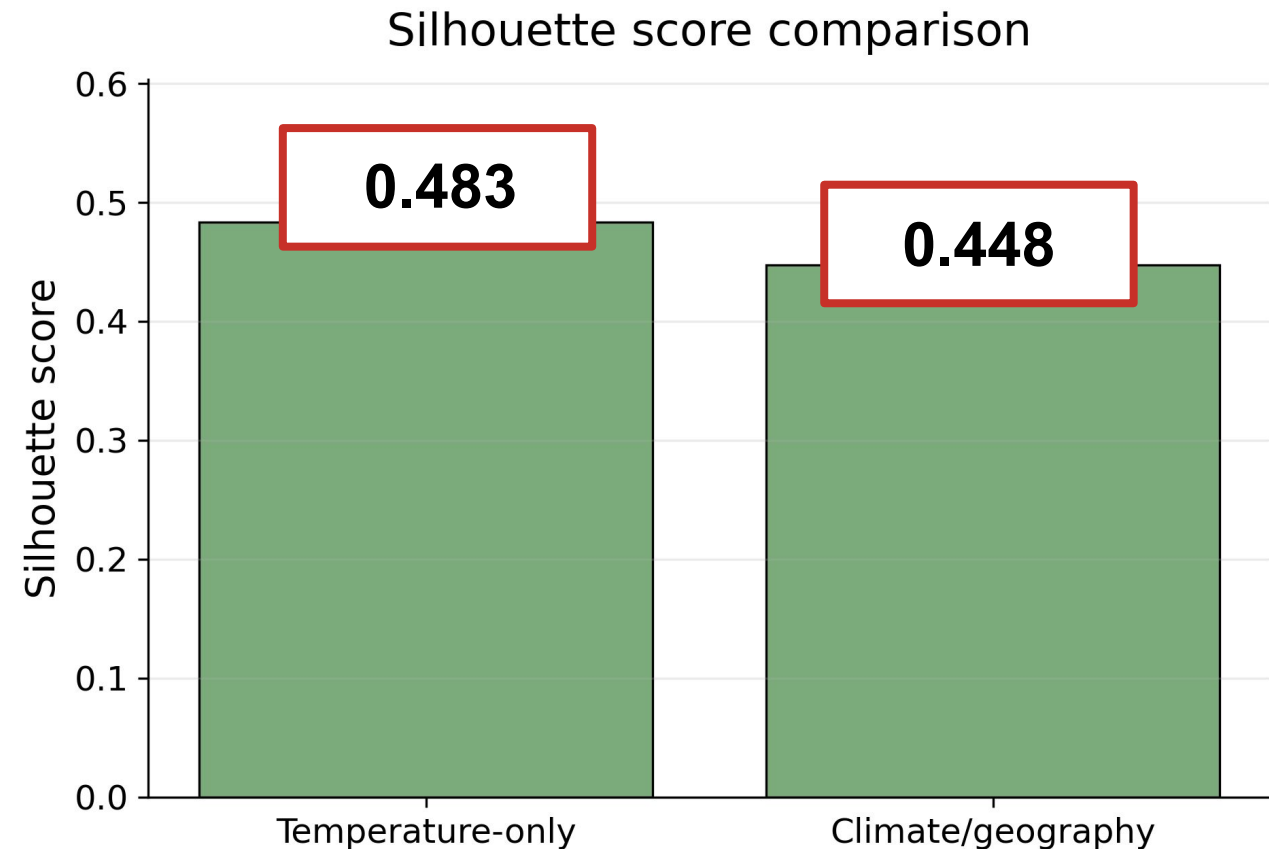
Comparing clustering

Silhouette score comparinon

Measure of how well
separated the clusters are

Higher score = more compact
and clearly separated clusters

Temp only > Climate/geography

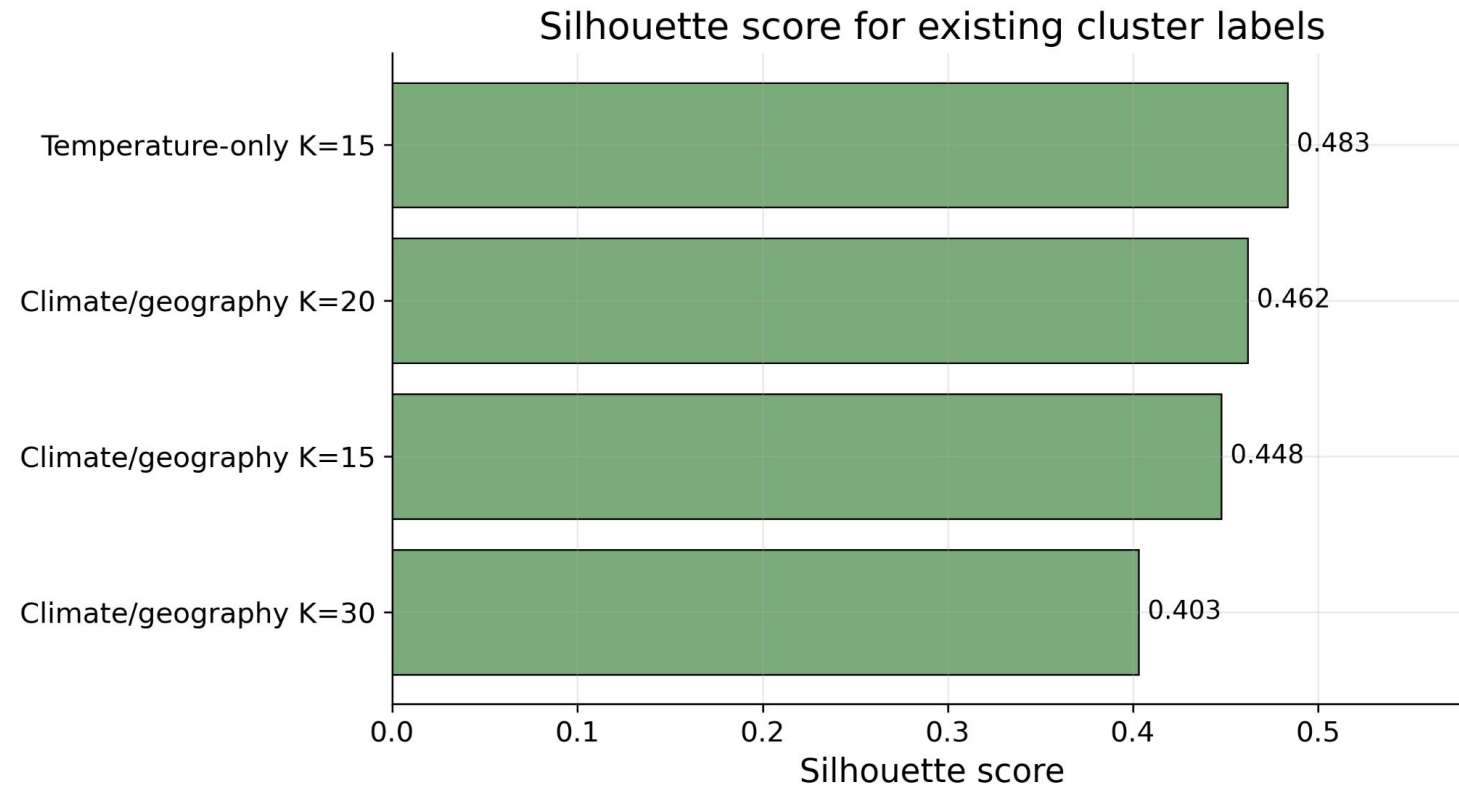


Cluster

Silhouette score for cluster label systems

Silhouette score is used to measure how well separated the clusters are. It compares how close each country is to

A higher score indicates more compact and better separated clusters. **The temperature-only clustering obtains the highest silhouette score,** suggesting that countries are more clearly separated when only January and July temperatures are used. However, the climate/geography clusters may still be more meaningful for image classification, because they also include spatial information through latitude and longitude

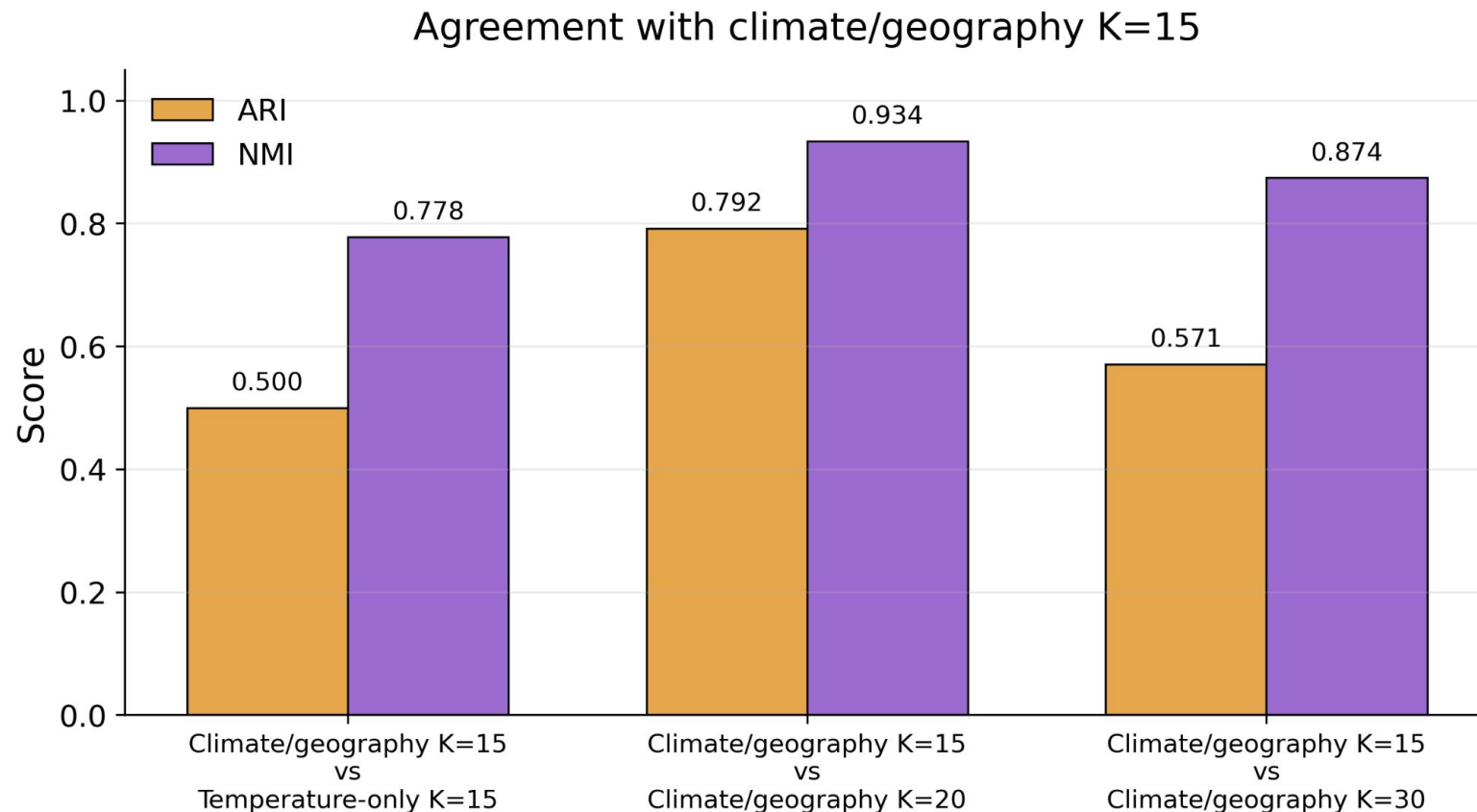


Cluster - Agreement with climate/geography K=15

ARI and NMI

ARI and NMI compare how similar two cluster label systems are. Here, the climate/geography clustering with K=16 is used as the reference and compared with the other clusterings.

ARI measures exact agreement in how countries are assigned to clusters, adjusted for random agreement. NMI measures how much information the two clusterings share.

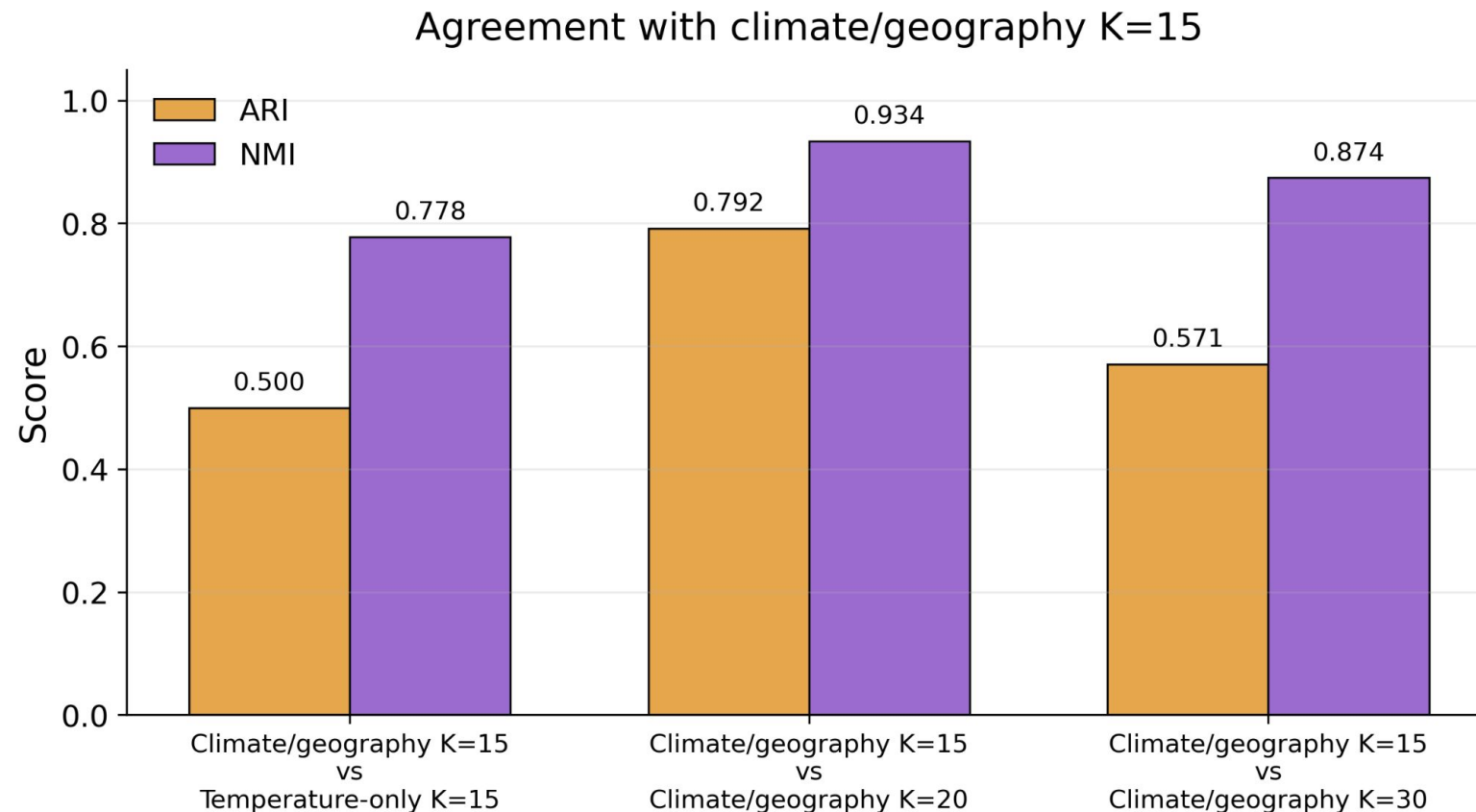


Cluster - Agreement with climate/geography K=15

ARI and NMI

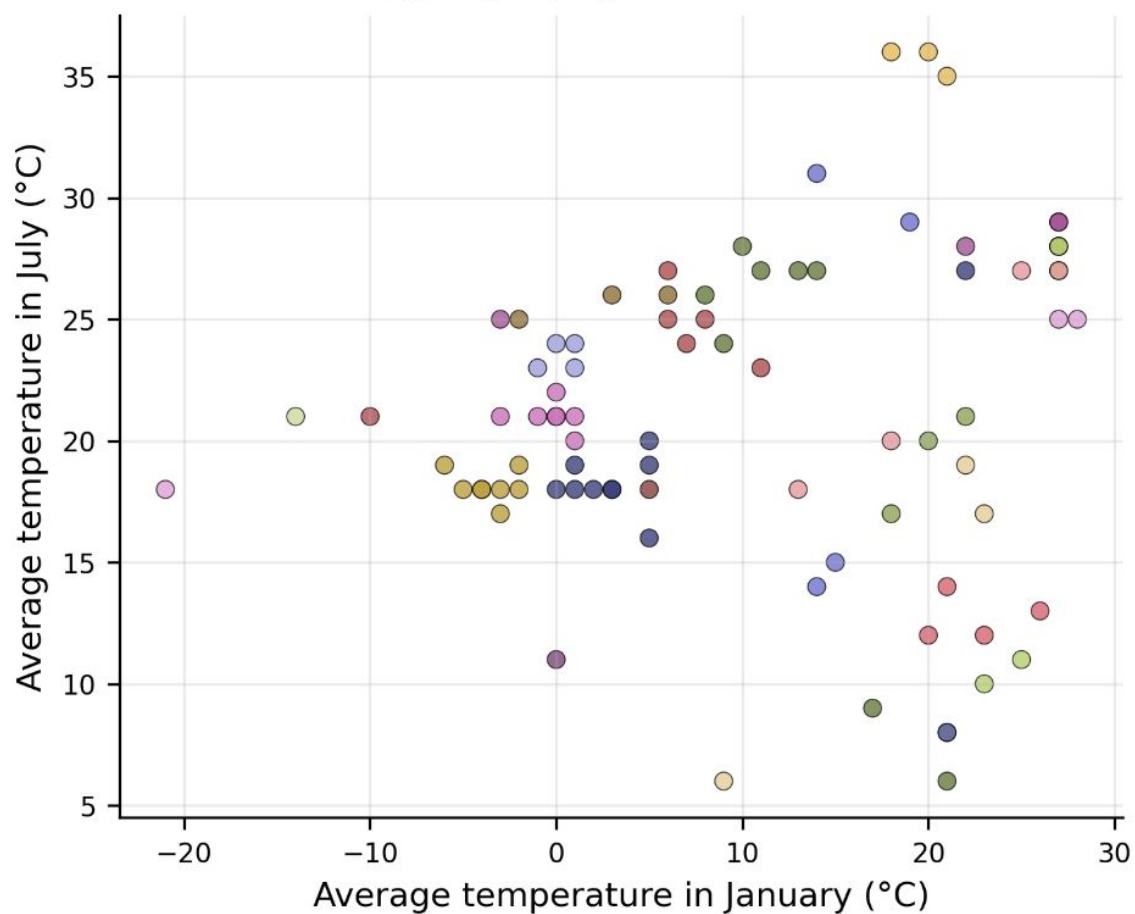
The climate/geography K=20 clustering has the strongest agreement with K=15 which is expected because both are based on the same climate/geography features. The temperature only clustering has lower

ARI but still relatively high NMI, meaning that temperature explains part of the structure, but does not give exactly the same country groupings.



Cluster for K=30

Climate/geography clusters for K = 30

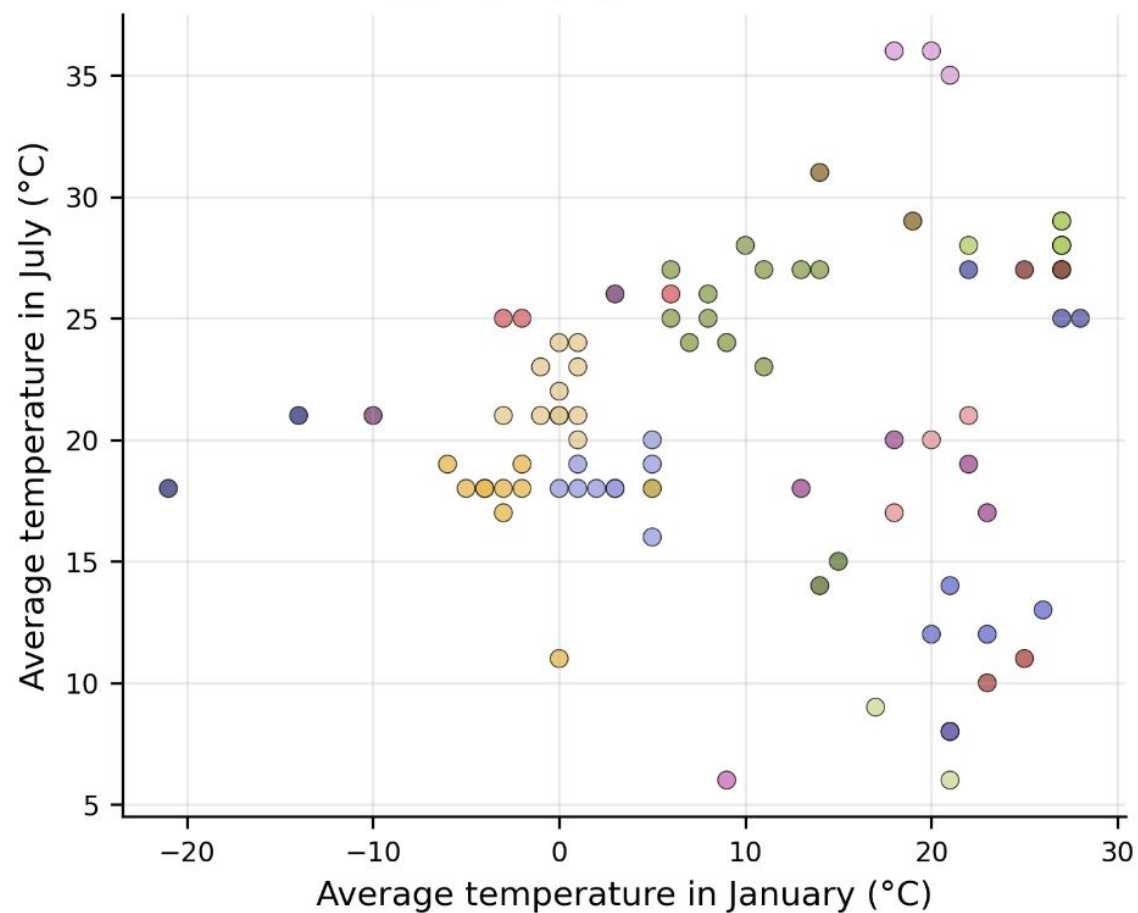


Cluster overview

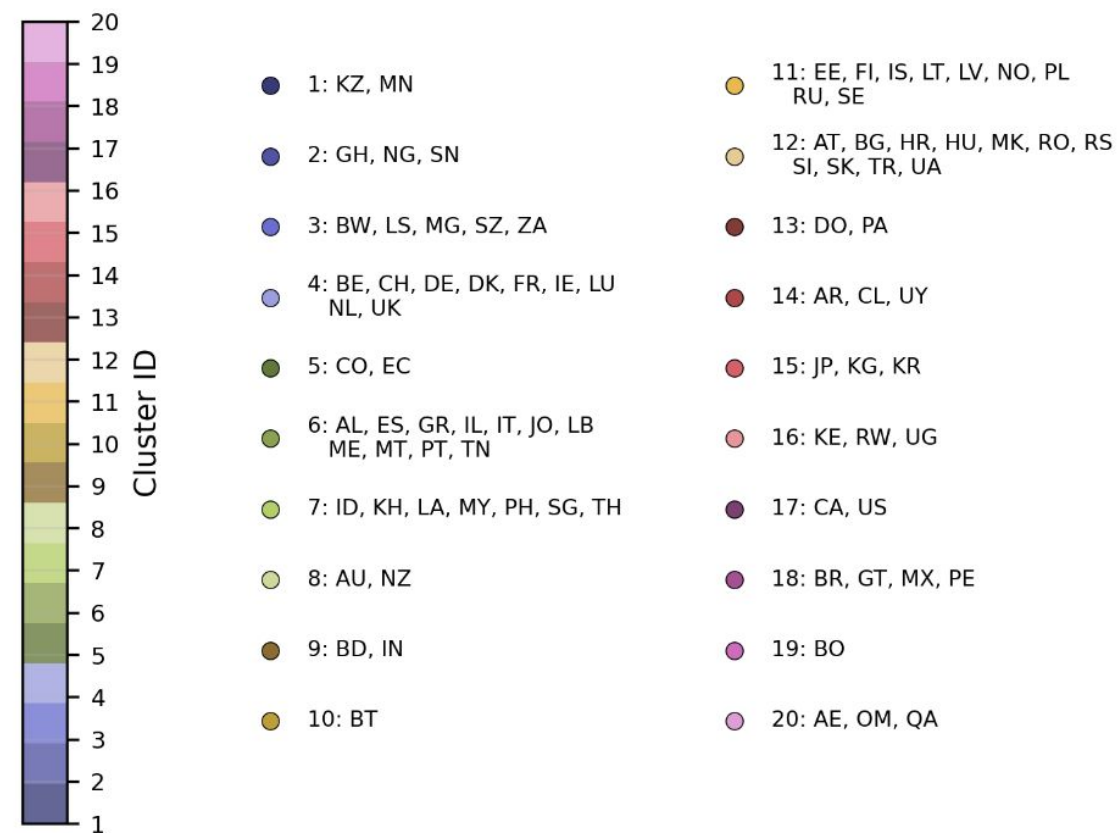


Cluster for K=20

Climate/geography clusters for K = 20



Cluster overview



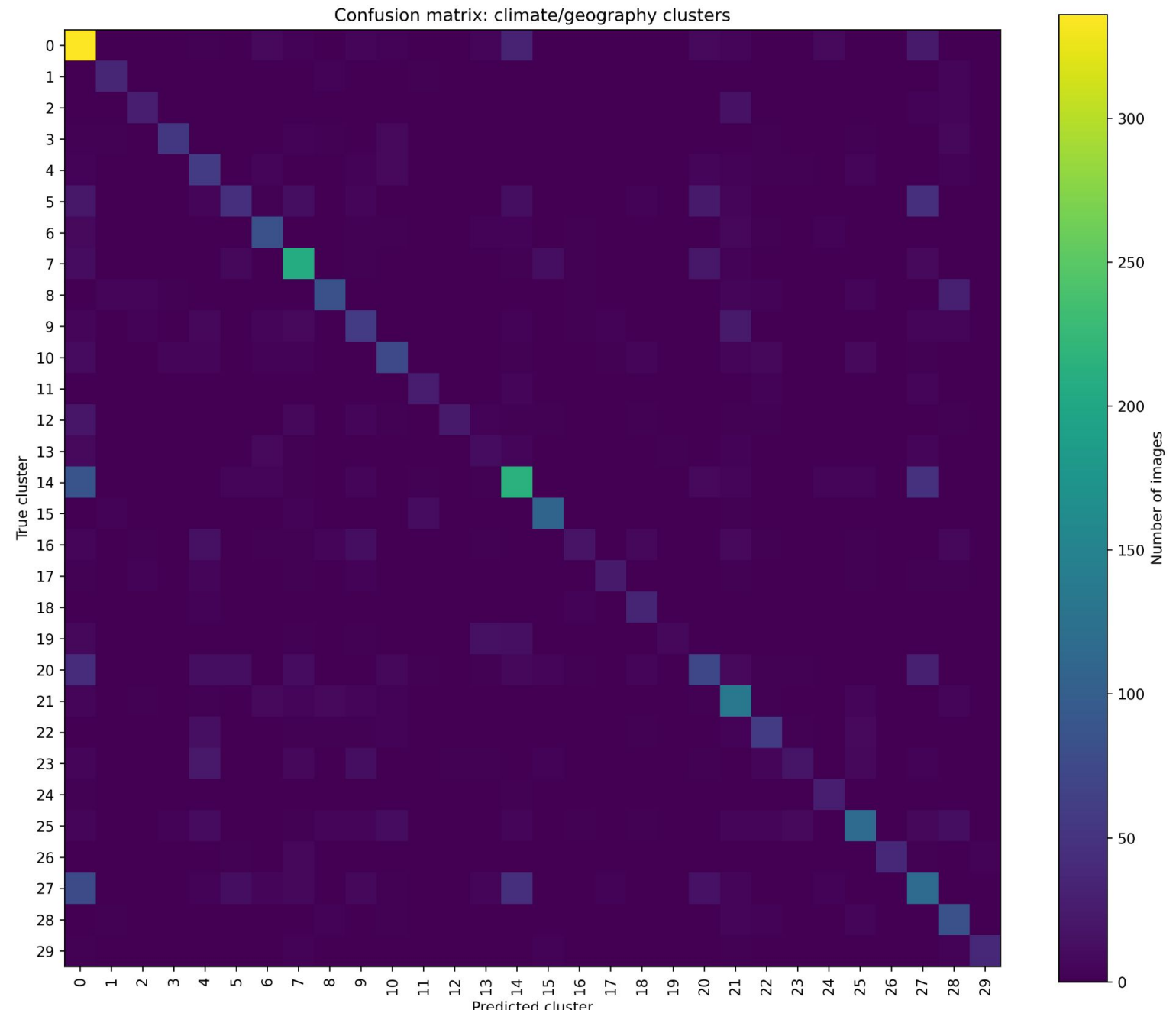
Confusion matrix for climate/geography

For the K=30 model

Most predictions lie on the diagonal, showing that the model generally predicts the correct cluster.

The off-diagonal cells indicate remaining confusion between some clusters.

Takeaway: The cluster model works overall, but not all clusters are equally separable

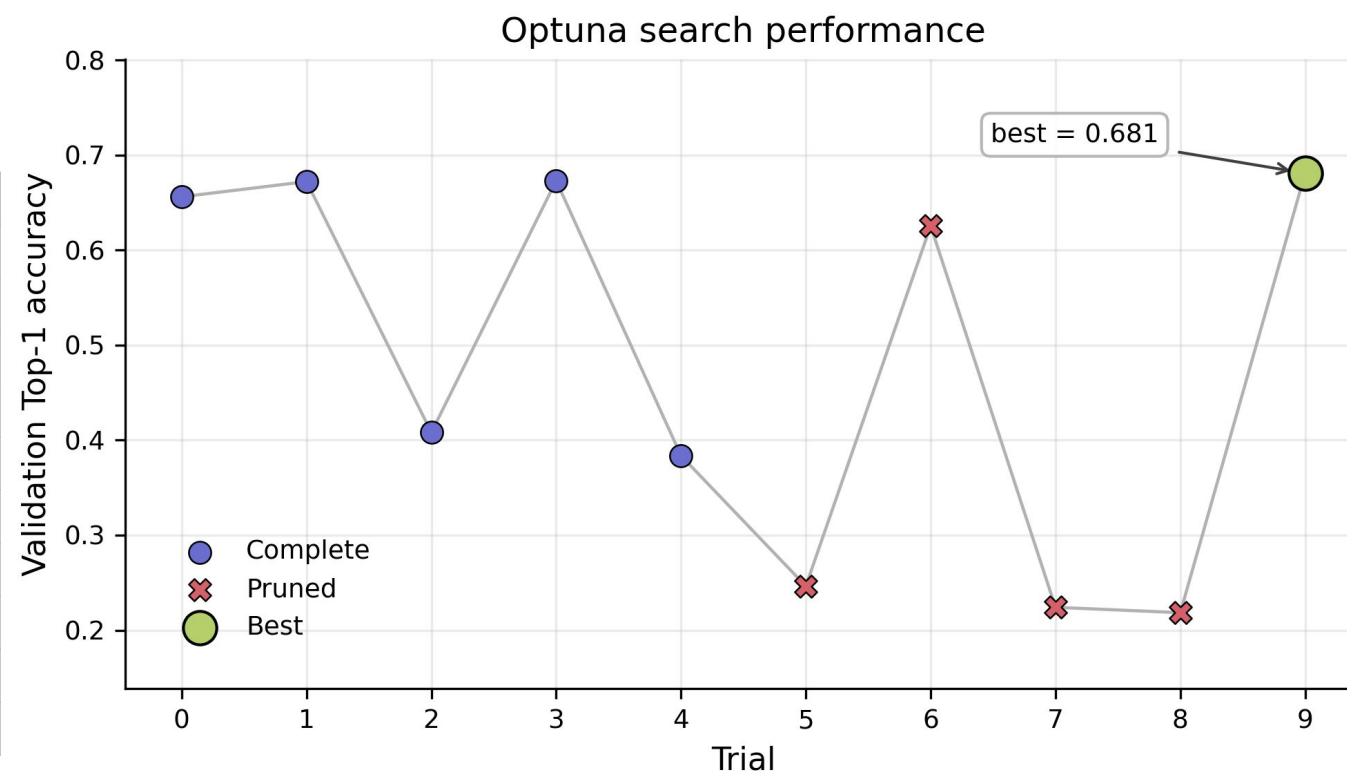


Hyperparameter optimization with Optuna

Optuna was used to tune the ResNet34 climate/geography (K=15) model. The objective was to maximize validation Top-1 accuracy. The search tested combination of learning rate, dropout, optimizer, weight decay and whether to freeze the backbone. The best complete trial reached validation **Top-1 accuracy (=0.681)**. This optimized model was used for the final cluster based evaluation.

Best Optuna hyperparameters
ResNet34 climate/geography K=15

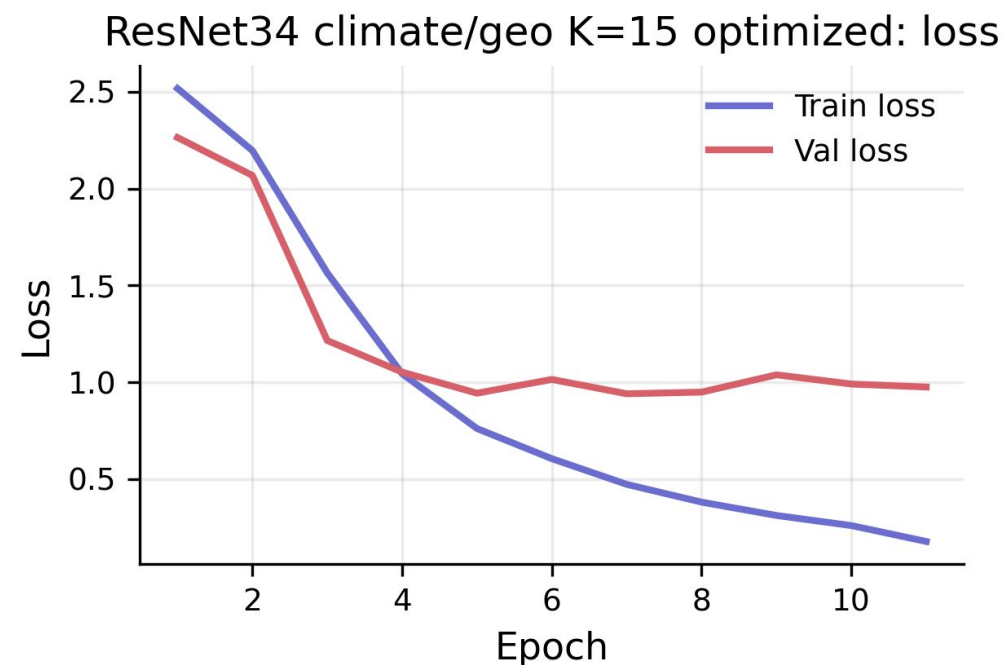
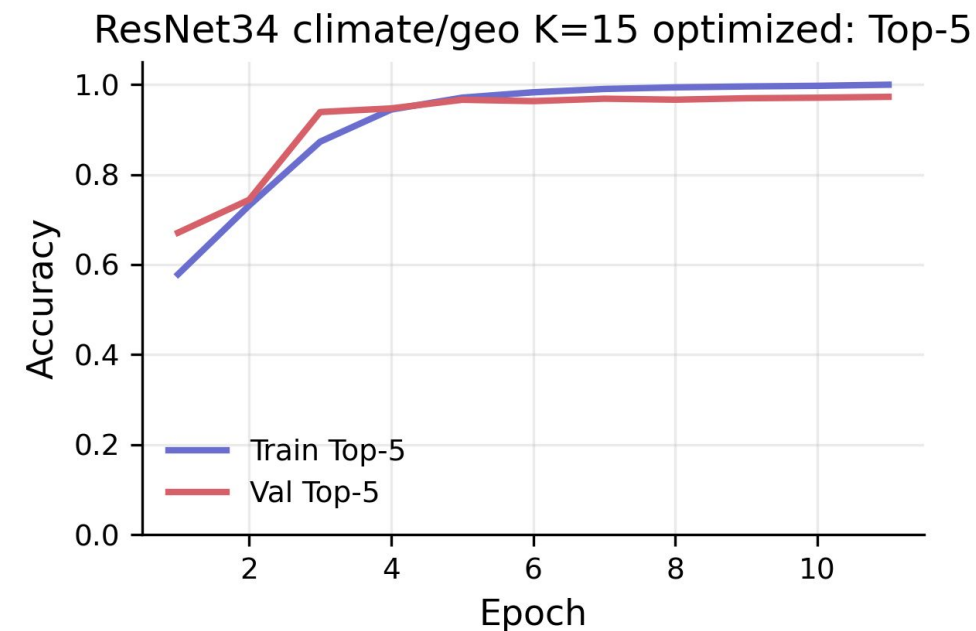
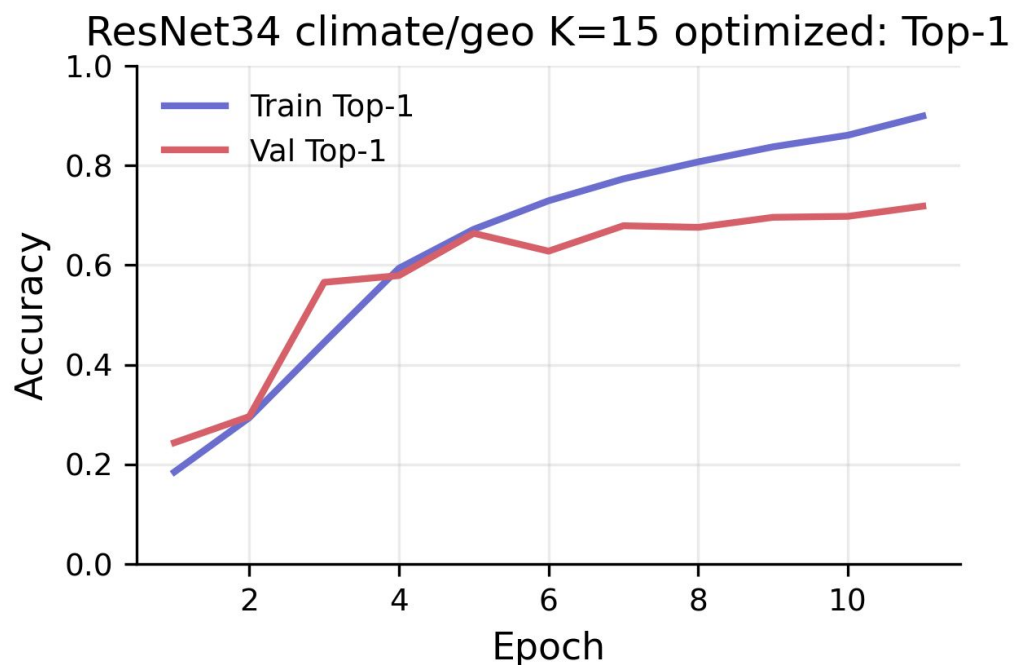
Quantity	Value
Best trial type	Best complete trial
Trial number	9
Validation Top-1	0.6810
State	COMPLETE
batch size	32
dropout	0.4951
freeze backbone	False
lr	3.39e-05
optimizer	AdamW
weight decay	1.31e-04



Training curves for optimized cluster model

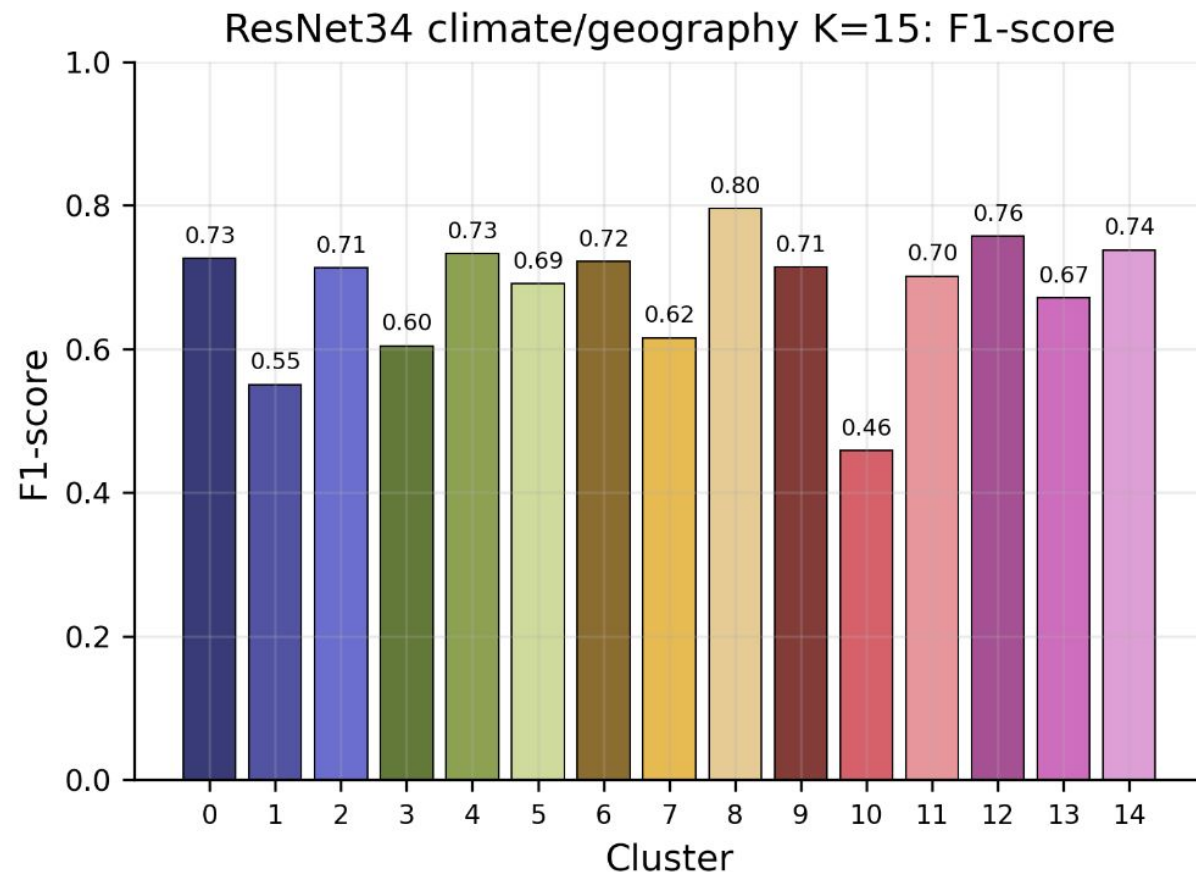
The optimized ResNet34 climate/geography (K=15) model improves during training. Top-1 validation accuracy increases and then kind of stabilizes, while Top-5 accuracy becomes very high.

The training loss continues to decrease, while validation loss levels off, indicating some overfitting but stable validation performance.



F1-score per cluster

The optimized climate/geogrpagy K=15 model does not perform equally well across all clusters. Most clusters have relatively high F1-scores, but a few clusters are noticeably lower. This suggests that some clusters are more visually coherent, while others constrain countries that are harder to separate. Ex. USA/Canda cluster may score lower because it's visually very diverse.



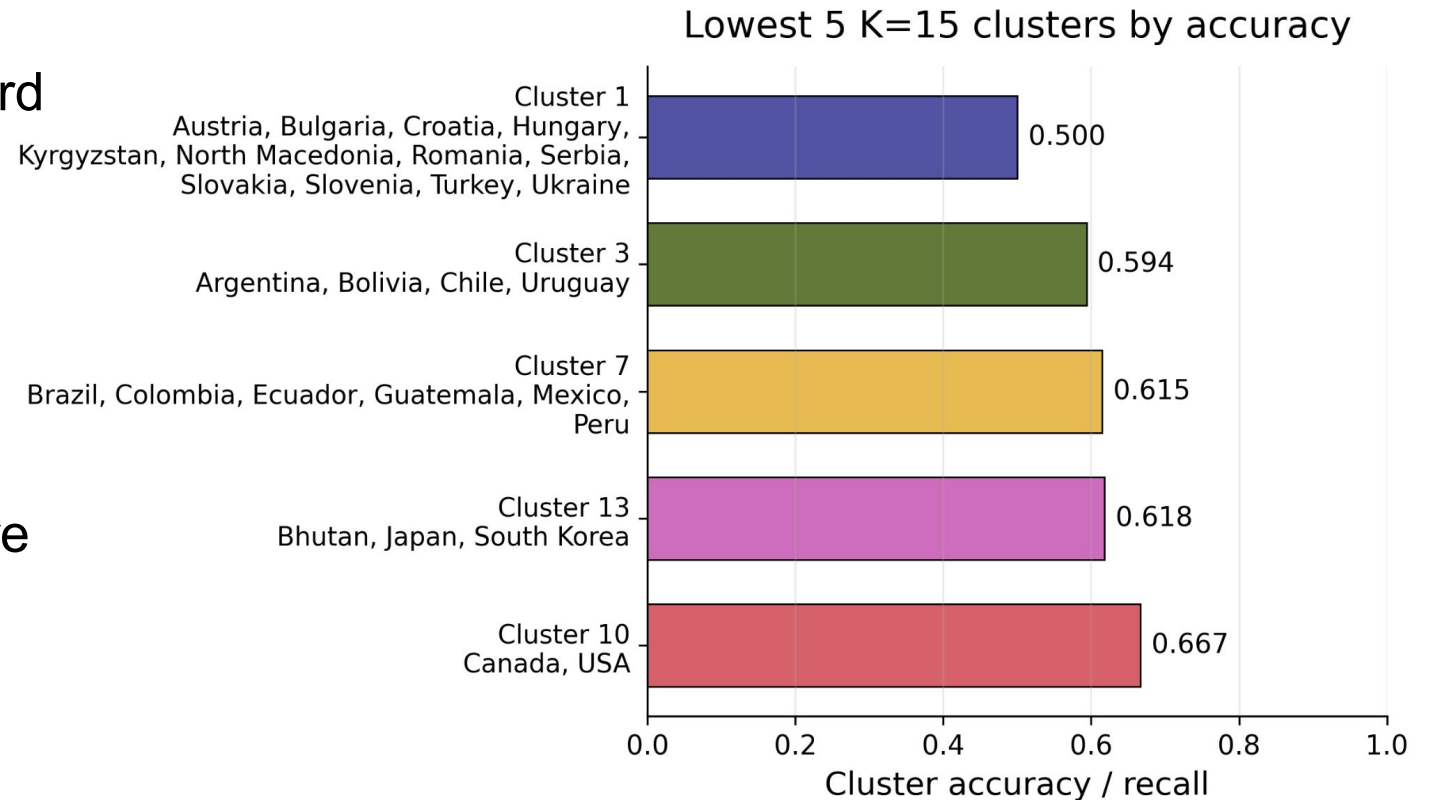
Cluster overview

- 0: ID, KH, LA, MY, PH, SG, TH
- 1: AT, BG, HR, HU, KG, MK, RO, RS, SI, SK, TR, UA
- 2: BW, LS, MG, SZ, ZA
- 3: AR, BO, CL, UY
- 4: BE, CH, DE, DK, EE, FI, FR, IE, IS, LT, LU, LV, NL, NO, PL, RU, SE, UK
- 5: AL, ES, GR, IL, IT, JO, LB, ME, MT, PT, TN
- 6: GH, NG, SN
- 7: BR, CO, EC, GT, MX, PE
- 8: AE, BD, IN, OM, QA
- 9: KZ, MN
- 10: CA, US
- 11: AU, NZ
- 12: KE, RW, UG
- 13: BT, JP, KR
- 14: DO, PA

Cluster

Lowest 5 clusters by accuracy

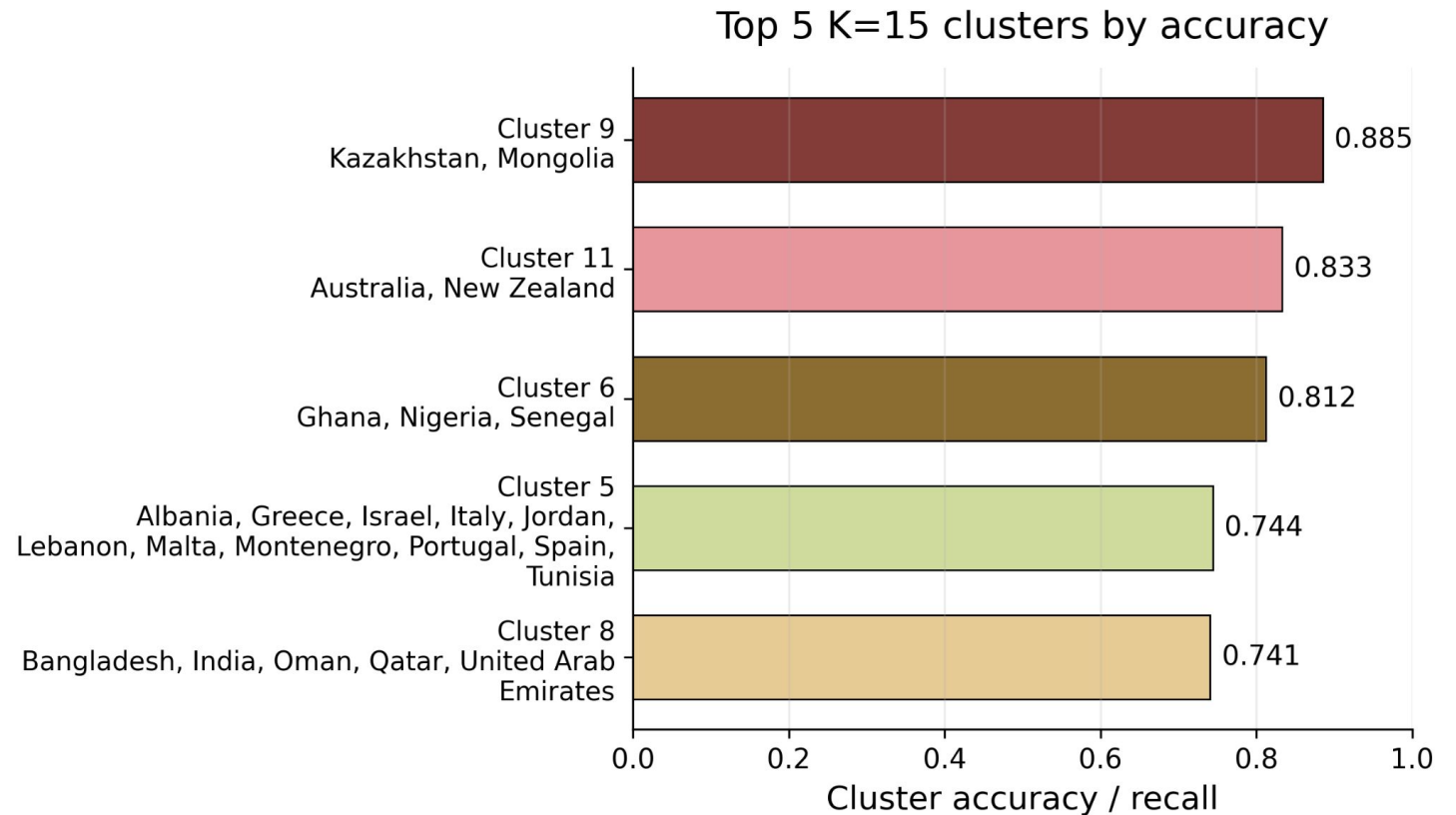
- Several low performing clusters contain countries that were also hard individually
- E.g. Croatia, Slovakia, Brazil, and Mexico
- Larger or visually diverse lusters are harder for the CNN to learn
- Some country-level difficulties disappear when countries are grouped into broader regions



Cluster

Best 5 performing clusters

- Several high performing clusters contain countries that were also easier in the country level model
- Eg. Mongolia, Ghana, Senegal, Bangladesh, India and Qatar
- Broader clusters labels can be easier than exact country prediction



Cluster - Conclusion

Overall, the clustering approach reduces the complexity of the classification task by grouping countries according to climate and geographical similarity. The temperature-only clusters are slightly more compact in metadata space according to the silhouette score, while the climate/geography clusters provide more regional structure.

The final comparison is therefore based on mainly on CNN performance, especially validation accuracy, Top-1 accuracy, F1-score and confusion matrices. Since the cluster based model performs better than the original country level model, this suggests that climate/geography based labels provide a useful structure for image classification.

However, the clustering is still based on simplified country level averages. For example, large countries such as the USA can contain very different landscapes, climates and visual environments, which may not be well represented by a single average temperature or coordinate value

Cluster - Future work

A possible improvement would be to include more detailed country metadata, such as precipitation, vegetation, elevation and regional climate variation. This could make the clusters more visually meaningful and potentially improve classification performance.

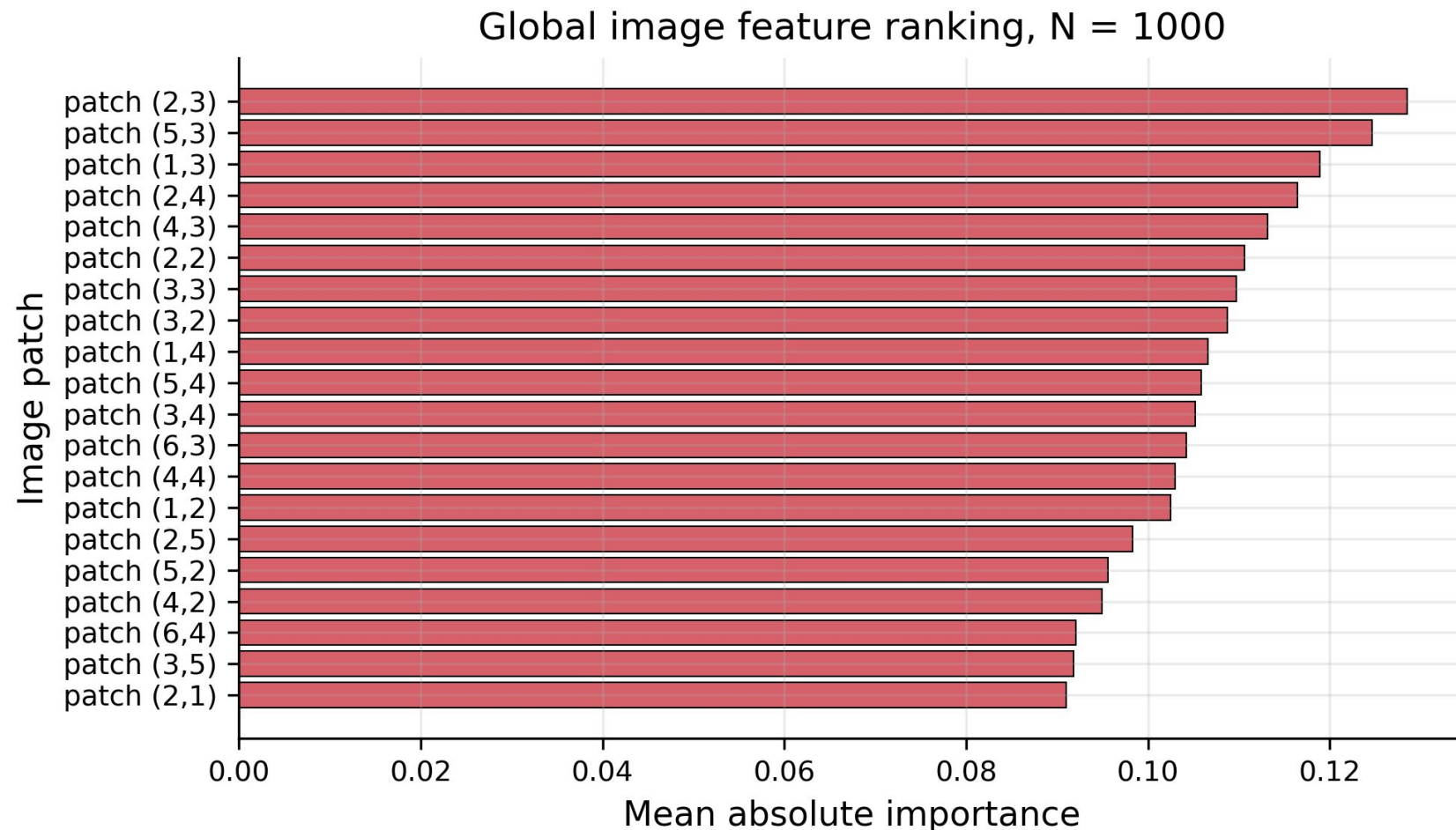
Dataset pollution: Car artefacts Rwanda



Patch importance

Which patches are most important

- Computed across (N=1000) validation images
- Each image was divided into patches
- One acth was removed at a time
- Importance = how much the rendition confidence changed
- The most important patches are mainly in the central part of the image



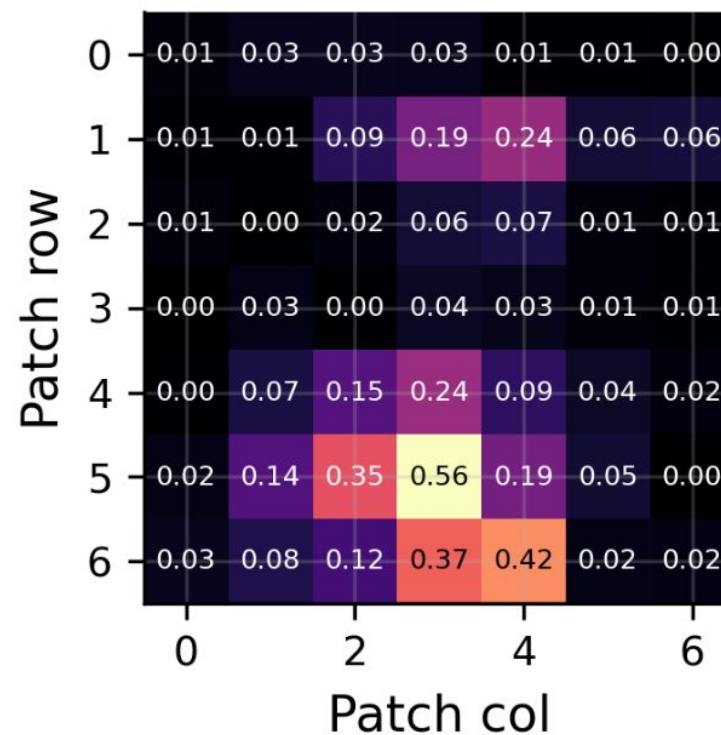
Cluster

Mexico | true cluster 8 | pred cluster 8

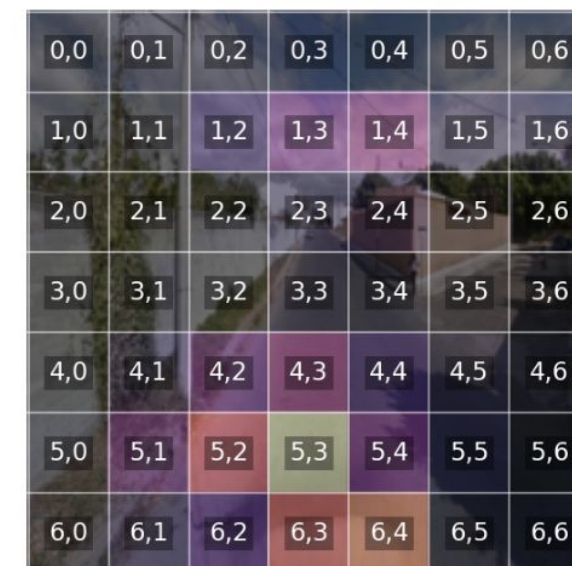
Original



Patch importance



Overlay + patch IDs



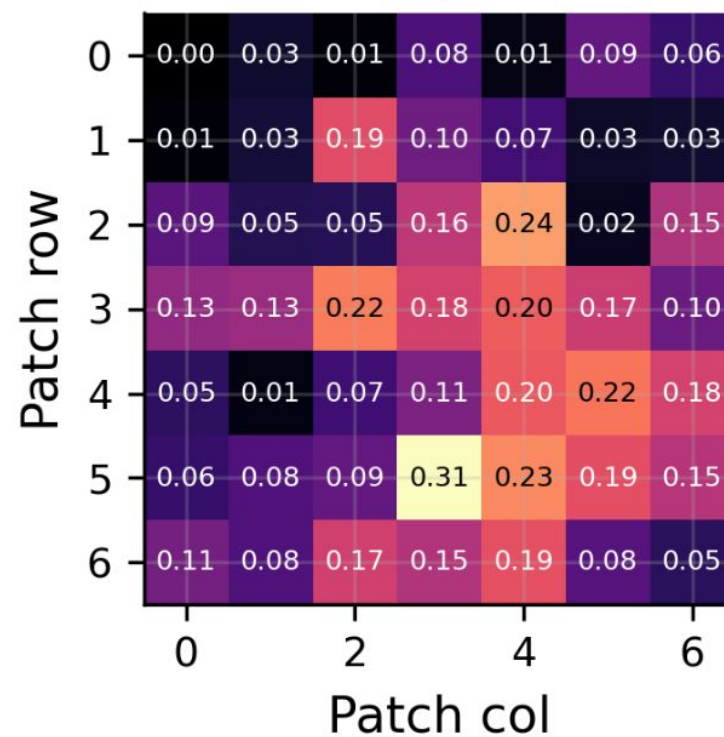
Cluster

Denmark | true cluster 5 | pred cluster 5

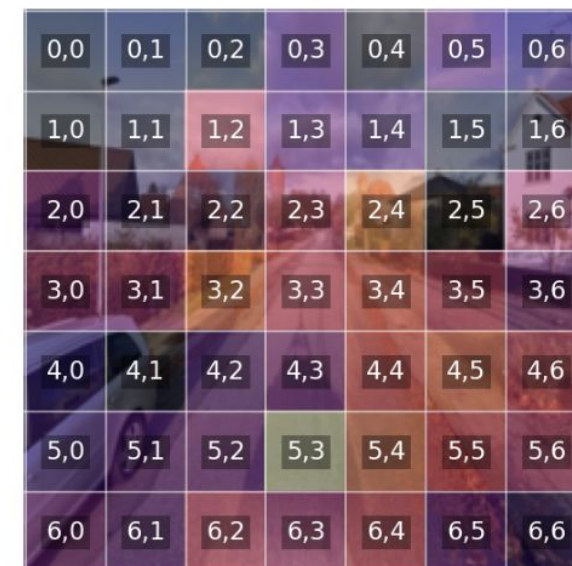
Original



Patch importance



Overlay + patch IDs



Cluster

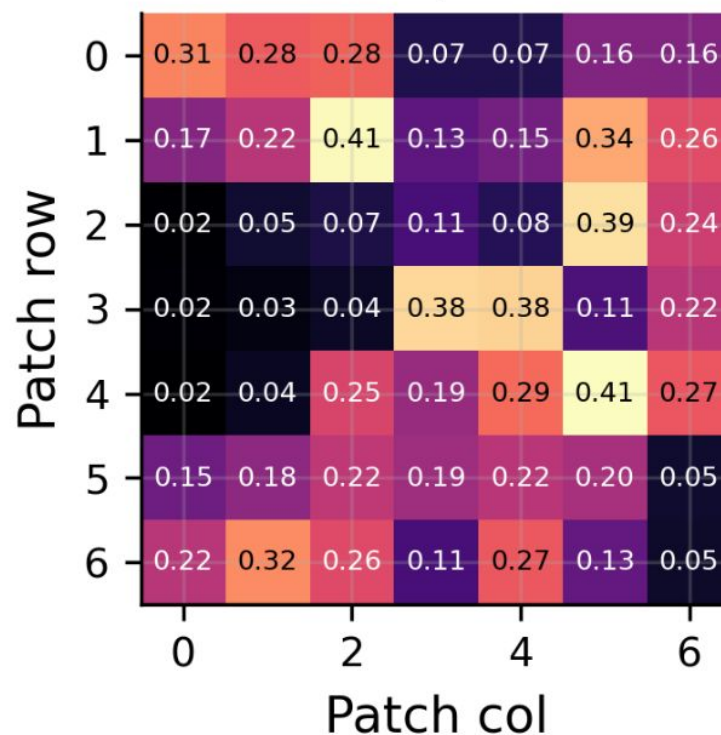
Looks at palm tree, road, house, vegetation

Bolivia | true cluster 4 | pred cluster 8

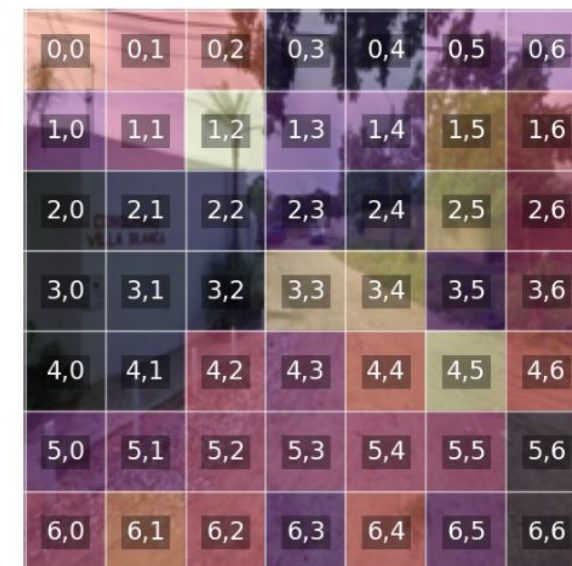
Original



Patch importance



Overlay + patch IDs



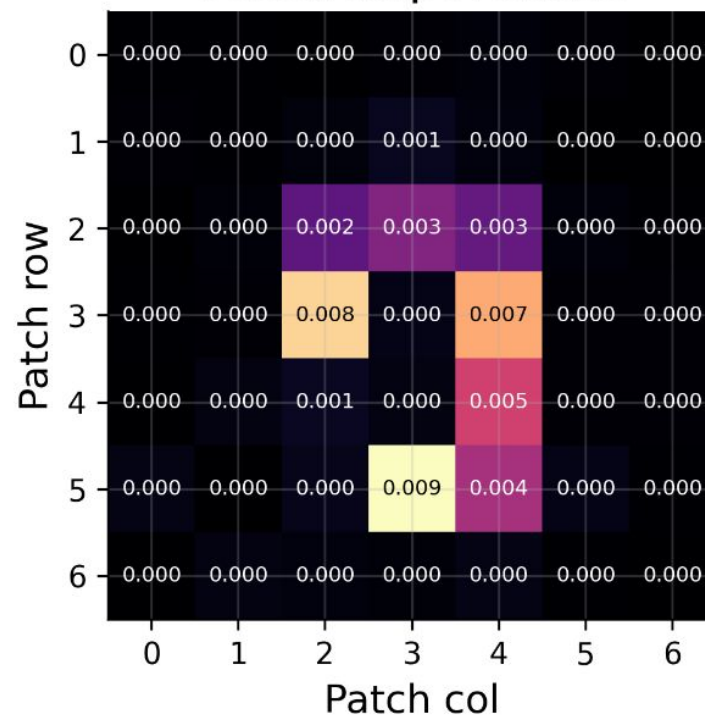
Cluster

Ghana | true cluster 7 | pred cluster 7

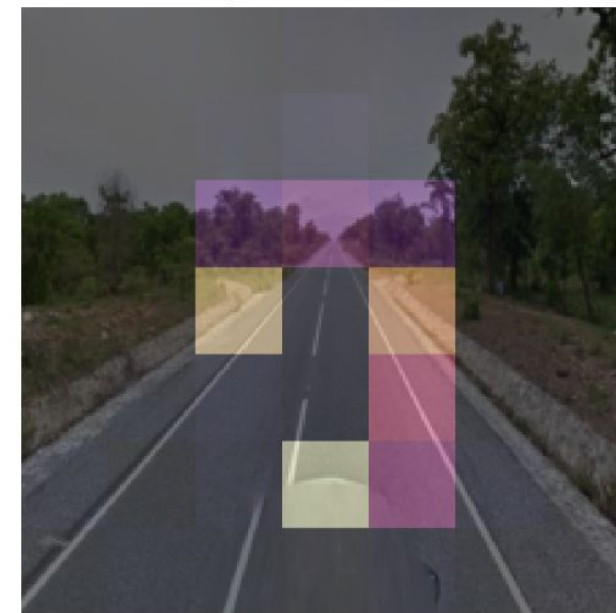
Original



Patch importance



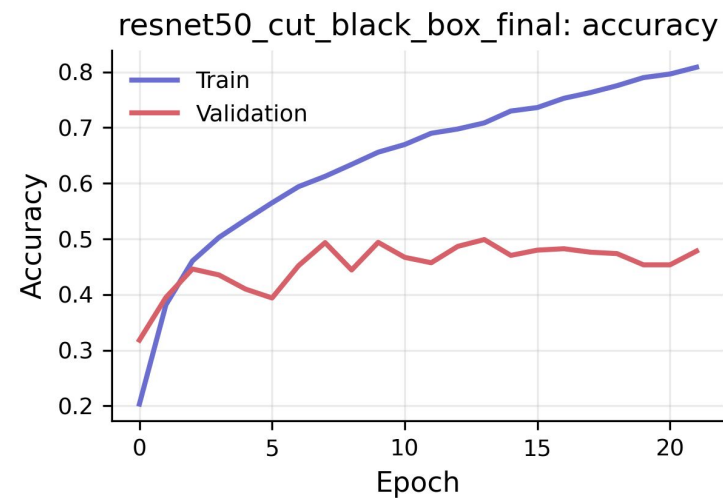
Patch importance overlay





Cluster

Additional experiment with black box



ResNet50 Test Accuracy

Top 1	Top 5
0.502	0.813

